PEDIATRIC INFECTIOUS DISEASE UNIT
FACULTY OF MEDICINE, CHIANG MAI UNIVERSITY

PREVENT-ID
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Breaking the Respiratory Cycle

Understanding the burden, recognizing the signs, and protecting our little ones

The Triple Burden:

Protecting Kids from Respiratory Illness

RSV (Respiratory Syncytial Virus)

A common respiratory virus that usually causes severe respiratory illness in young infants

Influenza (The Flu)

A contagious viral infection that attacks the nose, throat, and lungs.

Pertussis (Whooping Cough)

A highly contagious respiratory tract infection characterized by a severe hacking cough.

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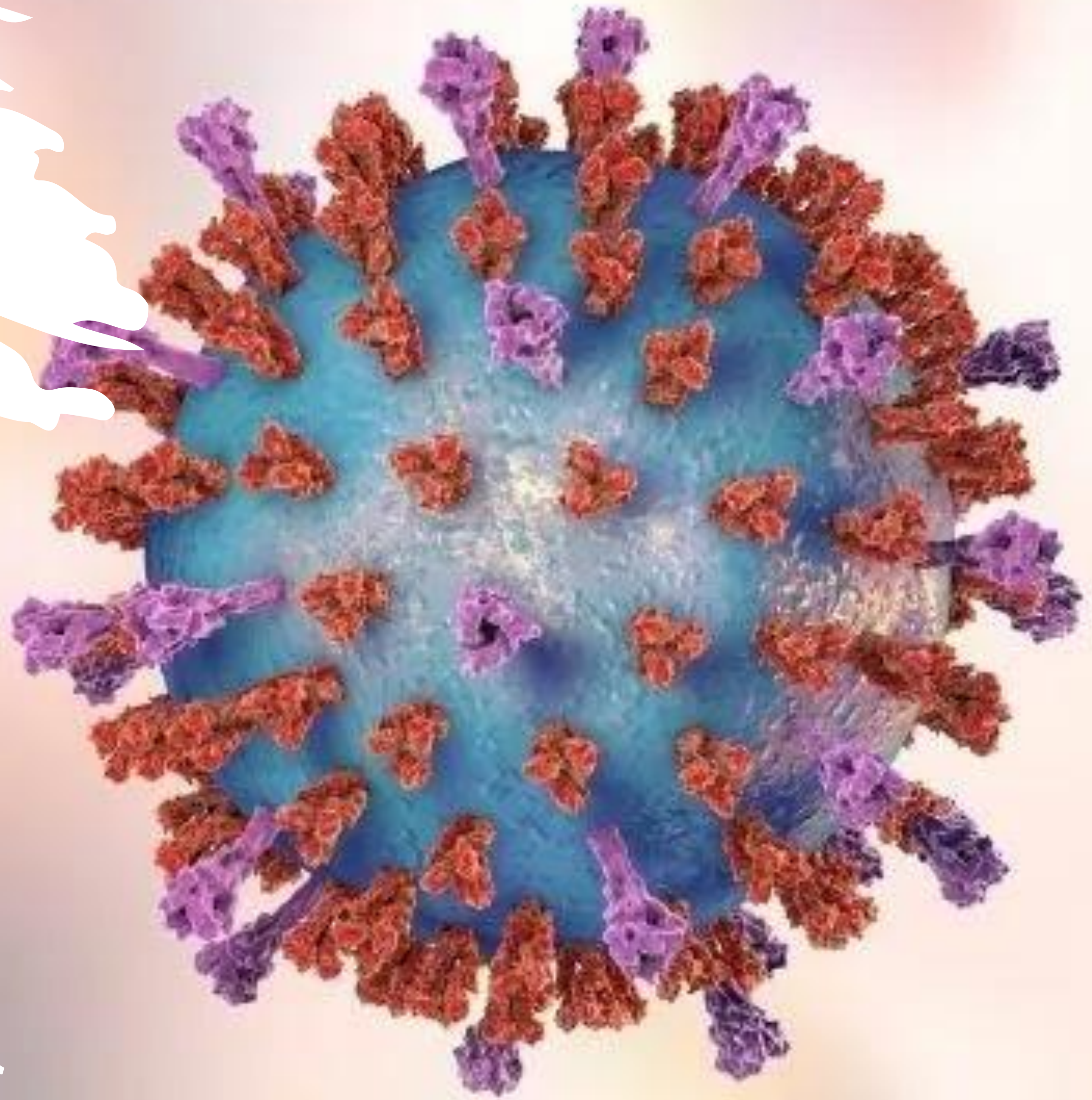
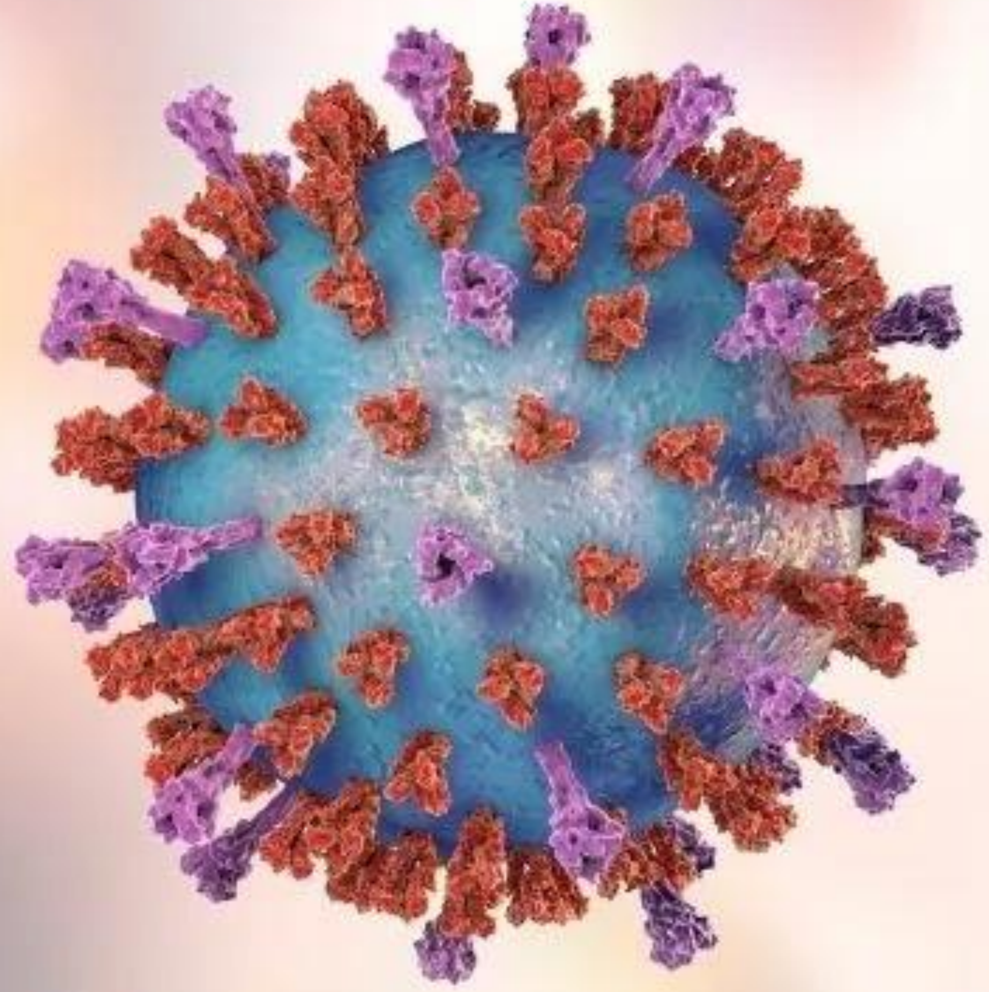


Respiratory syncytial virus

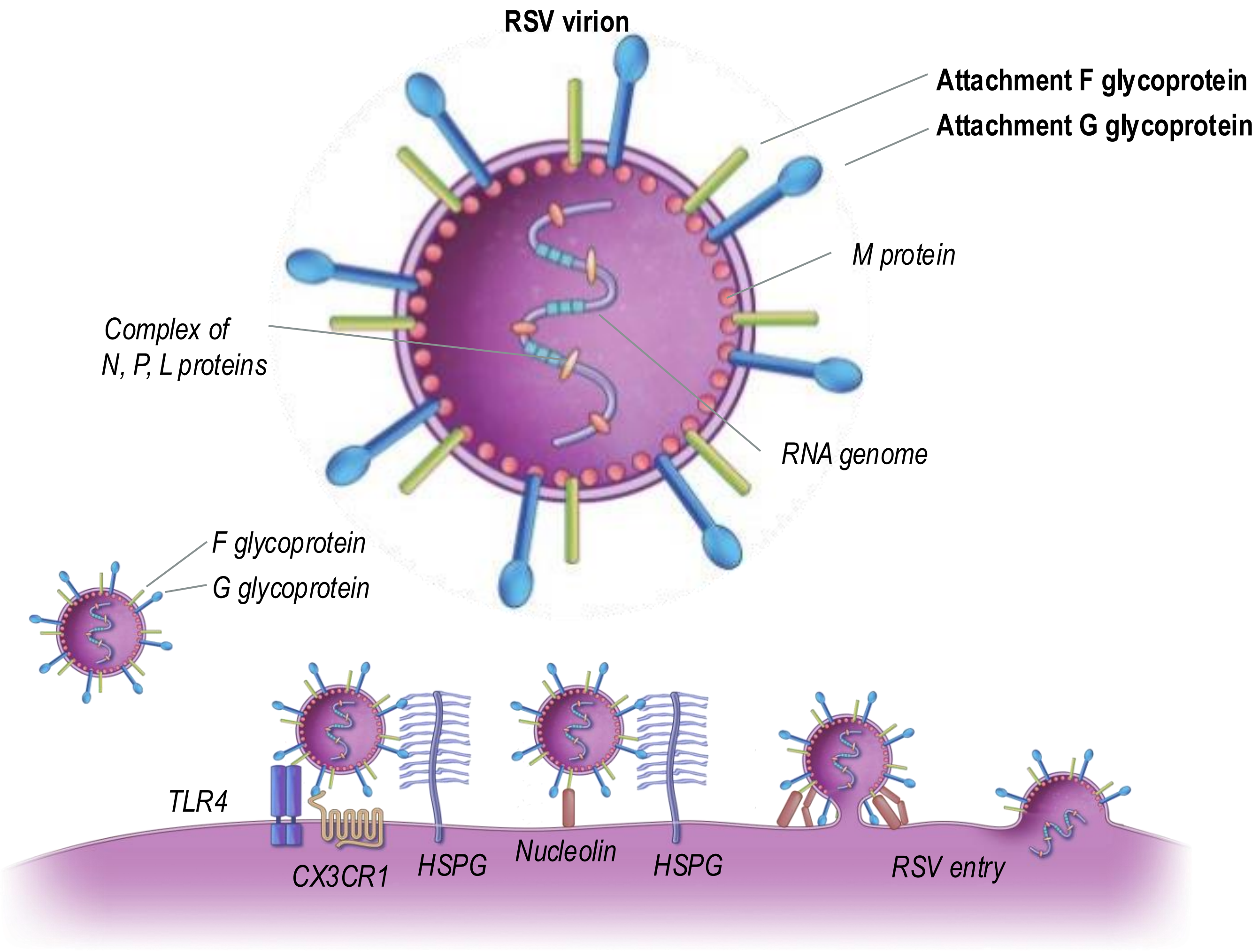


What is RSV?

- **RSV is a respiratory virus** and one of the most common causes of childhood illness, infecting almost all children by 2 years of age
  - An important cause of LRTIs during infancy and early childhood
  - Responsible for the "common cold" in adults and healthy children
- **RSV-associated LRTI rates** vary and account for ~60-80% of infant bronchiolitis in the Northern Hemisphere and ~30% infant pneumonia globally

1. Karron R. In: Plotkin's Vaccines. 7th ed. Elsevier; 2018:943-949. 2. Piedimonte G, Perez MK. Pediatr Rev. 2014;35(12):519-530. 3. Wat D. Eur J Intern Med. 2004;15(2):79-88. 4. Domachowske JB, Rosenberg HF. Clin Microbiol Rev. 1999;12(2):298-309. 5. Bermúdez Barrezueta L, et al. Infection. 2022;50(4):1001-1005. 6. Mazur NI, et al. Lancet Infect Dis. 2023;23(1):e2-e21.

RSV structure, function, and infection



Single-stranded RNA virus with 2 major glycoproteins on its surface known as:

- The attachment (G) glycoprotein
- The fusion (F) glycoprotein

F and G glycoproteins are mediators of RSV infection

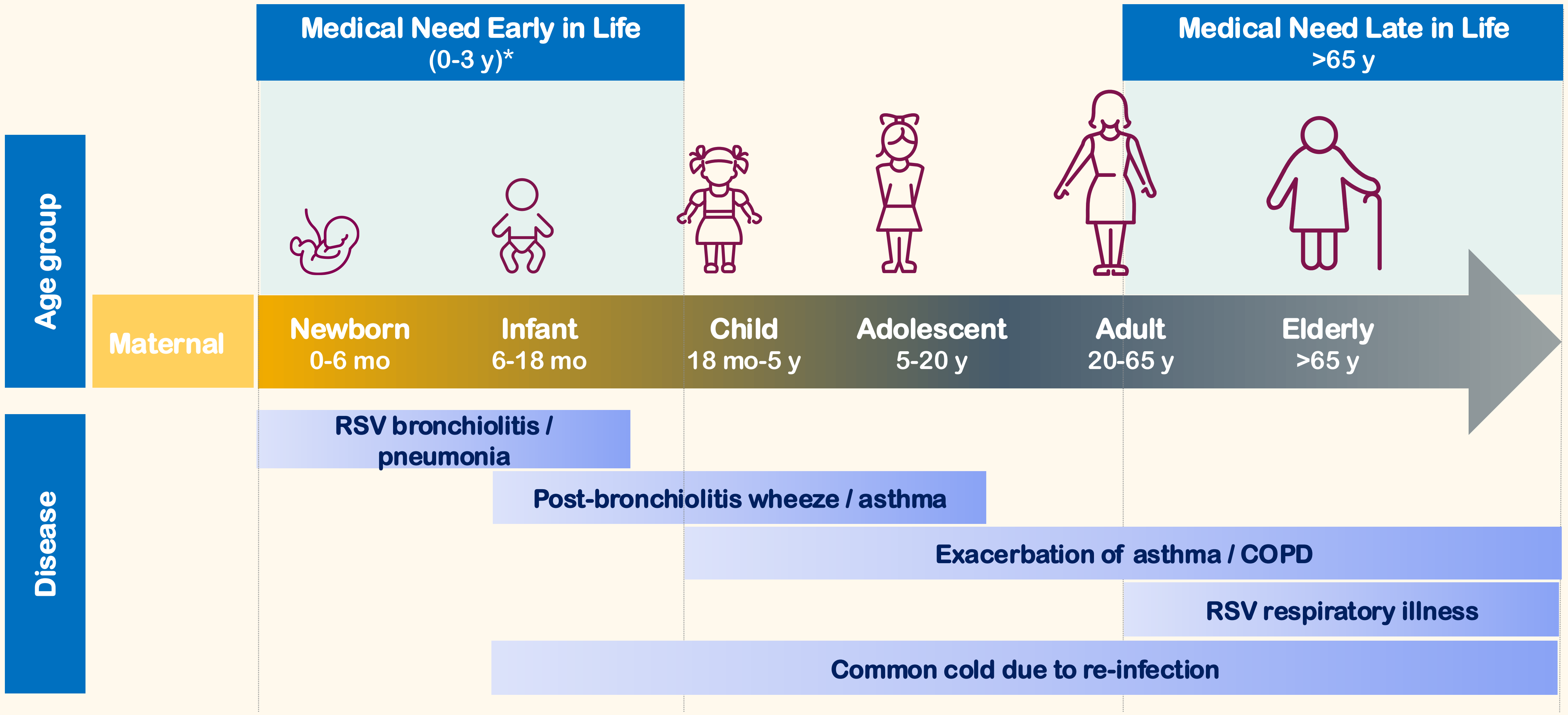
Adapted from Rezaee F, et al. Curr Opin Virol. 2017;24:70-78 and Griffiths C, et al. Clin Microbiol Rev. 2017;30(1):277-319.
CX3CR1: CX3 chemokine receptor 1; HSPG: heparan sulphate proteoglycans; RNA: ribonucleic acid; RSV: respiratory syncytial virus; TLR4: toll-like receptor 4.

1. Rezaee F, et al. Curr Opin Virol. 2017;24:70-78. 2. Griffiths C, et al. Clin Microbiol Rev. 2017;30(1):277-319.

The Lifecycle of an RSV Infection

Redbook 2024.
Principles and practice of pediatric infectious diseases. 5th ed.

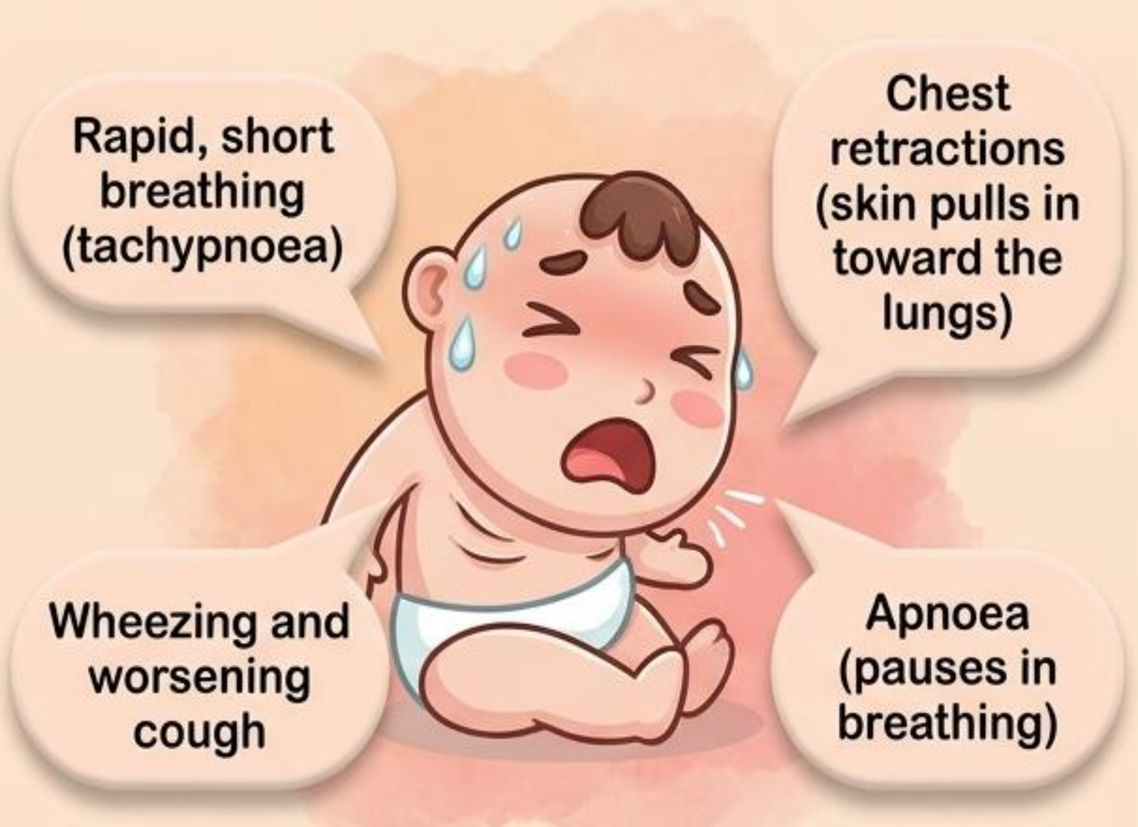
Clinical manifestations of RSV

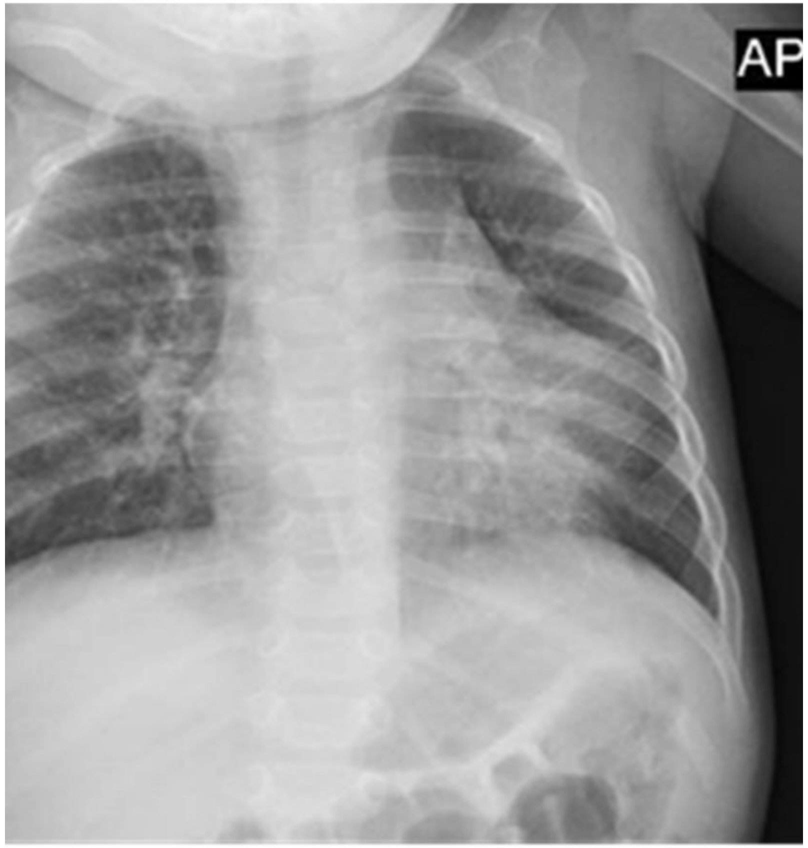


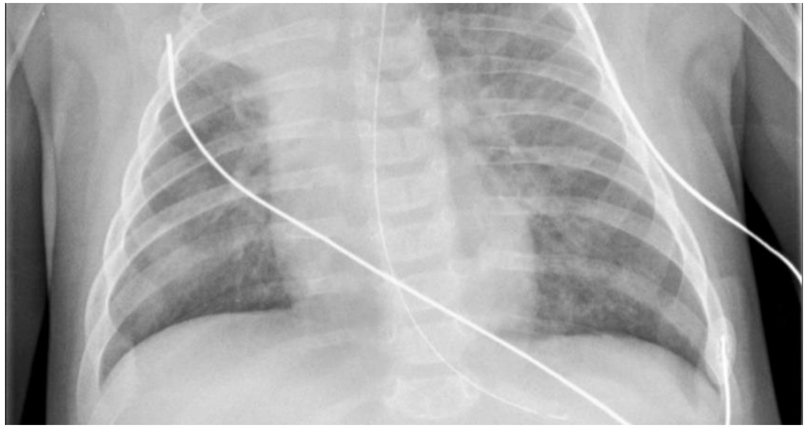
Recognizing the Warning Signs

Upper Respiratory (Mild)

Lower Respiratory (Severe)







CXR findings

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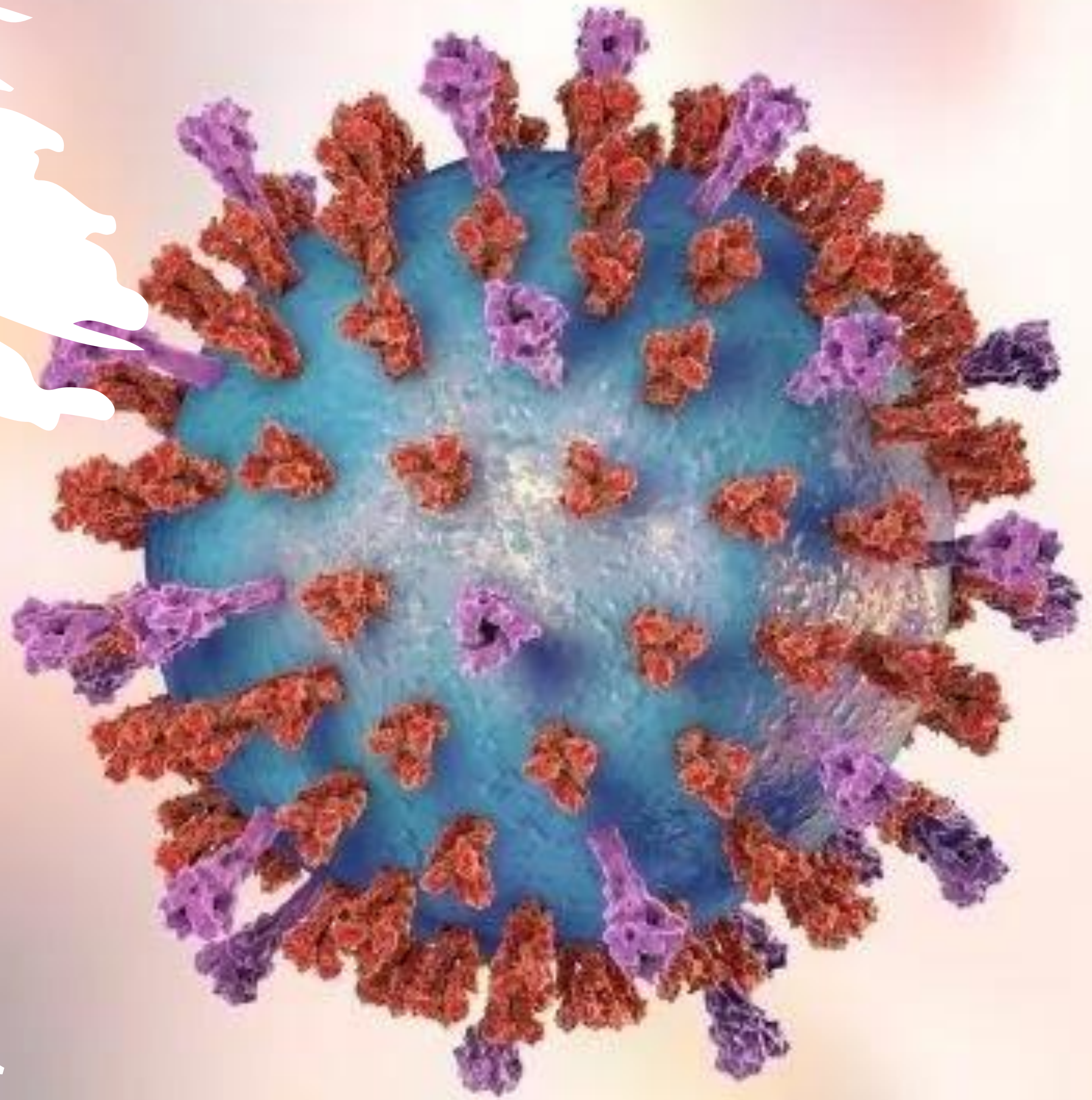


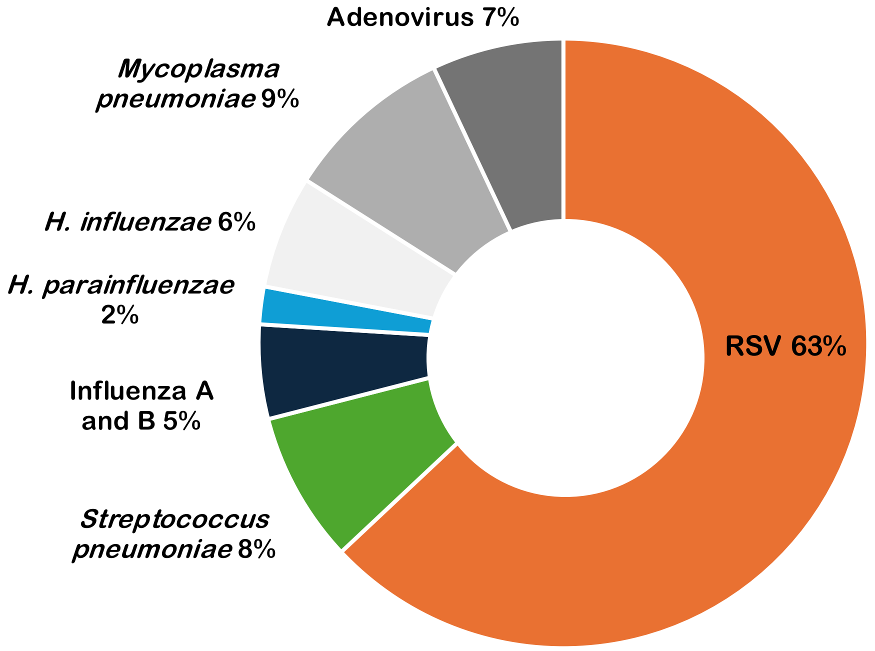
Burden of RSV infection in children

RSV is a leading cause of respiratory disease in infants[1]

RSV is a common and highly contagious RNA virus[2,3]

RSV is the most common cause of pediatric bronchiolitis and pneumonia globally[1]

Etiology of acute respiratory infections in infants and young children*,[1]



63%

The World Health Organization estimates that RSV accounts for more than 60% of acute respiratory infections in infants and young children worldwide[1]

* Figure showing etiology of acute respiratory infections as researched by The World Health Organization (RSV) in infants younger than 1 year and at the peak of viral season. RSV, respiratory syncytial virus; RNA, ribonucleic acid.

1. Piedimonte G, Perez MK. *Pediatr Rev.* 2014;35(12):519–530. 2. Drajac C, et al. *J Immunol Res.* 2017;2017:8734504. 3. CDC. RSV transmission. 2022. Available at: https://www.cdc.gov/rsv/about/transmission.html. Accessed: August 2024.

Nearly ***all children*** are infected with RSV[1] ***by 5 years old***

90%

*of children have been infected with **RSV** at least once[2] **before age of 2**.*

1. Phanthila Sitthikarnkha et al, Influenza and other respiratory viruses, Volume16, Issue1, January 2022 Pages 142-150 https://doi.org/10.1111/irv.12911
2. Piedimonte G, Perez MK. Respiratory syncytial virus infection and bronchiolitis. Pediatr Rev. 2014;35(12):519-530. doi:10.1542/pir.35-12-519

Burden of RSV infection in Thailand (2025)

New suspected case

49,417 cases (76.129)

Incidence per 100,000 population

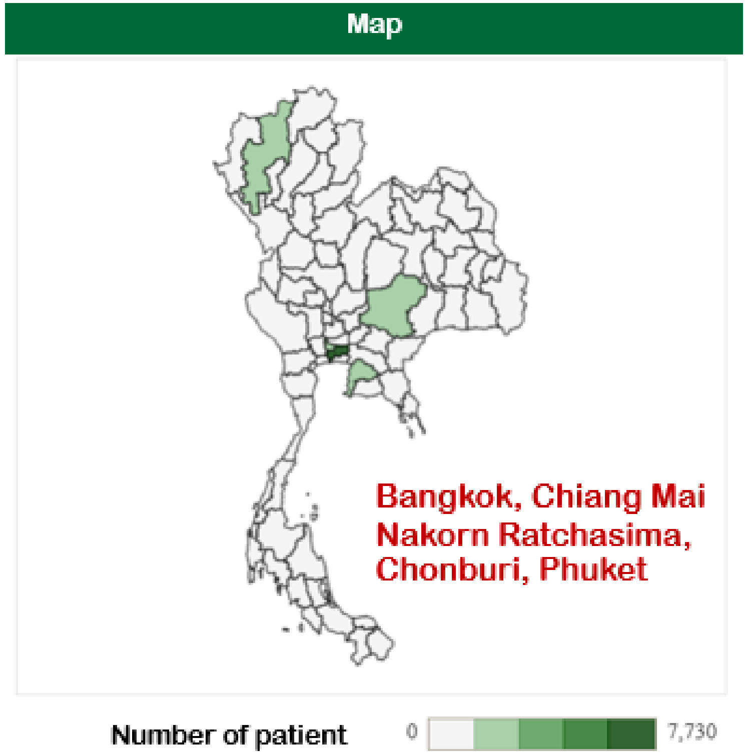
4 out of 10 were children <5 years

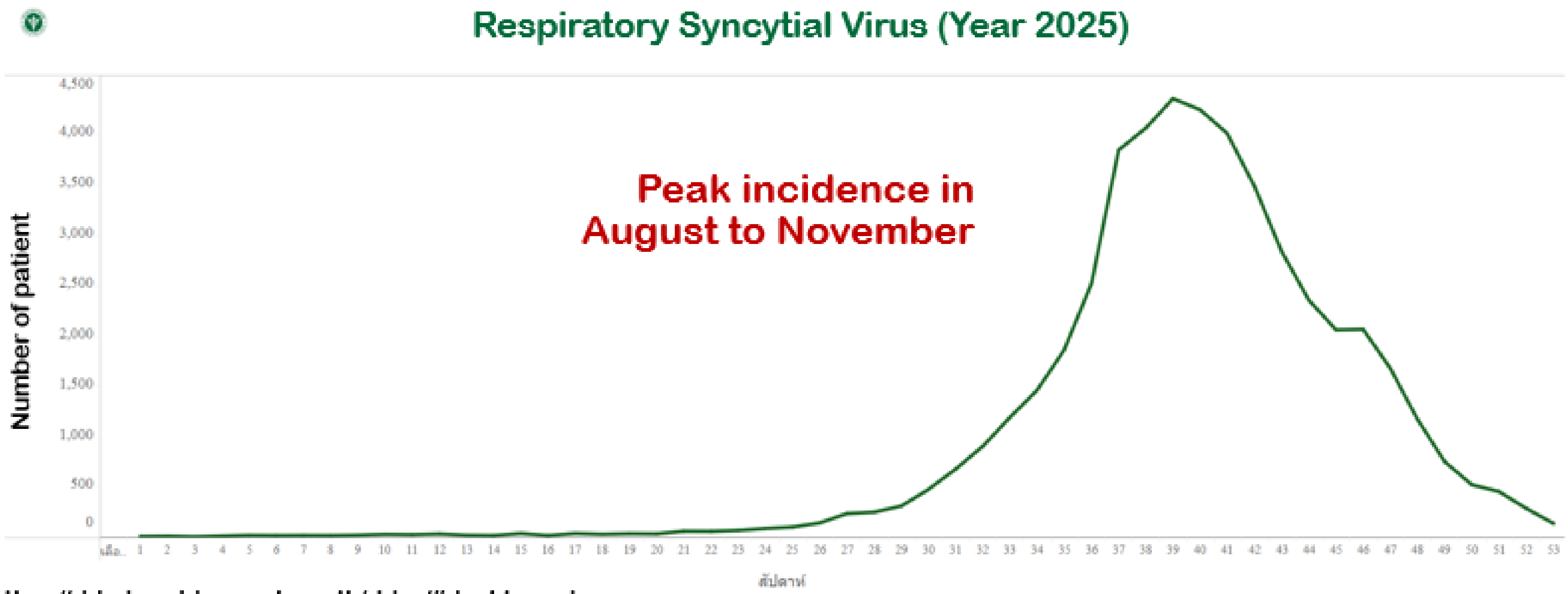
New dead case

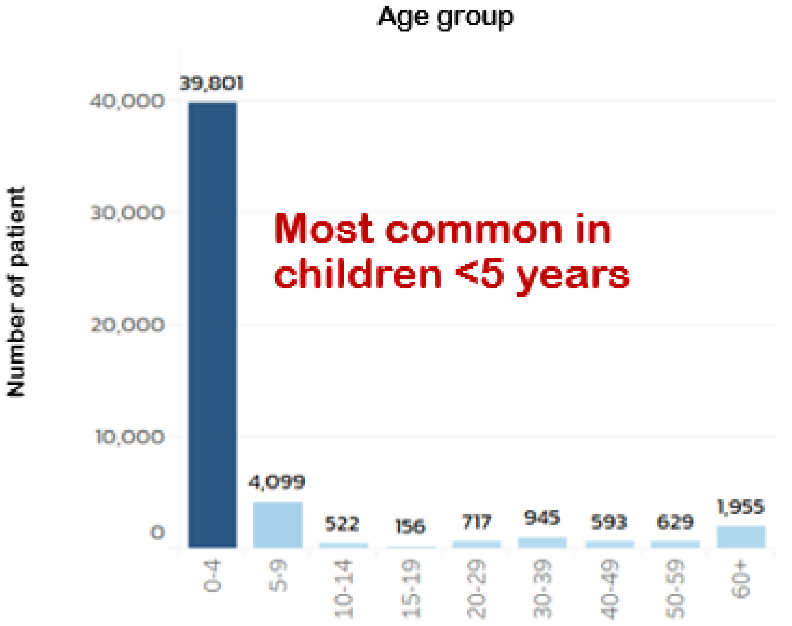
10 cases (0.015)

Mortality per 100,000 population

https://ddsdoe.ddc.moph.go.th/ddss/#dashboard







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รายงานผลการเฝ้าระวังไข้หวัดใหญ่และเชื้อสาเหตุโรคติดเชื้อทางเดินหายใจ ปี 2569

ระหว่างสัปดาห์ที่ 17 - 18 (19 เมษายน - 2 พฤษภาคม 2569)

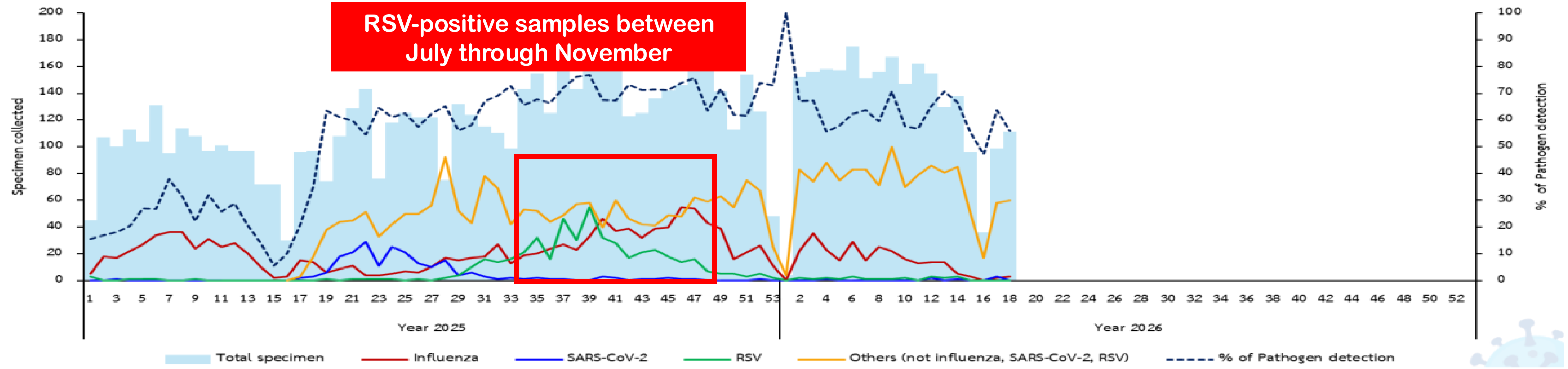
รายงาน ณ วันที่ 12 พฤษภาคม 2569

กองระบาดวิทยา กรมควบคุมโรค ร่วมกับสถาบันวิจัยวิทยาศาสตร์สาธารณสุข กรมวิทยาศาสตร์การแพทย์ ดำเนินงานโครงการพัฒนาการเฝ้าระวังโรคไข้หวัดใหญ่เพื่อตรวจจับและตอบโต้การระบาดของโรคไข้หวัดใหญ่และโรคติดเชื้อระบบทางเดินหายใจในประเทศไทย (Flu-RightSize) โดยจัดตั้งระบบเฝ้าระวังโรคไข้หวัดใหญ่ในโรงพยาบาล 8 จังหวัด ได้แก่ เชียงราย ตาก นครพนม จันทบุรี พระนครศรีอยุธยา กรุงเทพมหานคร ระนอง และสุราษฎร์ธานี ภายใต้การสนับสนุนงบประมาณจากศูนย์ป้องกันและควบคุมโรคแห่งชาติประเทศสหรัฐอเมริกา (U.S.CDC)

ดำเนินการเฝ้าระวังโดยการเก็บตัวอย่างจากระบบทางเดินหายใจของผู้ที่มาโรงพยาบาลด้วยกลุ่มอาการ Influenza-like illness (ILI) และ Severe acute respiratory infection (SARI) จำนวน 120 - 200 ตัวอย่างต่อสัปดาห์ ตรวจหาเชื้อโดยวิธี PCR ด้วยชุดตรวจ Multiplex PCR respiratory panel ซึ่งครอบคลุมเชื้อ 23 ชนิด*

[*Influenza A, A/H1N1 pdm2009, A/H1, A/H3, B, Coronavirus 4 strains, SARS-CoV-2, RSV, HMPV, Adenovirus, Parainfluenza virus 1,2,3,4, Bocavirus, Rhinovirus/Enterovirus and Bacteria (*M. pneumoniae*, *L. pneumophila*, *B. pertussis*, *C. pneumoniae*) ตั้งแต่วันที่ 21 เมษายน 2568]

กราฟเส้นแสดงร้อยละของไข้หวัดใหญ่ อาร์เอสวี โควิด-19 และเชื้อก่อโรคทางเดินหายใจอื่น ๆ ระหว่างวันที่ 1 มกราคม 2568 - 2 พฤษภาคม 2569



https://ddsdoe.ddc.moph.go.th/ddss/#dashboard

https://ddc.moph.go.th/doe/pagecontent.php?page=1496&dept=doe

RSV: The Global Burden on Infants

RSV is a primary cause of lower respiratory tract infections (LRTI) in infants under 12 months, leading to millions of hospitalizations and tens of thousands of deaths annually. While certain groups are at higher risk, a significant majority of RSV-related fatalities occur in otherwise healthy infants.

The Scale of Global Impact

12.9 Million

Annual LRTI Episodes

Global data shows millions of infants suffer lower respiratory infections from RSV each year.

57% of Deaths occur in Healthy Infants

Most RSV-related deaths worldwide occur in infants without pre-existing medical conditions.

2.2 Million

Hospitalizations

RSV causes a substantial burden on healthcare systems through high infant admission rates.

Identifying the Symptoms

Mild to Moderate Warning Signs

Includes nasal congestion, cough, fever, irritability, and significant difficulty feeding or sleeping.

Severe Respiratory Distress

Progresses to bronchiolitis, pneumonia, wheezing, and dangerous breathing interruptions known as apnoea.

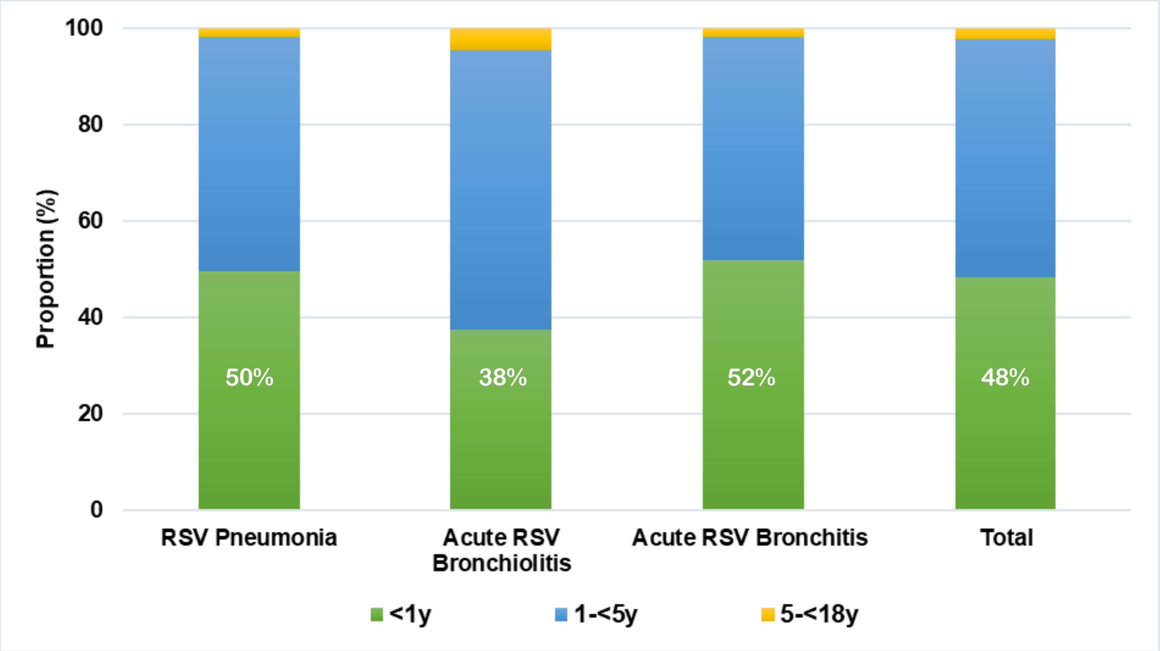
Severity is Difficult to Predict

Clinical progression can be unpredictable, making early monitoring essential for all infants.

Burden of RSV related acute LRTI in hospitalized Thai children: A 6-year national data analysis

- During 2015-2020: approximately 1.6 million hospital admissions due to LRTI in children <18 years
- RSV-related LRTI accounted for 19,340 admissions (1.2%)
- RSV pneumonia (70.8%) was the most common diagnosis, followed by RSV bronchiolitis (14.7%) and RSV bronchitis (14.5%)

Proportion of RSV LRTI in hospitalized children by age group



48% hospitalized children with RSV LRTI were <1 y of age

Admission and mortality rates of RSV LRTI by age group

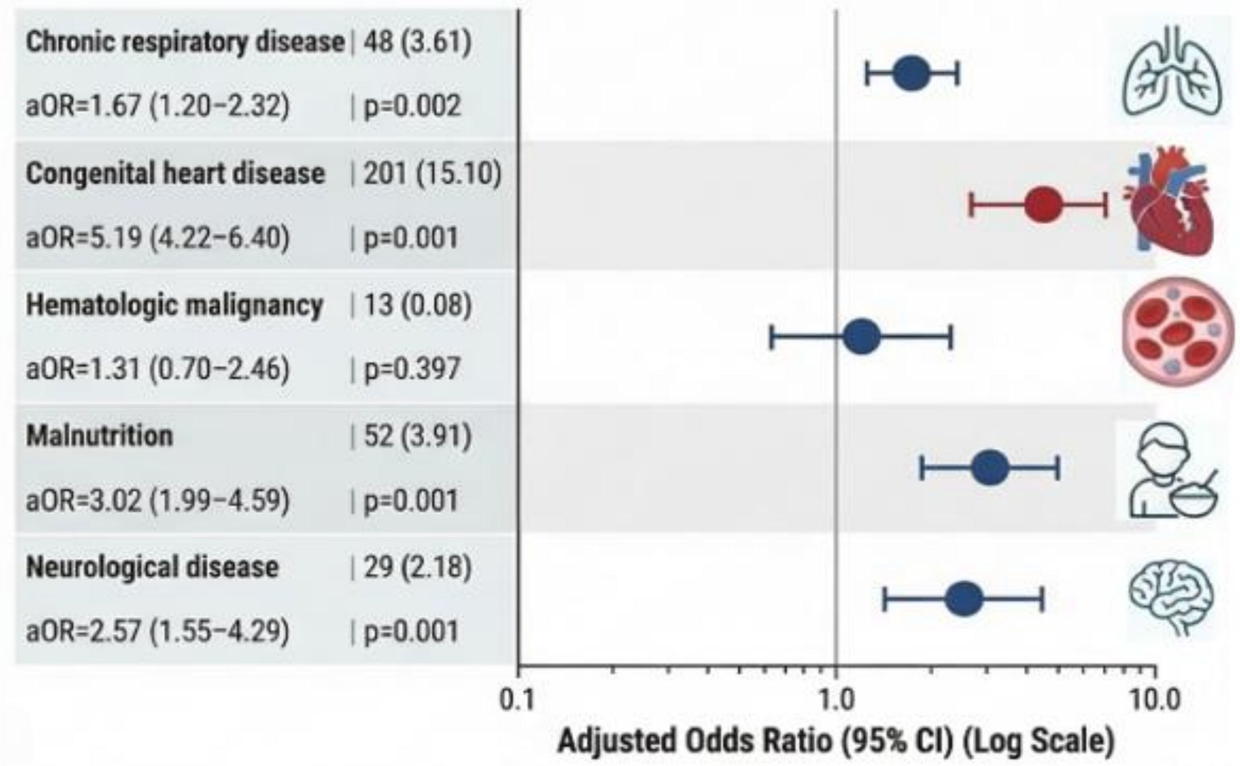
Age group (years)	Admission		Mortality	
	Number	Admission rate (/1000 Person Time)	Number	Mortality rate (/100,000 Person Time)
<1	9,335	2.60	65	1.75
1-<5	9,581	0.58	35	0.21
5-<18	424	0.01	6	0.01

- 7% had respiratory failure, the highest prevalence were among children with RSV pneumonia
- Most children with respiratory failure (67%) were infants <1 year
- Overall mortality was 0.6% with highest in infants <1 year at 0.7%

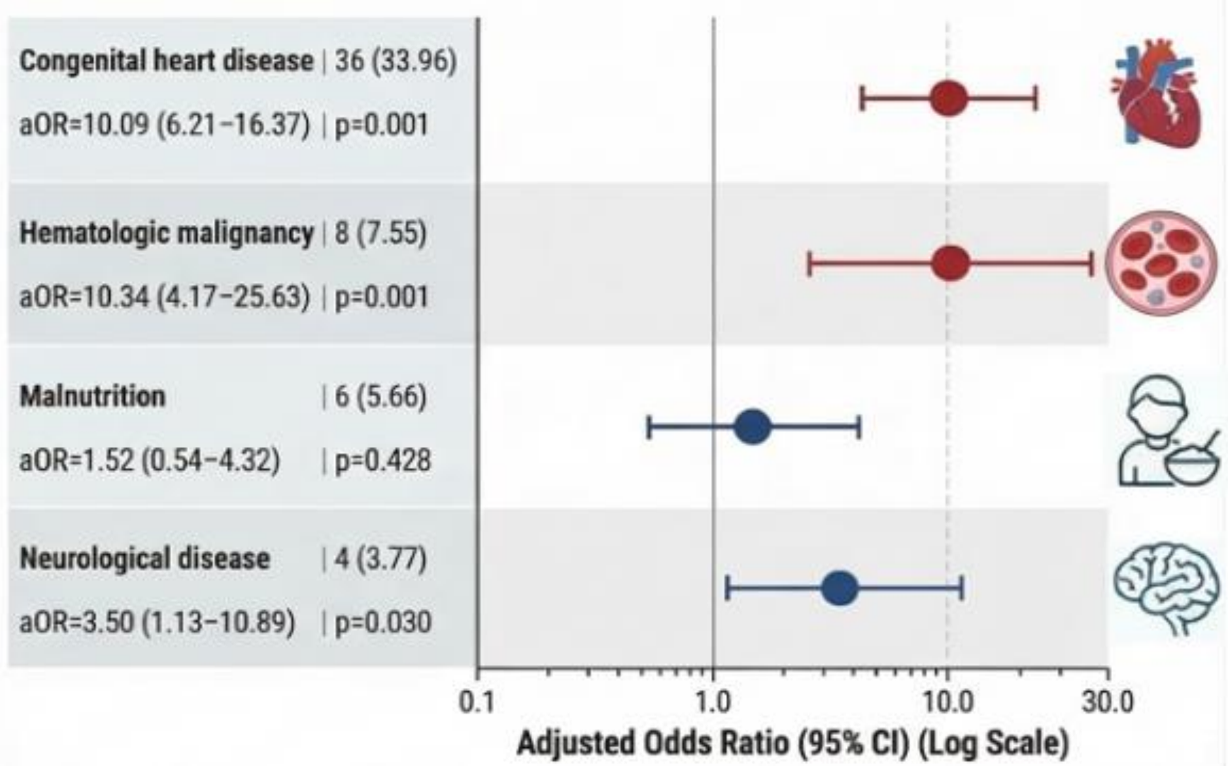
Sitthikarnkha P, et al. Children (Basel) 2022;9(12):1990.

Burden of RSV related acute LRTI in hospitalized Thai children: A 6-year national data analysis

Factors associated with respiratory failure among hospitalized children with RSV-related LRTI



Factors associated with mortality among hospitalized children with RSV-related LRTI



Congenital heart disease, malnutrition, neurological disease, and chronic respiratory disease were associated with respiratory failure in hospitalized children with RSV-related LRTI

Congenital heart disease, hematologic malignancy, and neurological disease were associated with mortality in hospitalized children with RSV-related LRTI

Sitthikarnkha P, et al. Children (Basel). 2022;9(12):1990.

Economic and clinical outcomes of RSV infection among Thai children <2 years of age

CLINICAL SEVERITY & OUTCOMES

50% hospitalized within 3 days

Half of diagnosed patients require hospitalization within 72 hours of their diagnosis.

22.5% received critical care

Nearly one in four hospitalized children required intensive medical intervention.

1.5% in-hospital mortality

6-day median hospital stay

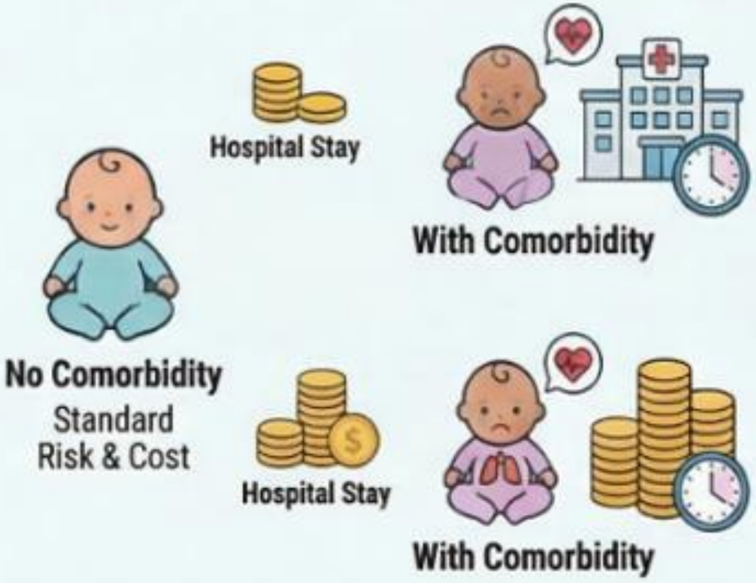
The typical duration of hospitalization ranges between 4 to 9 days.

ECONOMIC BURDEN & COMORBIDITY RISK

$2,112 Median Hospitalization Cost

Median total cost per RSV episode: $539

Non-hospitalized patients: $167



Comorbidities triple the risk

Patients with comorbidities have a 2.83 times higher risk of hospitalization.

And 5.5 days longer length of stay

higher medical cost

Pre-existing conditions significantly increase the overall medical expenses per episode. ($3,384)

Tan CJ. Pediatr Infect Dis J. 2023;42(10):883-887.

Early-life RSV LRTI associated with long-term respiratory sequelae

Recurrent LRTI

- Children with early-life RSV LRTI **3x risk of subsequent LRTI** than non-RSV LRTI[1]
- Infants with RSV LRTI **1.36 increased odds for pneumonia**[2]

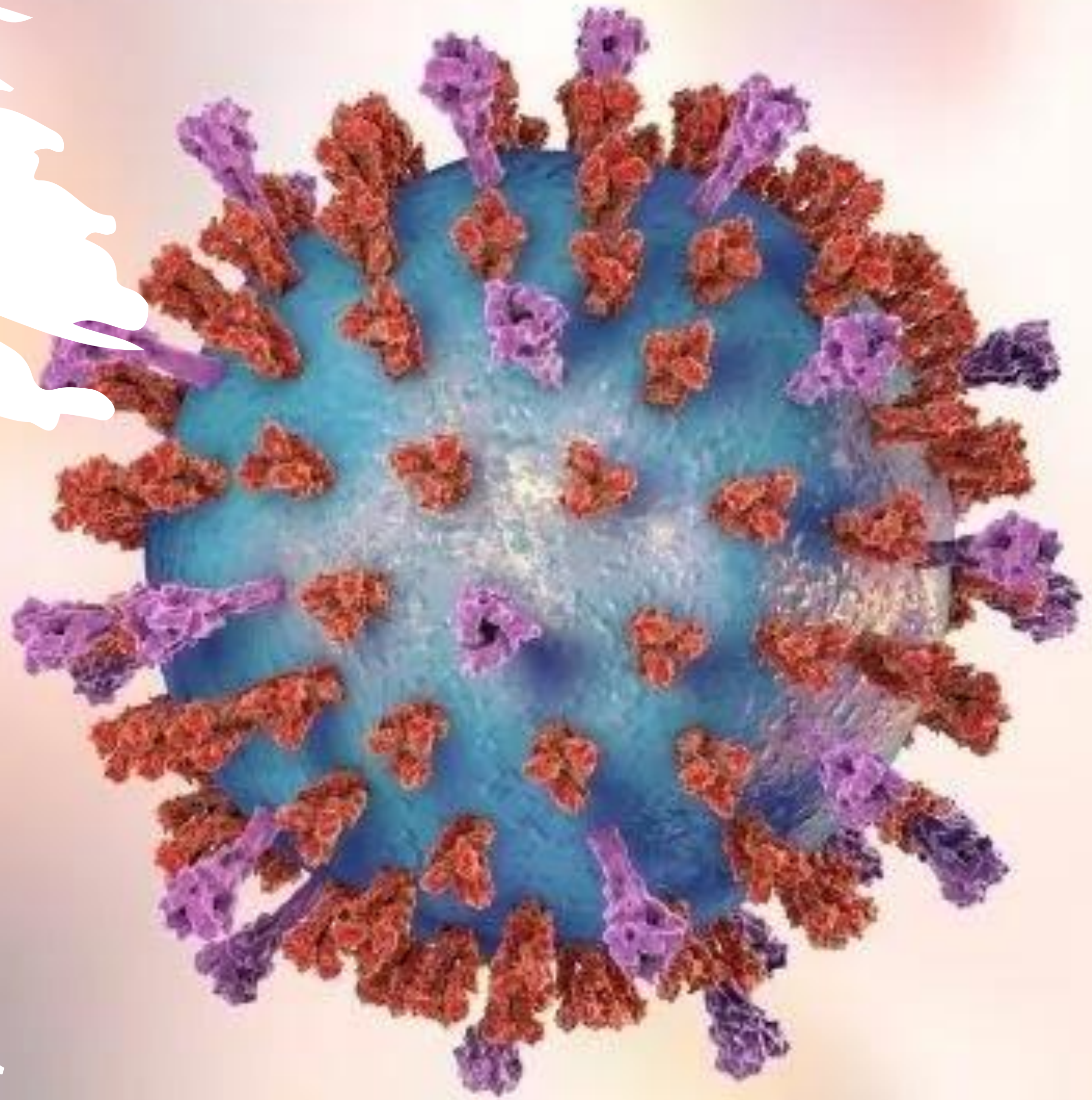
Recurrent wheeze & asthma

- RSV infection an exogenous risk factor **consistently shown associated with asthma**[3,4]
- Avoiding RSV infection during the first year of life has been associated with a lower risk of developing childhood asthma at 5 years old[5]

Lung function impairment

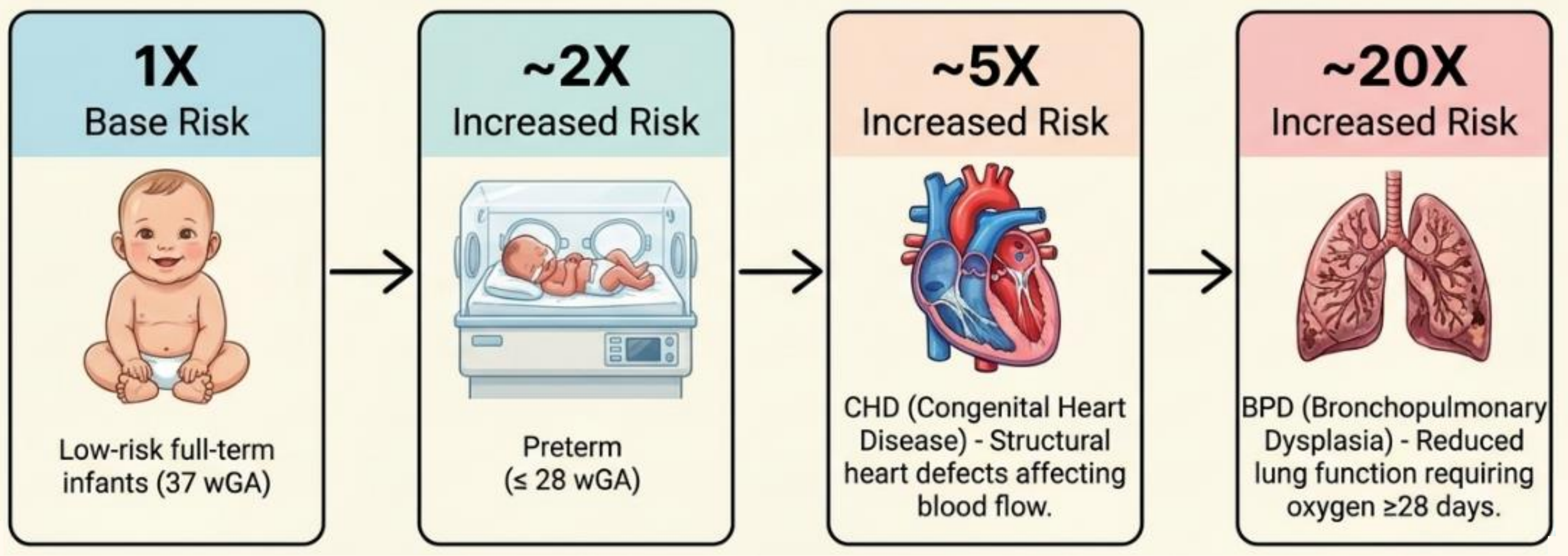
- The EU Child Cohort Network has observed that lower respiratory tract infections (LRTI) are associated with **reduced lung function parameters (FEV1, FEV1/FVC, and FEF75%) during school-age years.**[6]
- RSV LRTI associated with dev of **obstructive pattern on lung function**[7]

1. Zar HJ, et al. Lancet Glob Health 2020; 8: e1316–25. 2. Abreo A, et al. Clin Infect Dis 2020; 71: 211–14. 3. McCready C, et al. Lancet Child Adolesc Health 2023; 7: 127–35. 4. Régnier SA, Huels J. Pediatr Infect Dis J 2013; 32: 820–26. 5. Rosas-Salazar C, et al. Lancet 2023; 401: 1669–80. 6. van Meel ER, et al. Eur Respir J 2022; 60: 2102395. 7. McCready C, et al.. Lancet Child Adolesc Health 2024; 8: 400–12.



High risk children for RSV infection

High risk groups of severe RSV infection



Other high-risk groups vulnerable to RSV are children with neuromuscular impairment, immunodeficiency, and Down syndrome[2]

1. Piedimonte G, Perez MK. Pediatr Rev. 2014;35(12):519-530. 2. Sommer C, et al. Open Microbiol J. 2011;5(Suppl 2-M4):144-154. 3. World Health Organization. Preterm birth. February 19, 2018. Accessed June 29, 2023. https://www.who.int/news-room/fact-sheets/detail/preterm-birth. 4. Centers for Disease Control and Prevention. Accessed June 29, 2023. https://www.cdc.gov/ncbddd/heartdefects/facts.html. 5. Ibrahim J, et al. Pediatr Res. 2018;84:586-588. 6. Duijts L, et al. Eur Respir J. 2020;55:1-17. 7. Gien J, et al. Curr Opin Pediatr. 2011;23(3):305-313.

The Risk Paradox

Healthy Term Infants

Represents ~50% of RSV hospitalizations and ~57% of RSV-related deaths worldwide.

High-Risk / Preterm Infants

Represents the other ~50% of hospitalizations.

We cannot predict which infants will develop severe RSV disease. All infants are at risk.

Risk factors of RSV-associated acute LRTI in infants and young children

- Premature birth
- Less than 6 months of age at the start of RSV season, or being born during the RSV season
- Malnutrition/small size for gestational age
- Family history of atopy or asthma
- Male
- Maternal smoking and indoor smoke pollution
- Low cord serum RSV antibody titres
- Crowded living conditions
- Siblings and/or in day care

LRTI: lower respiratory tract infection; RSV: respiratory syncytial virus.

Resch B, et al. Int J Infect Dis. 2017;57:50-53.

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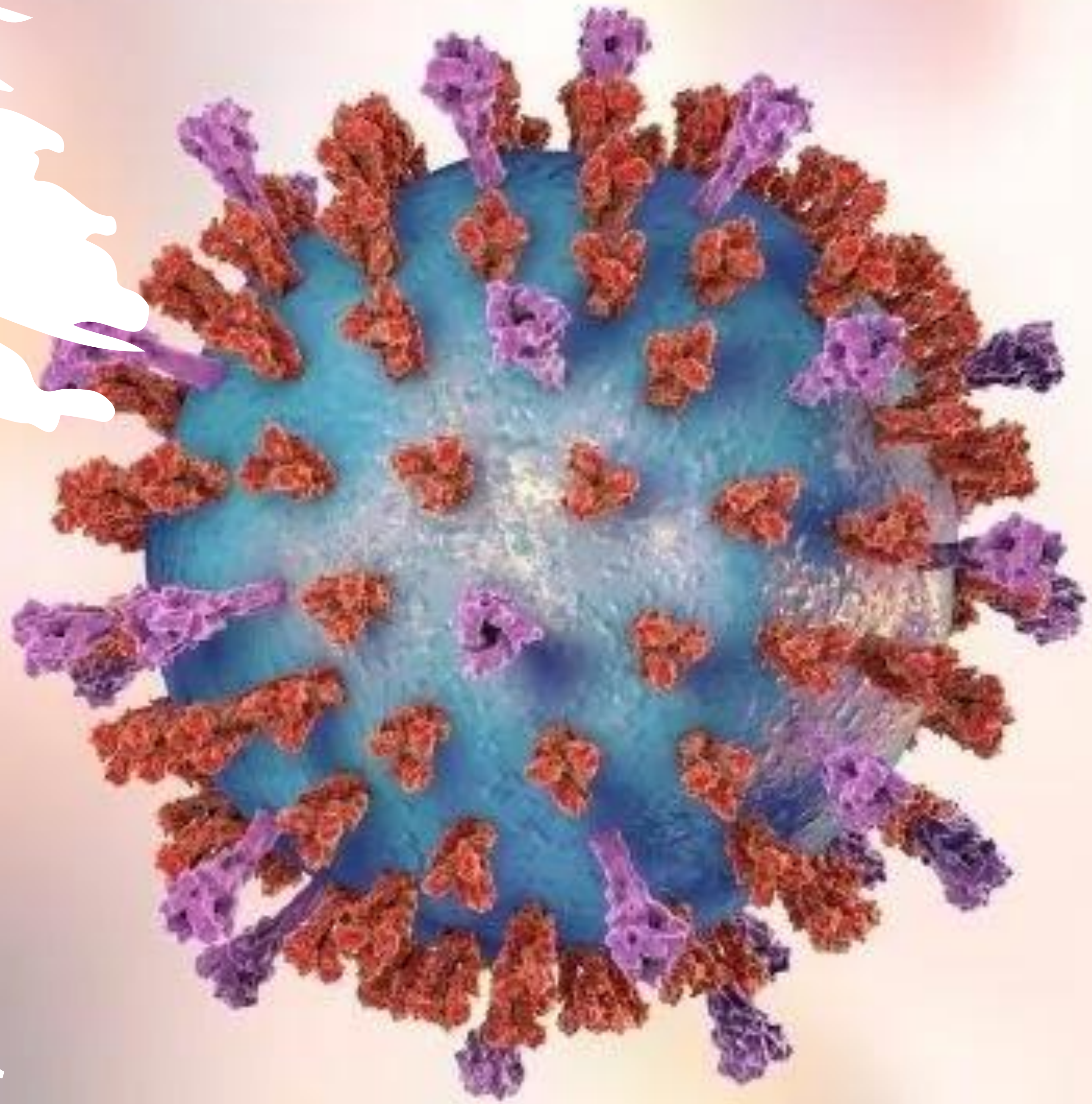
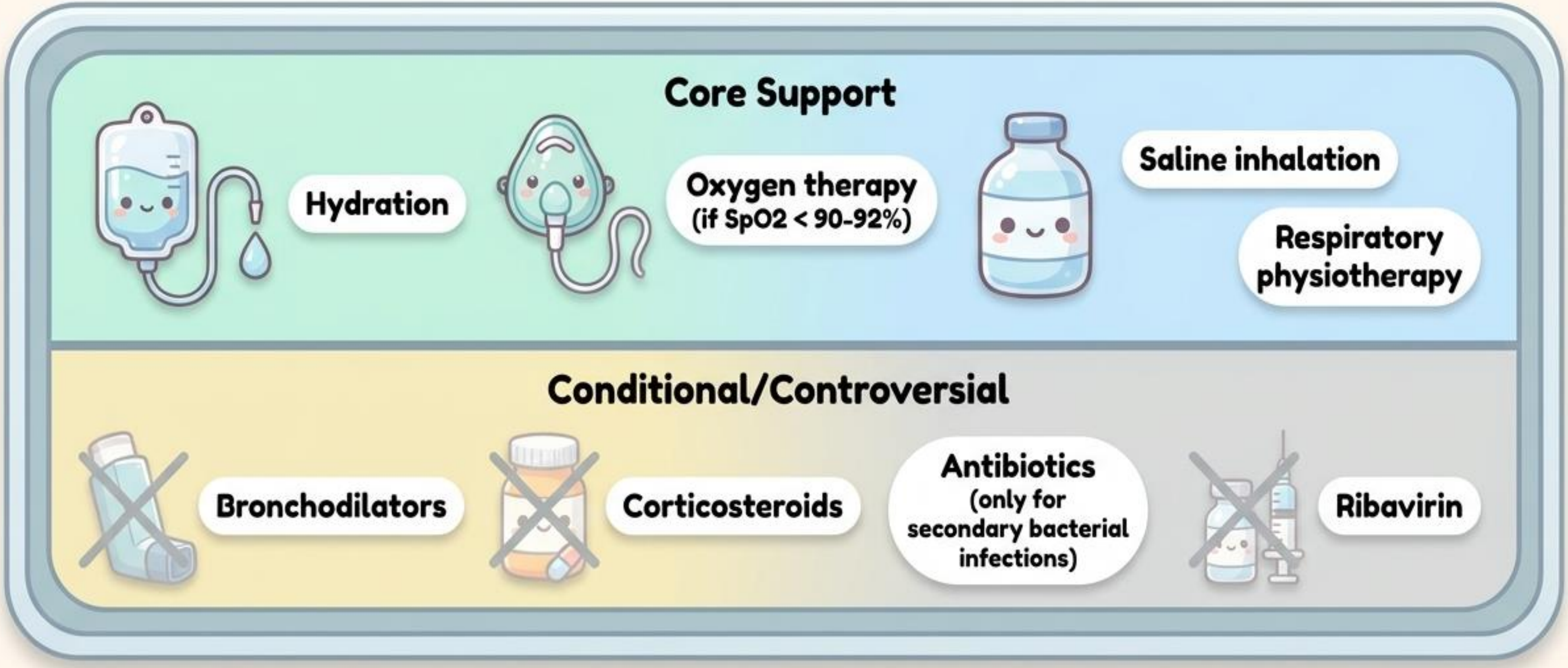


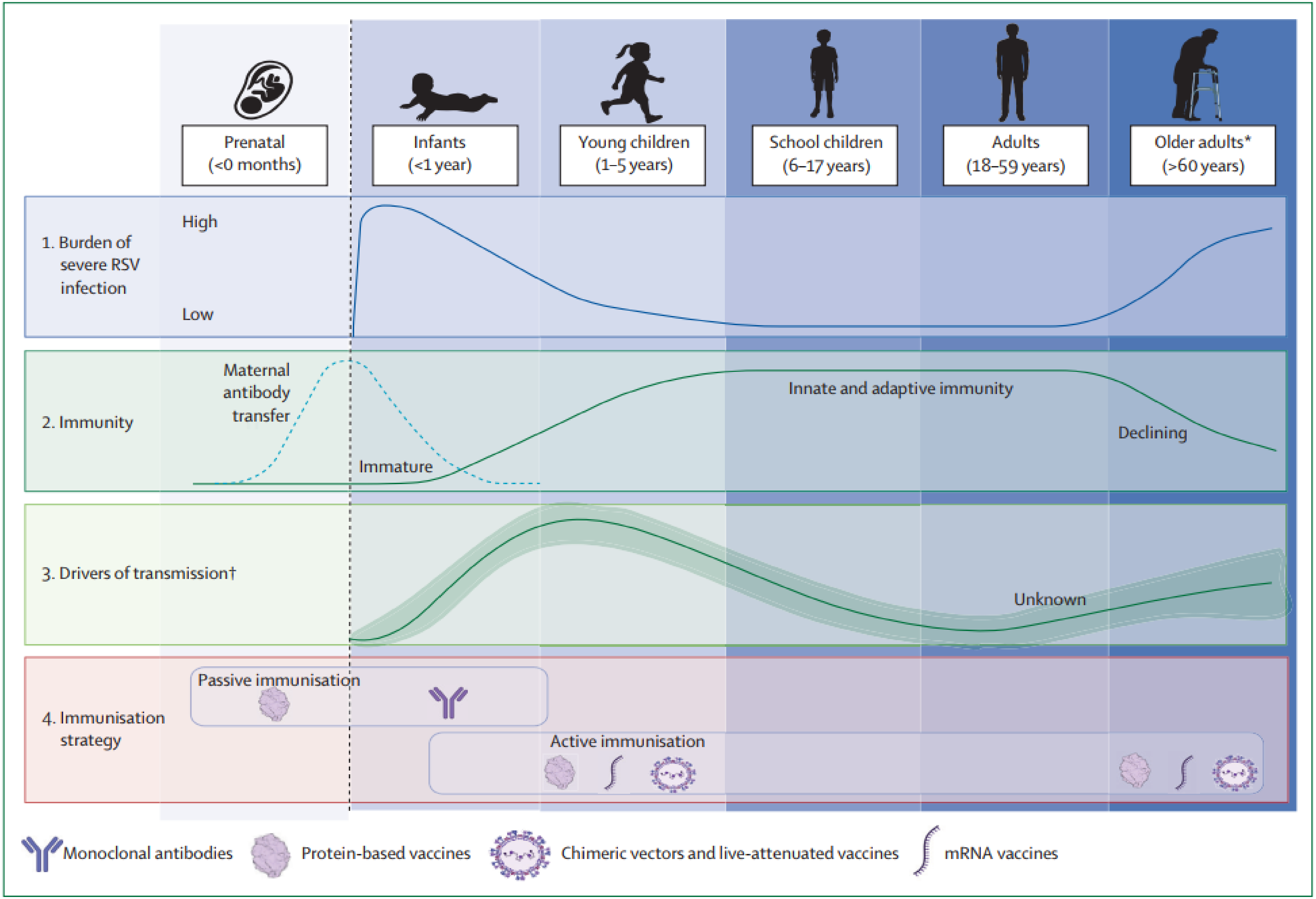
RSV treatment and prevention

Treatment Reality: A Lack of Cures

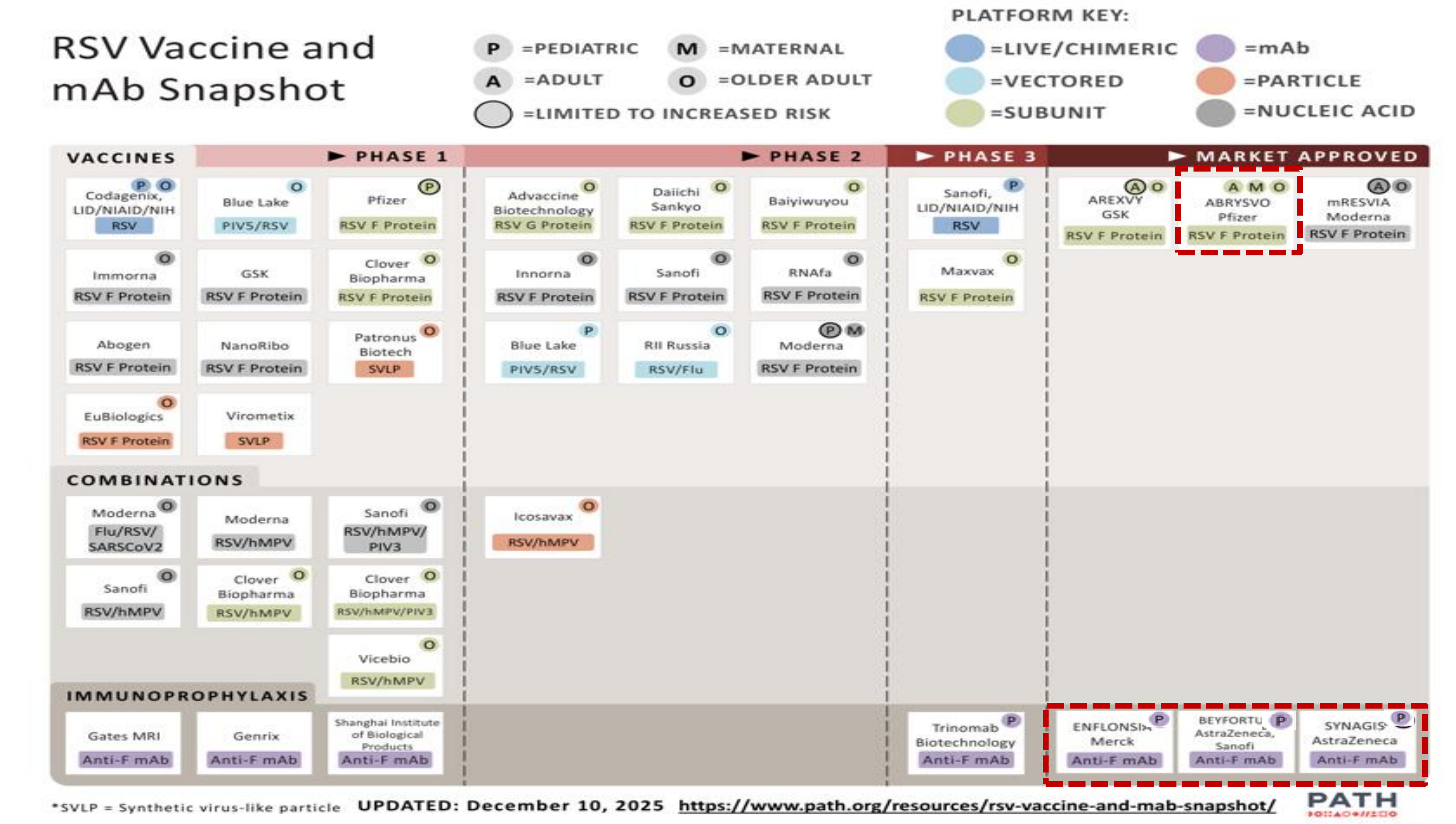


Because medical treatment is purely supportive, proactive prevention is our only true defense.

Conceptual overview of RSV prevention strategies



Terstappen J, et al. Lancet Infect Dis. 2024 Sep 23:S1473-3099(24)00455-9. doi: 10.1016/S1473-3099(24)00455-9. Epub ahead of print.


RSV Vaccine and mAb Snapshot
P =PEDIATRIC
M =MATERNAL
A =ADULT
O =OLDER ADULT
=LIMITED TO INCREASED RISK
PLATFORM KEY:
=LIVE/CHIMERIC
=mAb
=VECTORED
=PARTICLE
=SUBUNIT
=NUCLEIC ACID
VACCINES
PHASE 1
PHASE 2
PHASE 3
MARKET APPROVED
Codagenix, LID/NIAID/NIH RSV
Blue Lake PIV5/RSV
Pfizer RSV F Protein
Immorna RSV F Protein
GSK RSV F Protein
Clover Biopharma RSV F Protein
Abogen RSV F Protein
NanoRibo RSV F Protein
Patronus Biotech SVLP
EuBiologics RSV F Protein
Virometix SVLP
Advaccine Biotechnology RSV G Protein
Daiichi Sankyo RSV F Protein
Baiyiwuyou RSV F Protein
Innorna RSV F Protein
Sanofi RSV F Protein
RNAfa RSV F Protein
Blue Lake PIV5/RSV
RII Russia RSV/Flu
Moderna RSV F Protein
Sanofi, LID/NIAID/NIH RSV
Maxvax RSV F Protein
AREXVY GSK RSV F Protein
ABRYSVO Pfizer RSV F Protein
mRESVIA Moderna RSV F Protein
COMBINATIONS
Moderna Flu/RSV/SARSCoV2
Moderna RSV/hMPV
Sanofi RSV/hMPV/PIV3
Sanofi RSV/hMPV
Clover Biopharma RSV/hMPV
Clover Biopharma RSV/hMPV/PIV3
Vicebio RSV/hMPV
Icosavax RSV/hMPV
IMMUNOPROPHYLAXIS
Gates MRI Anti-F mAb
Genrix Anti-F mAb
Shanghai Institute of Biological Products Anti-F mAb
Trinomab Biotechnology Anti-F mAb
ENFLONSIA Merck Anti-F mAb
BEYFORTUS AstraZeneca, Sanofi Anti-F mAb
SYNAGIS AstraZeneca Anti-F mAb
*SVLP = Synthetic virus-like particle
UPDATED: December 10, 2025
https://www.path.org/resources/rsv-vaccine-and-mab-snapshot/
PATH

The Prevention Arsenal

Behavioral & Environmental

Disinfection, alcohol-based rubs, hand washing, and clinical use of gloves/gowns to limit transmission.

Maternal Vaccination

Administering the RSVpreF vaccine during pregnancy to provide passive, early protection for the newborn.

Infant Immunoprophylaxis

The use of Monoclonal Antibodies (mAbs) to provide immediate, passive immunity directly to the infant.

RSV prevention strategy in Thailand

Long-Acting Monoclonal Antibody

Healthy Infants <8 Months
Administer before the June–October RSV season for maximum protection.

High-Risk Pediatric Groups
Recommended for high-risk infants up to 12 months (1st season) or 19 months (2nd season).

Approved June 2024
Guidelines endorsed by five major Thai pediatric societies in February 2025.

Maternal RSV pre-F Vaccine

Vaccinate at 24-36 Weeks Gestation
Administer during the third trimester to ensure passive immunity transfer.

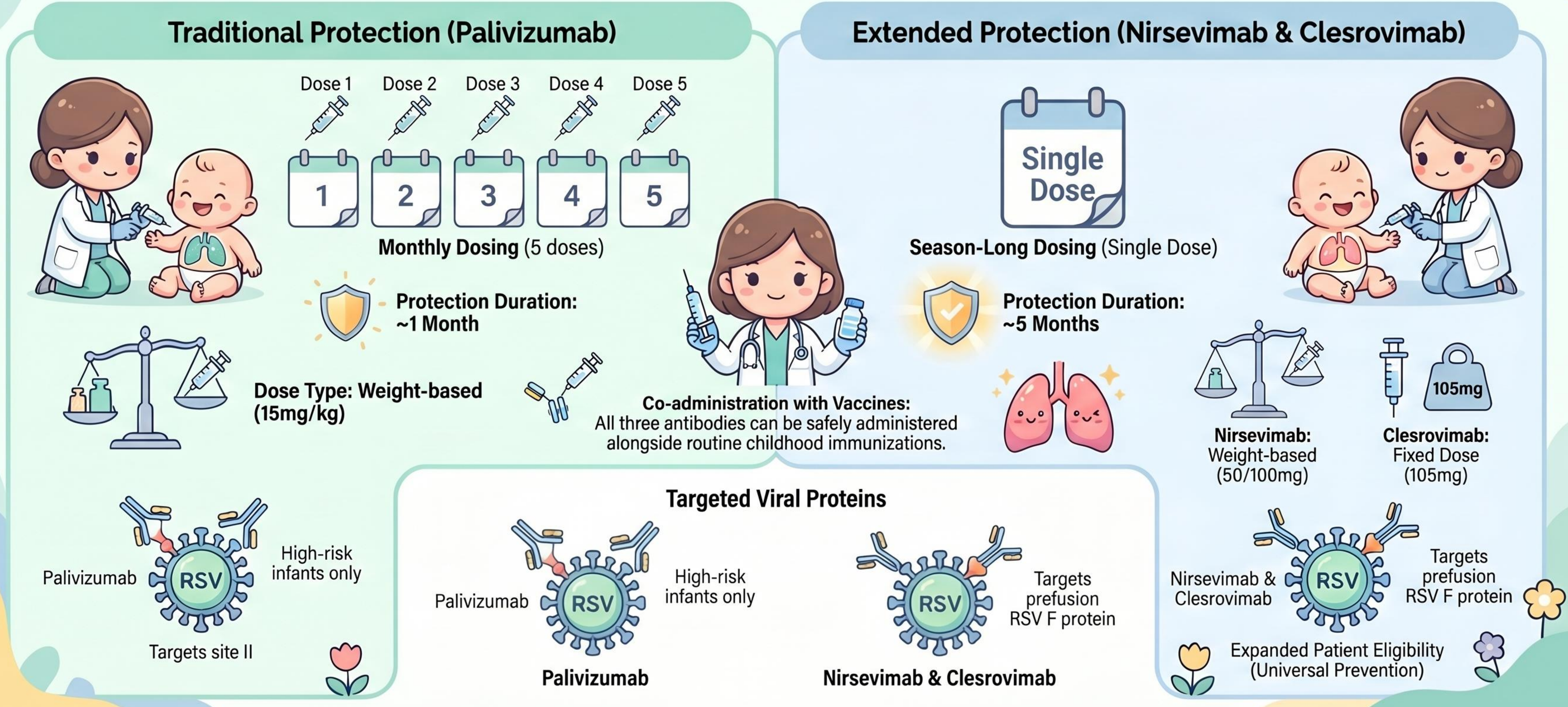
6 Months of Infant Protection
Provides critical defense against RSV-associated infections during the infant's first half-year.

Approved July 2024
Guidelines endorsed by The Infectious Disease Association of Thailand published recommendations in March 2025

Abrysvo is approved in Malaysia, Indonesia and the Philippines for the prevention of LRTD caused by RSV in infants through maternal immunization between weeks 24 and 36 of gestation.
Product licensing and approved indications may vary across countries. Please consult the locally approved prescribing informa tion.

Protecting Little Lungs: A Comparison of RSV Monoclonal Antibodies

Monoclonal antibodies provide passive immunity against RSV in infants. While traditional Palivizumab requires monthly doses, newer extended half-life options like Nirsevimab and Clesrovimab offer season-long protection with a single administration.



Nirsevimab Guidelines: Protecting Infants from Severe RSV

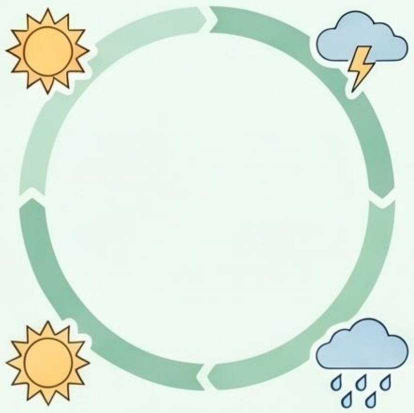
First RSV Season (Infants < 1 Year)

Primary Eligibility

Recommended for healthy infants ≤ 8 months and high-risk infants ≤ 12 months.

Weight-Based Dosing

- Give 50mg IM for infants ≤ 5kg
- Give 100mg IM for those > 5kg



Optimal Administration Window

Administer during the June–October peak season or immediately after birth.

Second RSV Season (High-Risk Children)

High-Risk Recommendation

Specifically recommended for children with severe RSV risk factors aged ≤ 19 months.

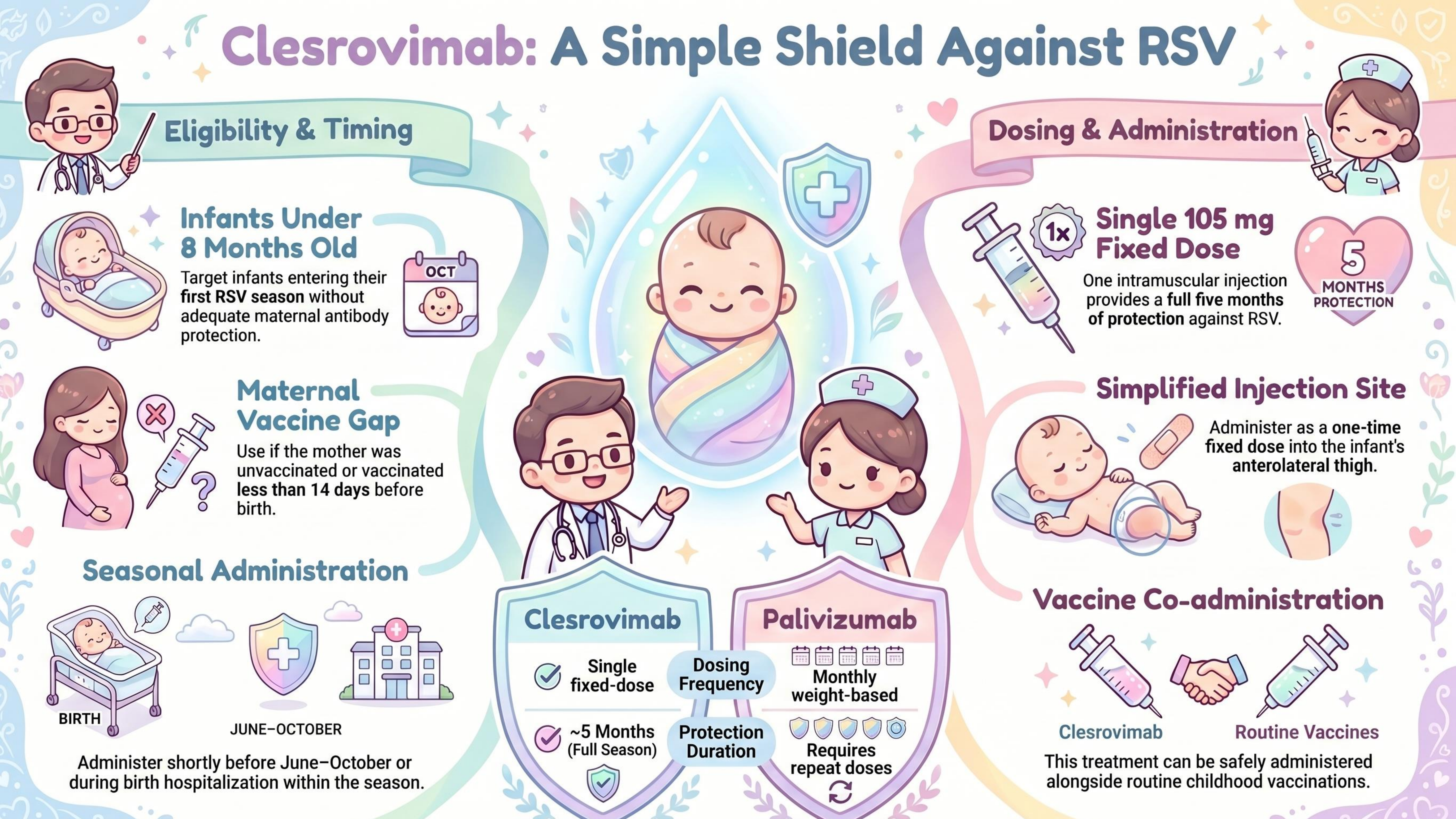
Dosage for Year Two

Administer 200mg IM via two separate 100mg injections at the same time.

Maternal Vaccination Factor

Prioritize if the mother was unvaccinated or vaccinated < 14 days before birth.

Target Group	Weight/Condition	Dosage Recommendation
Infants (Season 1)	Weight ≤ 5 kg	50 mg IM (Single Dose)
	Weight > 5 kg	100 mg IM (Single Dose)
Children (Season 2)	High-Risk Factor	200 mg IM (2x 100mg injections)


Clesrovimab: A Simple Shield Against RSV
Eligibility & Timing
Infants Under 8 Months Old
Target infants entering their first RSV season without adequate maternal antibody protection.
OCT
Maternal Vaccine Gap
Use if the mother was unvaccinated or vaccinated less than 14 days before birth.
Seasonal Administration
BIRTH
JUNE–OCTOBER
Administer shortly before June–October or during birth hospitalization within the season.
Clesrovimab
Palivizumab
Single fixed-dose
Dosing Frequency
Monthly weight-based
~5 Months (Full Season)
Protection Duration
Requires repeat doses
Dosing & Administration
1x
Single 105 mg Fixed Dose
One intramuscular injection provides a full five months of protection against RSV.
5 MONTHS PROTECTION
Simplified Injection Site
Administer as a one-time fixed dose into the infant's anterolateral thigh.
Vaccine Co-administration
Clesrovimab
Routine Vaccines
This treatment can be safely administered alongside routine childhood vaccinations.

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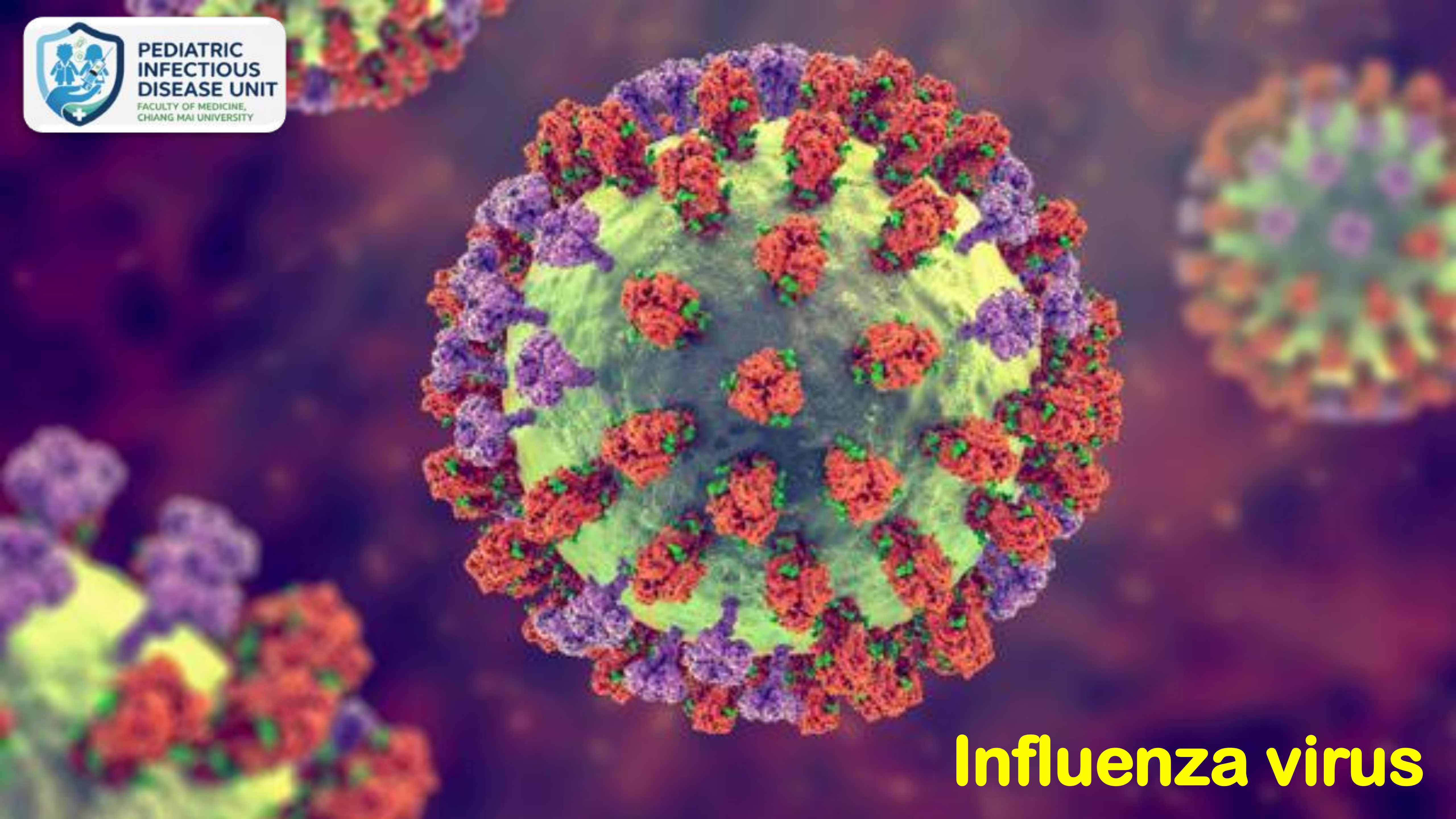
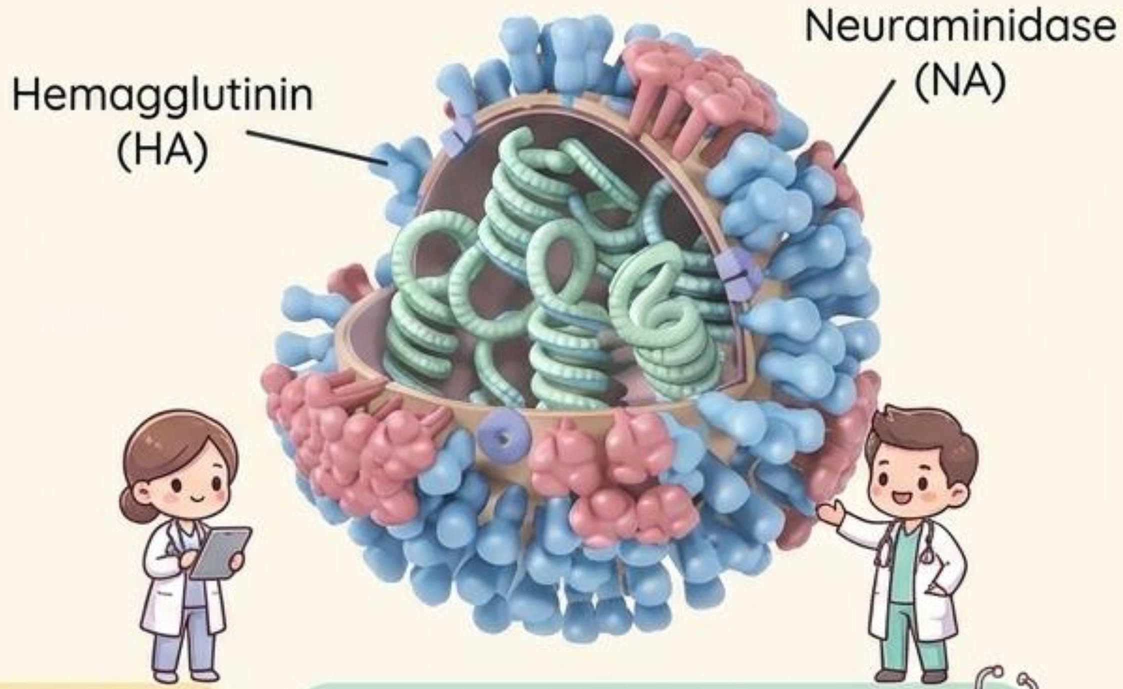


Influenza virus

Meet the Orthomyxovirus



Single-stranded RNA virus | **Family:** Orthomyxoviridae | **Genus:** Orthomyxovirus

Type A

More common, causes more severe disease. Sub-classified by HA and NA surface antigens.

Type B

Less common, causes less severe disease.

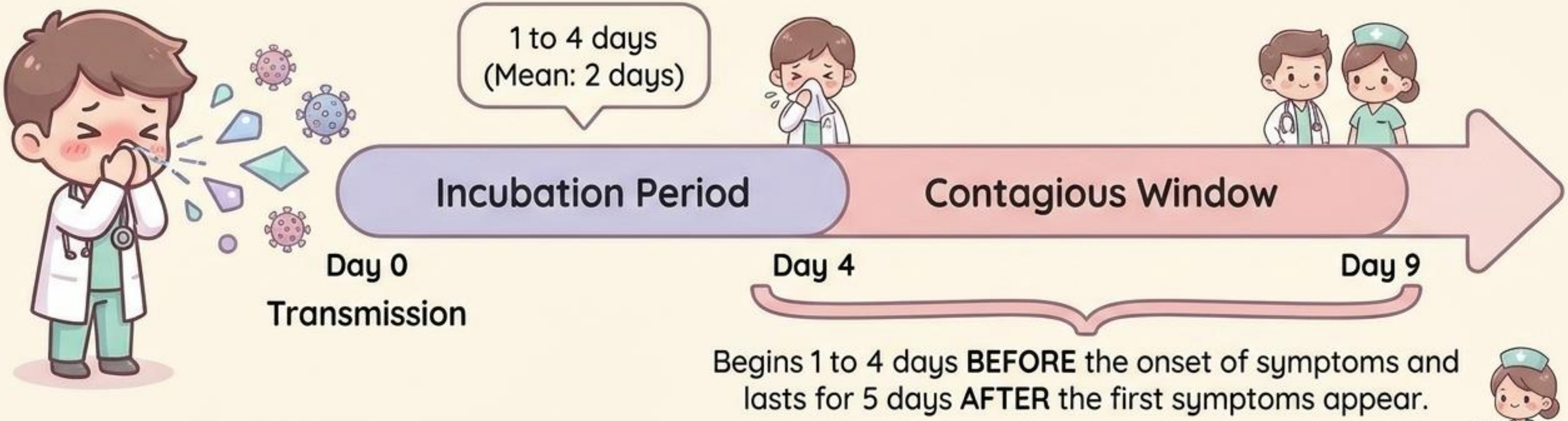
Type C

Sporadic occurrences, results in mild influenza-like illness (ILI).

American Academy of Pediatrics. Redbook 2024.

The Timeline of Infection

Mode of Transmission: Droplets and direct contact. Highly contagious.



The Invisible Threat:
50% of infected people without symptoms are still contagious.

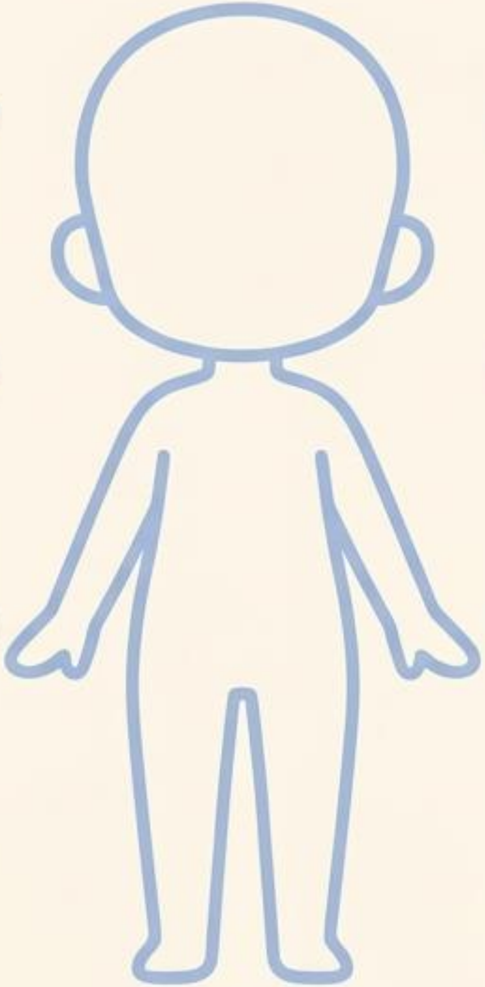
American Academy of Pediatrics. Redbook 2024.

Clinical Manifestations

Initial / Head
Sudden onset of fever, chills/rigors, headache, malaise, diffuse myalgia, nonproductive cough

Respiratory Tract
Sore throat, nasal congestion, rhinitis, cough

Constitutional / Systemic
Conjunctival injection, abdominal pain, nausea, vomiting, diarrhea



Initial / Head
Sudden onset of fever, chills/rigors, headache, malaise, diffuse myalgia, nonproductive cough

Respiratory Tract
Sore throat, nasal congestion, rhinitis, cough

Constitutional / Systemic
Conjunctival injection, abdominal pain, nausea, vomiting, diarrhea

Complications Note
Occasionally causes croup, bronchiolitis, or pneumonia.

Infant Alert Box
Infant Alert: In infants, influenza can produce a severe, sepsis-like picture.

American Academy of Pediatrics. Redbook 2024.

High-Risk Demographics


Adults aged >65 years


Unvaccinated infants aged 12–24 months

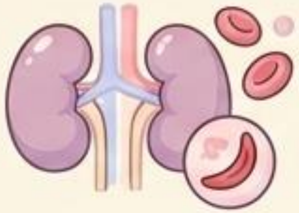

Pregnant women

Immunocompromised (including symptomatic HIV)


Chronic pulmonary disease (Asthma, COPD, Cystic Fibrosis)


Hemodynamically significant cardiac disease


Chronic metabolic disease (e.g., Diabetes mellitus)


Renal dysfunction & Hemoglobinopathies (Sickle cell)


Neuromuscular/Cognitive disorders compromising respiratory secretions

Residents of nursing homes/long-term care facilities

Patients on long-term aspirin therapy (Rheumatoid arthritis, Kawasaki disease)

American Academy of Pediatrics. Redbook 2024.

Diagnostic Criteria: Who Gets Tested?

During Influenza Season

- Outpatient immunocompetent at high risk for complications (febrile respiratory symptoms within 5 days of onset).
- Outpatient immunocompromised (febrile respiratory symptoms, ANY time from onset).
- Hospitalized persons with fever/respiratory symptoms (ANY time from onset).
- Children with fever/respiratory symptoms presenting for medical evaluation (ANY time from onset).

Any Time of the Year

- Healthcare personnel, residents, or visitors in an institution experiencing an influenza outbreak (symptoms within 5 days of onset).
- Persons epidemiologically linked to an outbreak (symptoms within 5 days of onset).

American Academy of Pediatrics. Redbook 2024.

The Diagnostic Arsenal

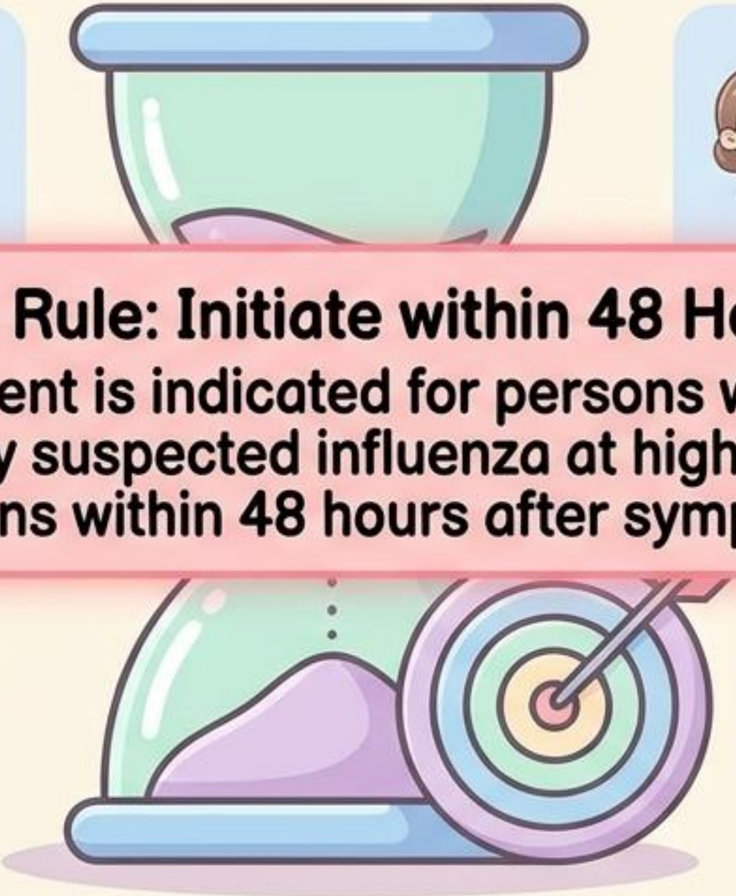
Test	Time to Results	Clinical Utility
RT-PCR (Gel, real-time, multiplex)	2 hours	High sensitivity & very high specificity. Highly recommended.
Immunofluorescence (Direct/Indirect)	2–4 hours	Moderately high sensitivity & high specificity. Distinguishes A/B and other viruses.
Rapid Antigen Detection (EIA)	10–20 mins	Low-to-moderate sensitivity & high specificity. Limitations must be recognized.
Rapid Neuraminidase Detection	20–30 mins	Detects but does not distinguish between A and B.
Viral Culture (Shell/Cell)	48-72h to 3-10 days	Highest specificity. Confirms screening & public health surveillance, not for timely clinical management.
Serologic Tests	Retrospective only	Used for surveillance/research, not clinical management.

Harper SA, et al. CID 2009;48:1003-32.

Treatment Initiation Guidelines

Why testing timing matters. Ideal tests yield timely results that influence:

- Decisions on initiation of antiviral treatment.



- Antibiotic treatment decisions.

Core Rule: Initiate within 48 Hours.

Antiviral treatment is indicated for persons with laboratory-confirmed or highly suspected influenza at high risk of developing complications within 48 hours after symptom onset.

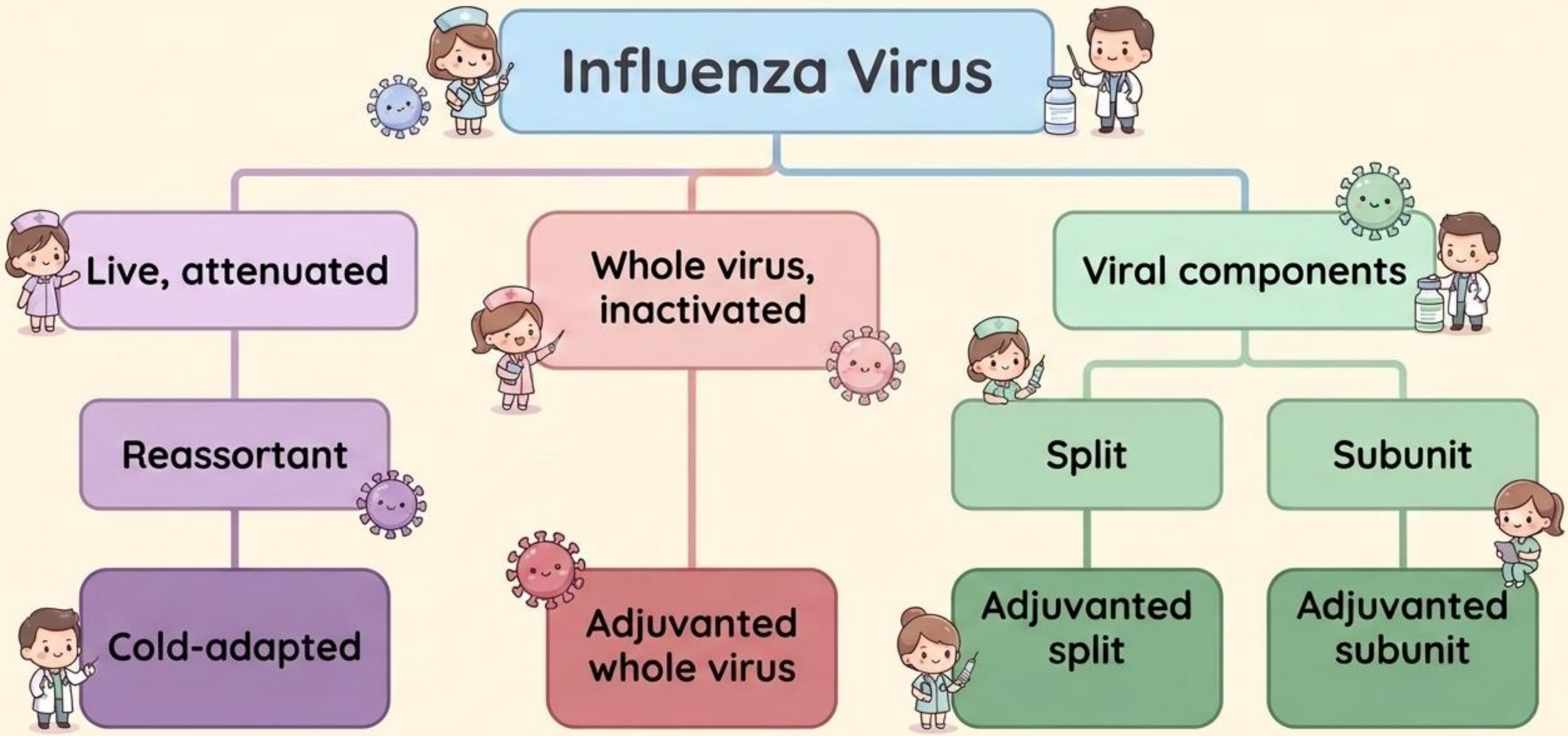
- Infection control practices.

Note: Results must be weighed against patient symptoms, test sensitivity/specificity, and community circulation.

Antiviral Therapy Matrix

Oral Oseltamivir (5 days)	Inhaled Zanamivir (5 days)	Intravenous Peramivir (Single dose)	Oral Baloxavir (Single dose)
Activity: A & B	Activity: A & B	Activity: A & B	Activity: A & B
Age: Any age	**Age >= 7 years**	**Age:** >=6 months	**Age:** >=5 years
	Avoid in: Underlying airway disease (asthma/COPD)		
Adverse Events: Nausea, vomiting, headache. Rare serious skin/neuropsychiatric events.	**Adverse Events:** Bronchospasm, sinusitis, dizziness.	**Adverse Events:** Diarrhea. Rare serious skin/neuropsychiatric events.	**Adverse Events:** None more common than placebo in trials.

The Vaccine Family Tree



Recommended Vaccine Strains (2026–2027)

Trivalent – Egg-based		Northern Hemisphere (2026/2027)	Southern Hemisphere (2026)
	A/H1N1	A/Missouri/11/2025	A/Missouri/11/2025
	A/H3N2	A/Darwin/1454/2025	A/Singapore/GP20238/2024
	B/Victoria	B/Tokyo/EIS13-175/2025	B/Austria/1359417/2021

Trivalent – Cell/Recombinant-based		Northern Hemisphere	Southern Hemisphere
	A/H1N1	A/Missouri/11/2025	A/Missouri/11/2025
	A/H3N2	A/Darwin/1415/2025	A/Sydney/1359/2024
	B/Victoria	B/Pennsylvania/14/2025	B/Austria/1359417/2021

Quadrivalent Addition	Northern Hemisphere	Southern Hemisphere
		B/Phuket/3073/2013-like virus

https://www.who.int/teams/global-influenza-programme/vaccines/who-recommendations

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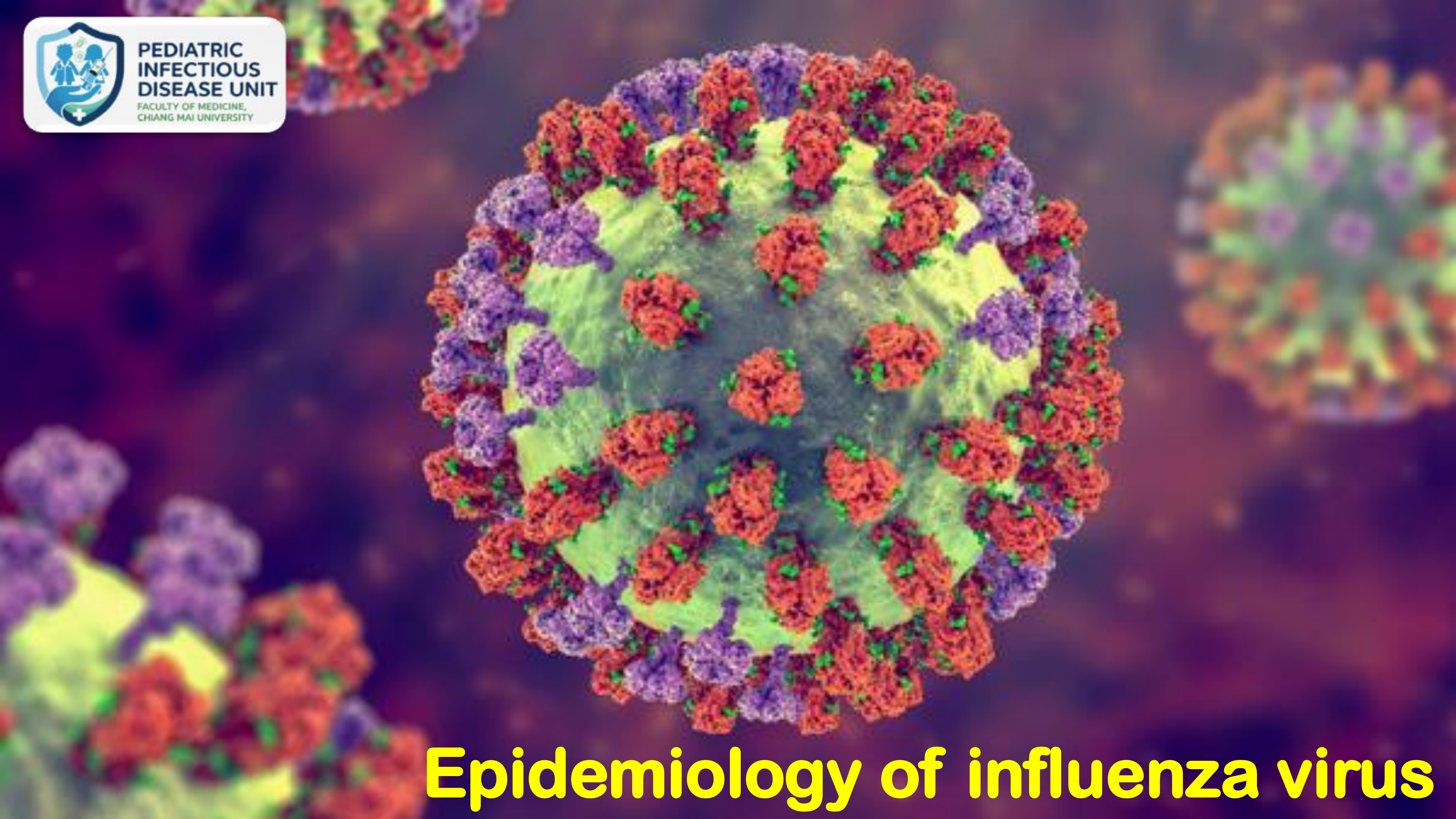

Epidemiology of influenza virus

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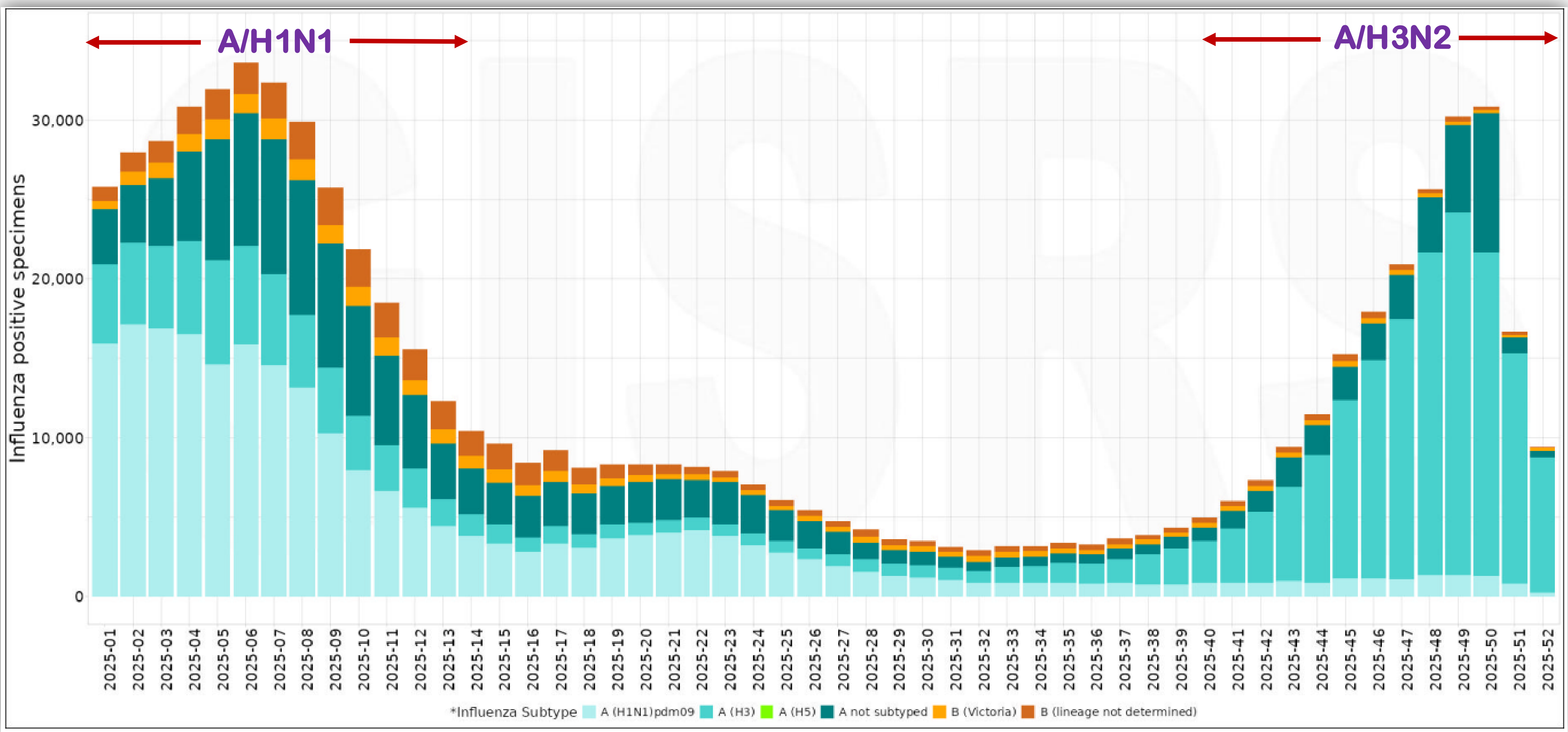
Global influenza situation in 2025

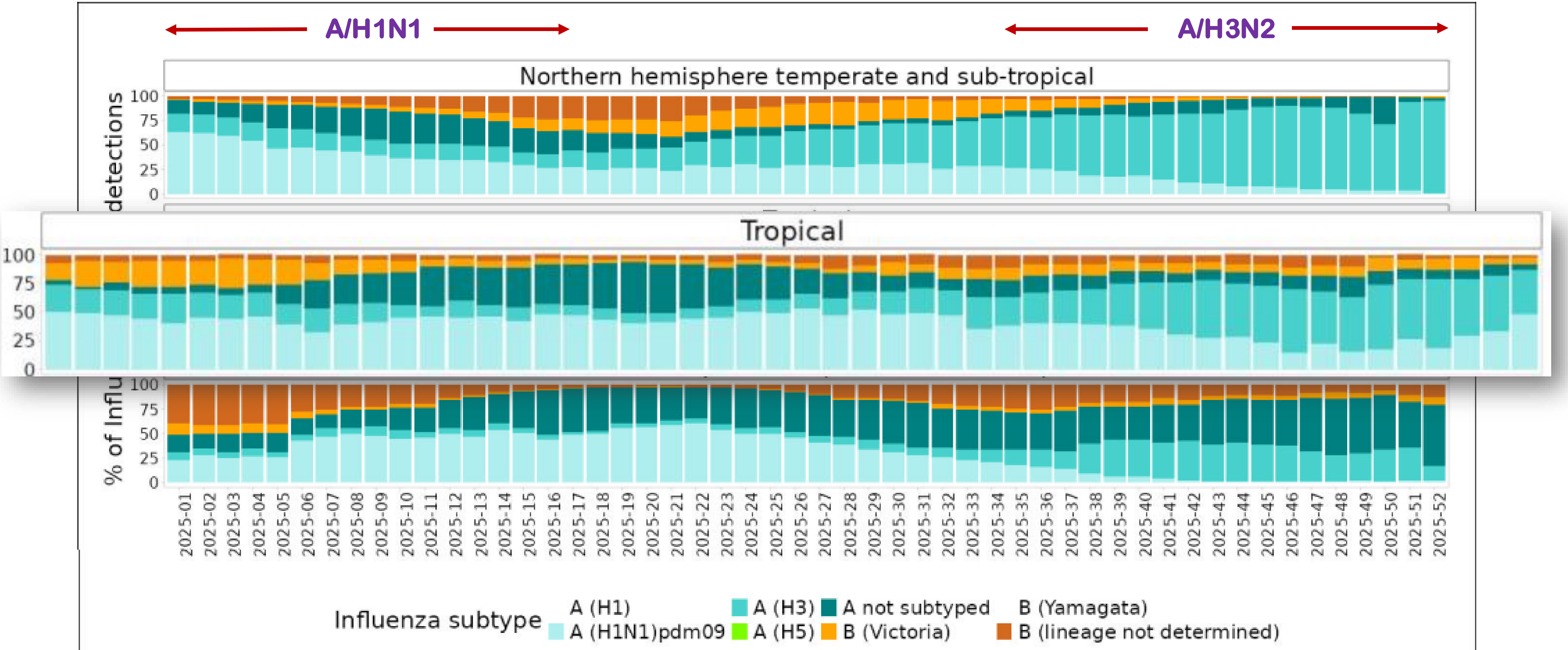
Numbers of influenza virus positive specimens by type and subtype at the global level in 2025



Influenza A viruses were predominant across all zones, with A/H1N1 circulating early in the season, followed by a shift to A/H3N2 as the dominant subtype later on

https://www.who.int/teams/global-influenza-programme/surveillance-and-monitoring/respiratory-viruses/influenza-updates-archive

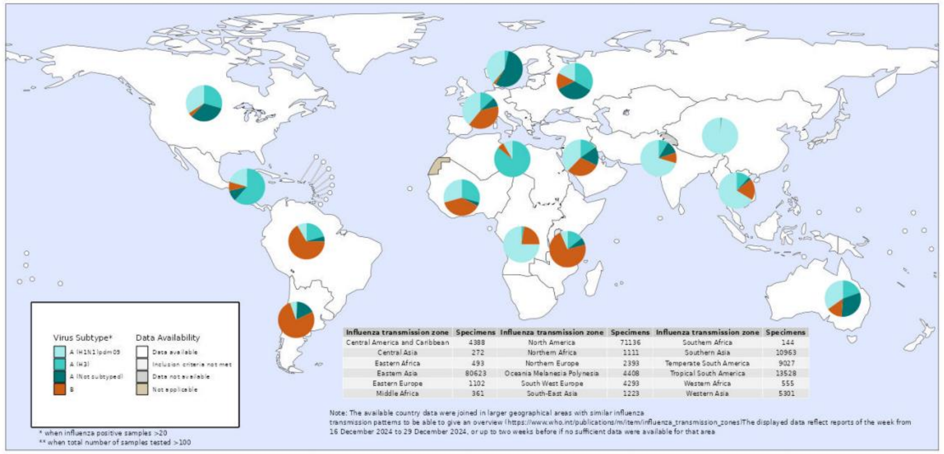
Distribution of influenza virus types and subtypes by geographic zone in 2025



The distribution of influenza viruses shifted to A/H3N2 during mid-July to August 2025

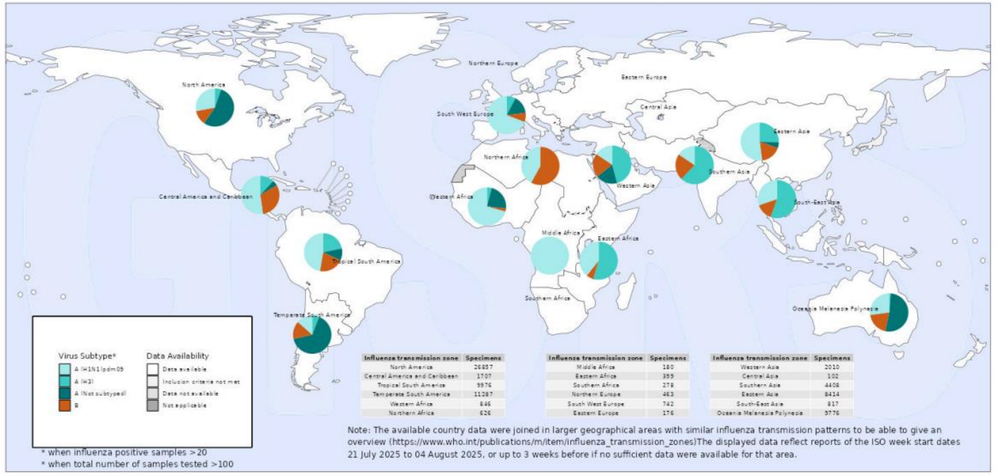
https://www.who.int/teams/global-influenza-programme/surveillance-and-monitoring/respiratory-viruses/influenza-updates-archive

Proportions of influenza virus types and subtypes by influenza transmission zones in 2025

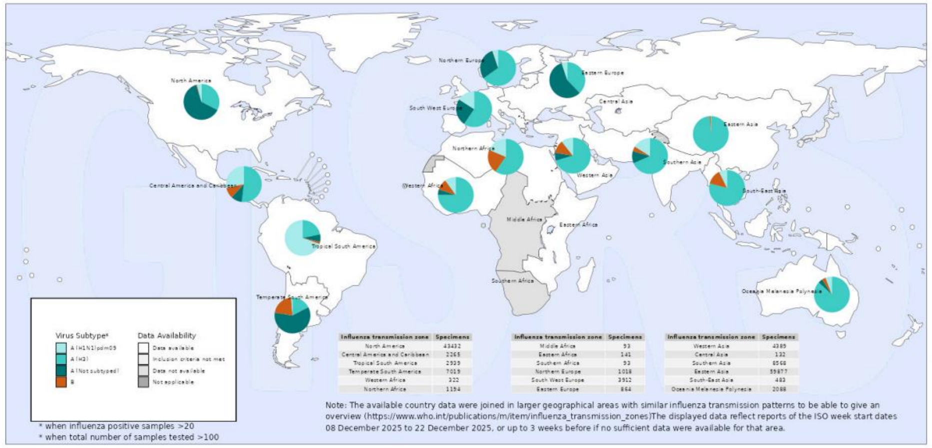


Week 1

Week 32



Week 52

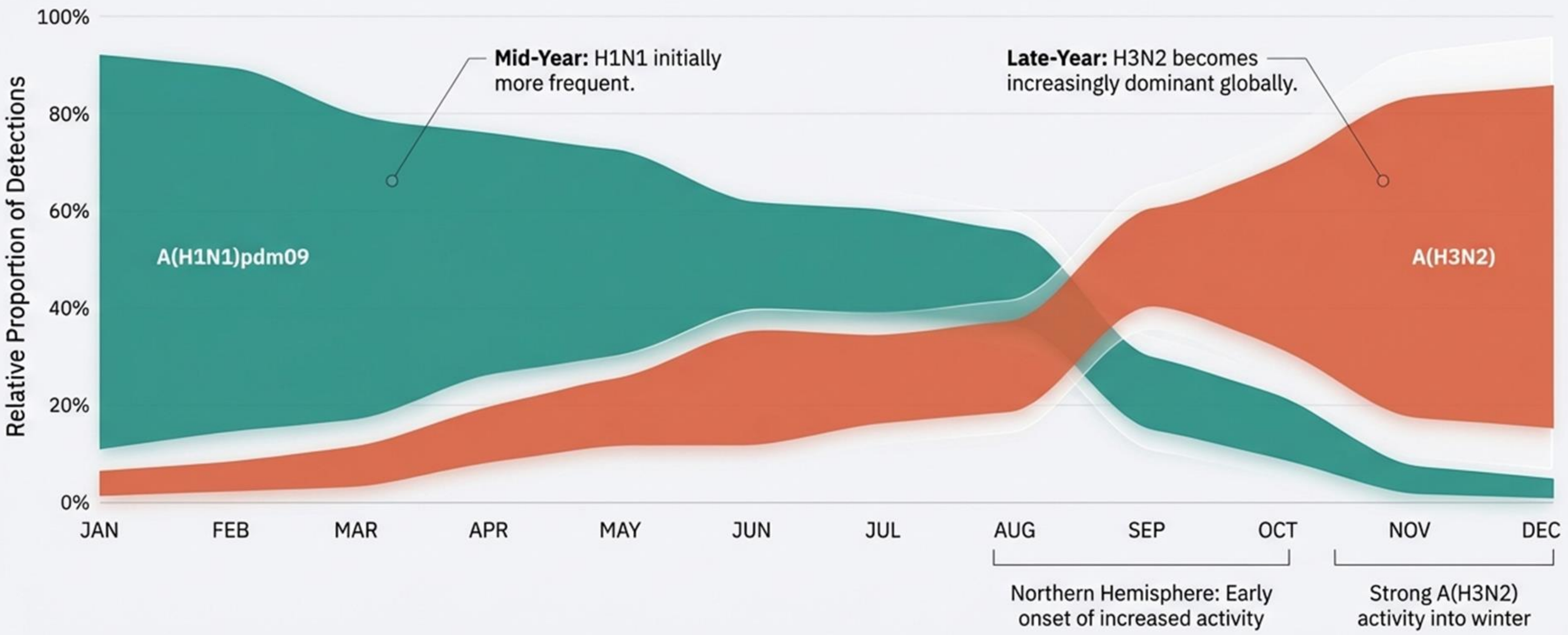


In the zones with elevated positivity, influenza A/H3N2 was predominant in all zones except Tropical South America where influenza A/H1N1(pdm09) was predominant

https://www.who.int/tools/flunet

Summary of global influenza situation in 2025

Key Takeaway: Influenza A viruses continued to predominate globally throughout 2025, but the specific strain composition inverted mid-year.



https://www.who.int/tools/flunet

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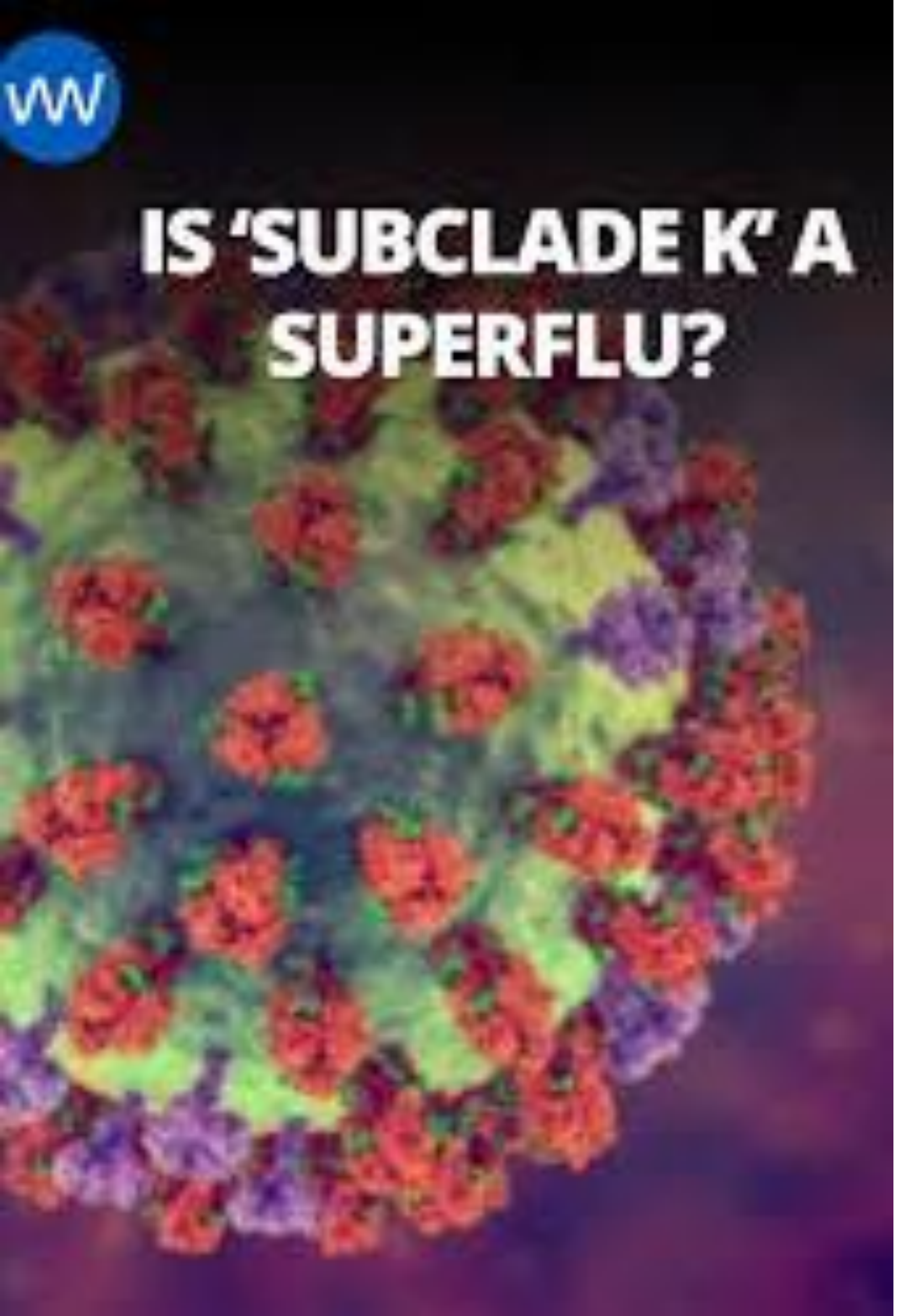
New flu variant Subclade K


IS 'SUBCLADE K' A
SUPERFLU?

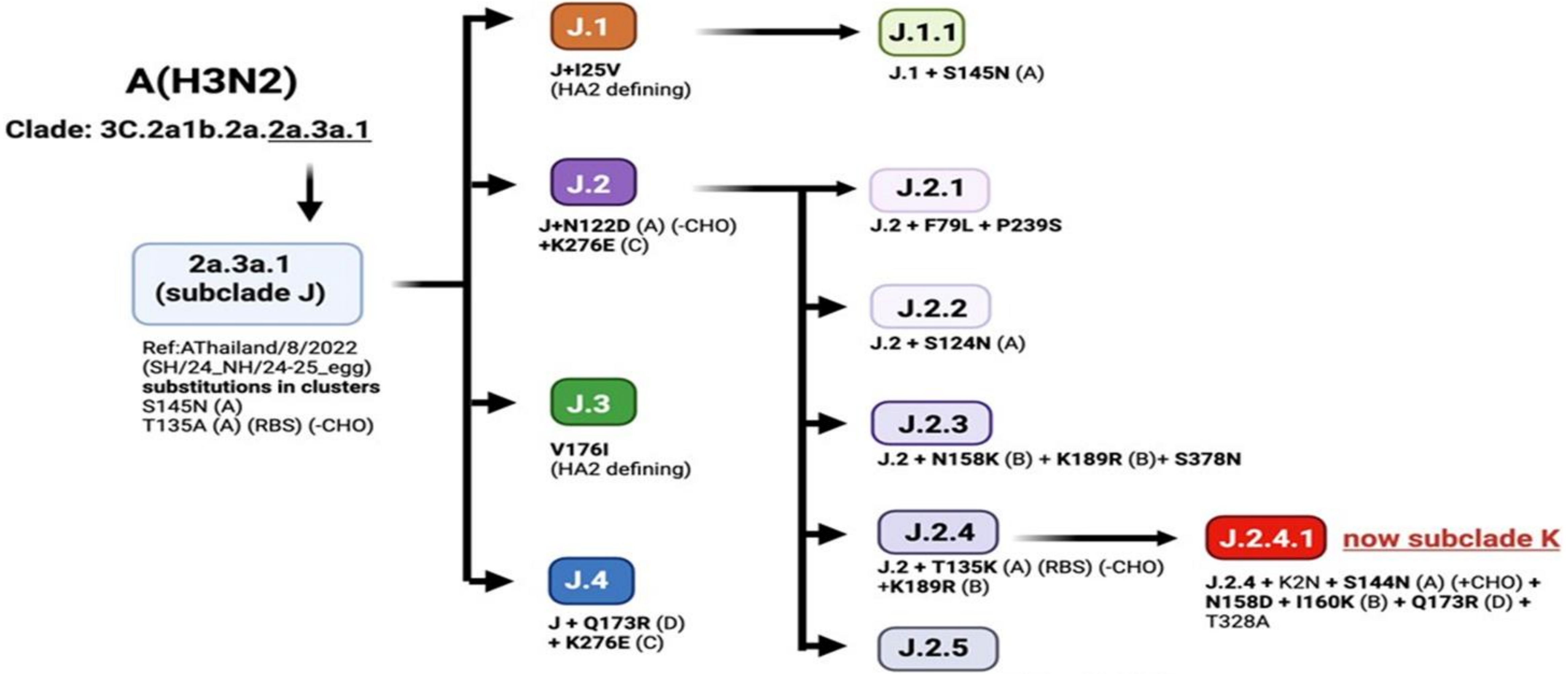

PENSACOLA, FLA.
WEAR
FLU SEASON WORSENS WITH
NEW 'SUBCLADE K' VARIANT


VARIANT
DE K

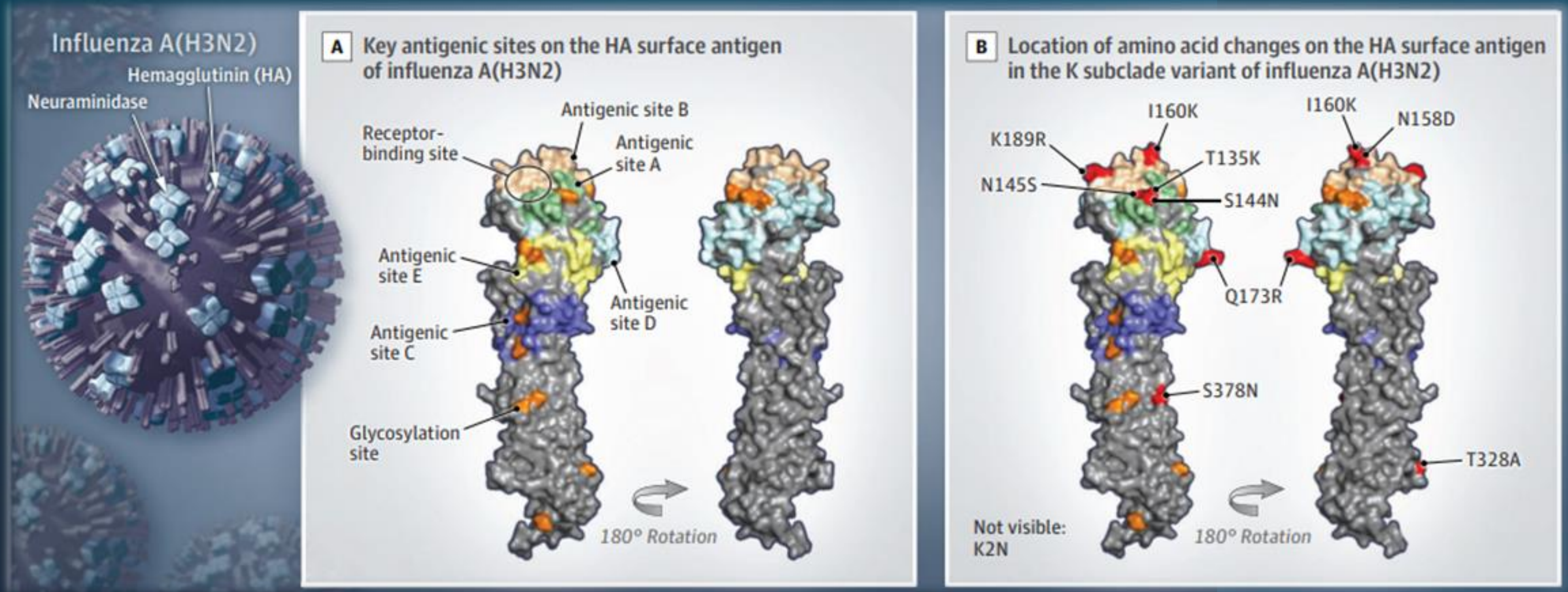

SCIENTISTS WARN THAT A NEW,
MUTATED "SUPER FLU" STRAIN IS
CIRCULATING WORLDWIDE.


Super Flu Risk
Increases Due to Lack
of Vaccine
www.propakistani.pk

Influenza A(H3N2) clade 2a.3a.1 (subclade J)



Subclade K variant of influenza A (H3N2)

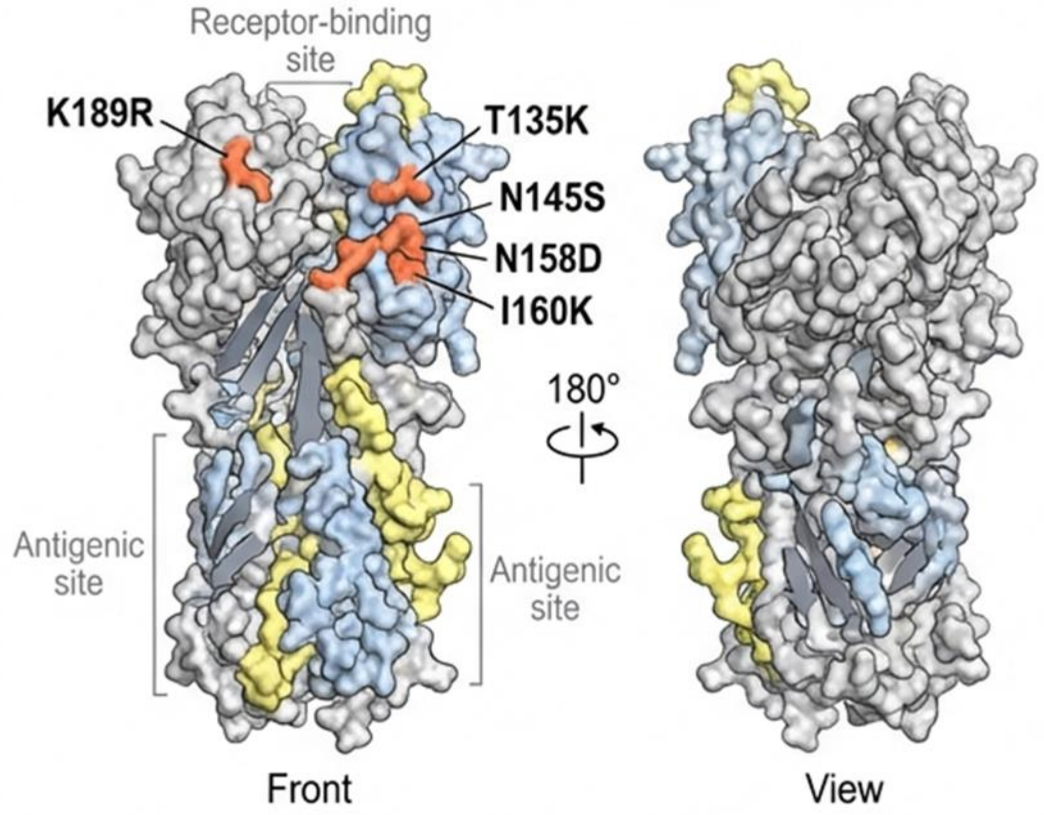


- Five antigenic sites (A-E) on the HA surface antigen encompassing 131 amino acids
- Changes in < 10 of the 131 amino acids are the key residues underpinning antigenic variation
- Substitutions occurring in these proteins may confer a selective advantage in evading existing antibodies, a process termed "antigenic drift"

Schweitzer K. JAMA. 2026;335(3):199-200.

A ‘Lock and Key’ Mismatch at the Receptor Binding Domain

Subclade K variant of influenza A(H3N2) hemagglutinin (HA) surface antigen



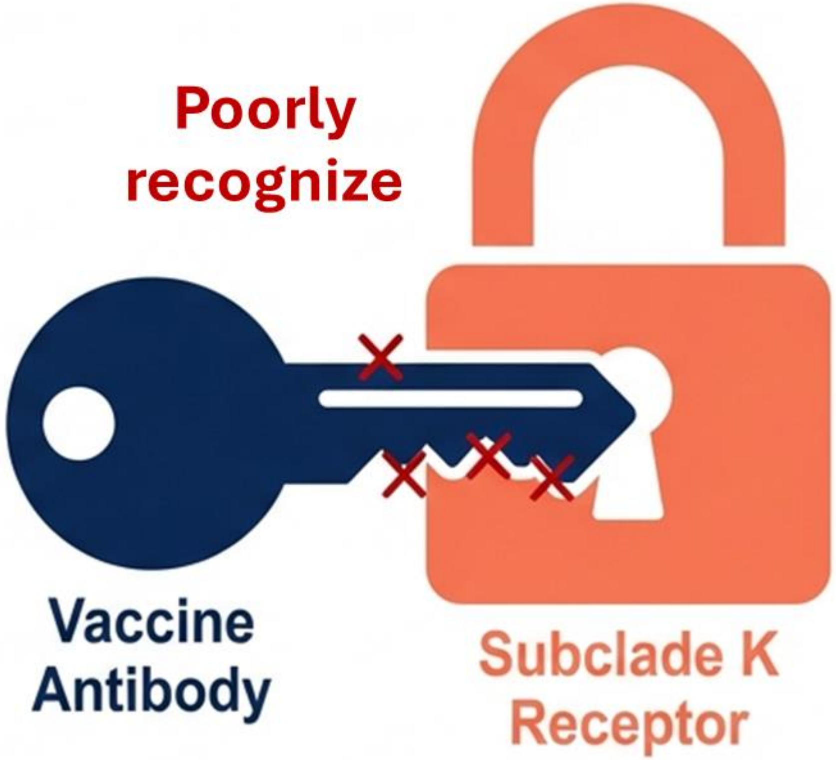
K189R

T135K[-CHO]
(Loss of carbohydrate side chain)

N145S

N158D

I160K



Ferret antisera and human post-vaccine sera show **reduced inhibition** of **Subclade K**, confirming it is an **Antigenic Drift** variant.

Schweitzer K. JAMA. 2026;335(3):199-200.

Dapat C, et al. Euro Surveill. 2025;30(49):2500894.

Global migration of influenza A(H3N2) subclade K virus

The variant emerged four months after selection of vaccine strain for northern hemisphere

The temporal gap between formulation and emergence allowed J.2.4.1 (K) to drift significantly from the J.2 target strain.

Schweitzer K. JAMA. 2026;335(3):199-200.

The 2025-2026 influenza vaccine targets Subclade J.2, creating a genetic mismatch

Vaccine Formulation
Target: Subclade J.2
Selected: Feb 2025
Status: Locked

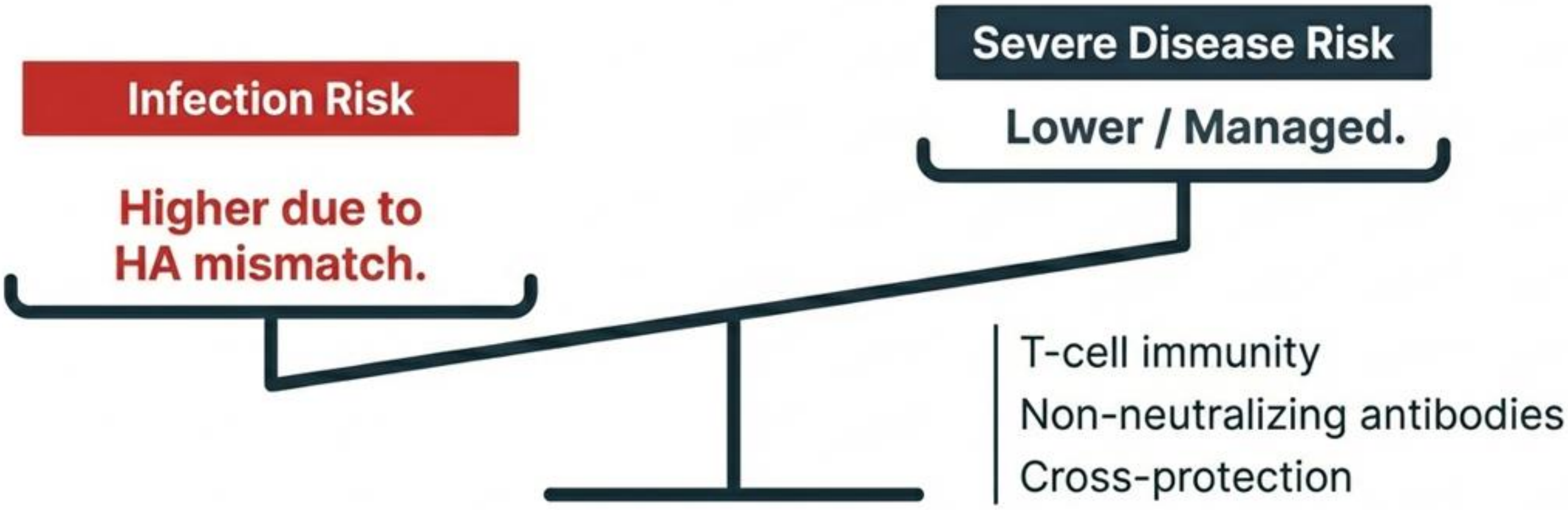
≠

Circulating Virus
Dominant Strain: Subclade J.2.4.1 **(K)**
Emergence: Post-Selection
Status: **Active Drift**

"Dominant circulating H3N2 strain... diverged from the vaccine strain **after the vaccine** was finalized."

Schweitzer K. JAMA. 2026;335(3):199-200.

Vaccine protection against severe illness likely remains robust.



Early data shows the vaccine still provides good protection against severe illness.

Schweitzer K. JAMA. 2026;335(3):199-200.

Executive Summary: The Emergence of Variant J.2.4.1 (Subclade K)

The Situation

H3N2 Subclade K has rapidly become dominant (**89% of characterized US viruses**). It is driving an early, severe season with **significant pediatric impact**.

The Complication

Rapid evolution in the Hemagglutinin (HA) protein has outpaced the 2025-26 Northern Hemisphere vaccine. Vaccine Effectiveness (VE) against infection is reduced, particularly in adults (**32-39%**).

The Resolution

The variant remains fully susceptible to Neuraminidase Inhibitors (**Oseltamivir) and Baloxavir**. **Early testing and immediate antiviral treatment** are the primary defense lines this season.

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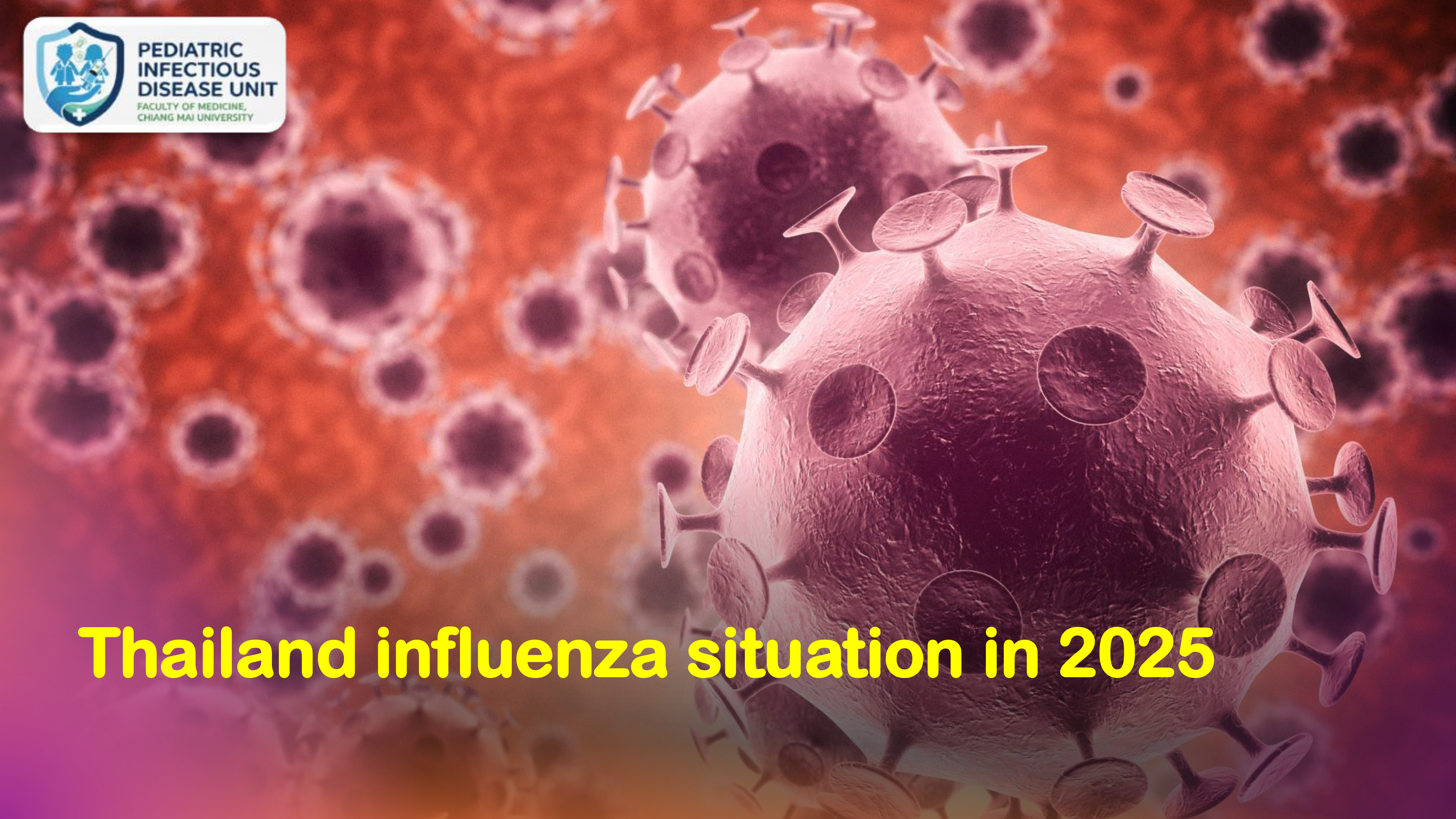
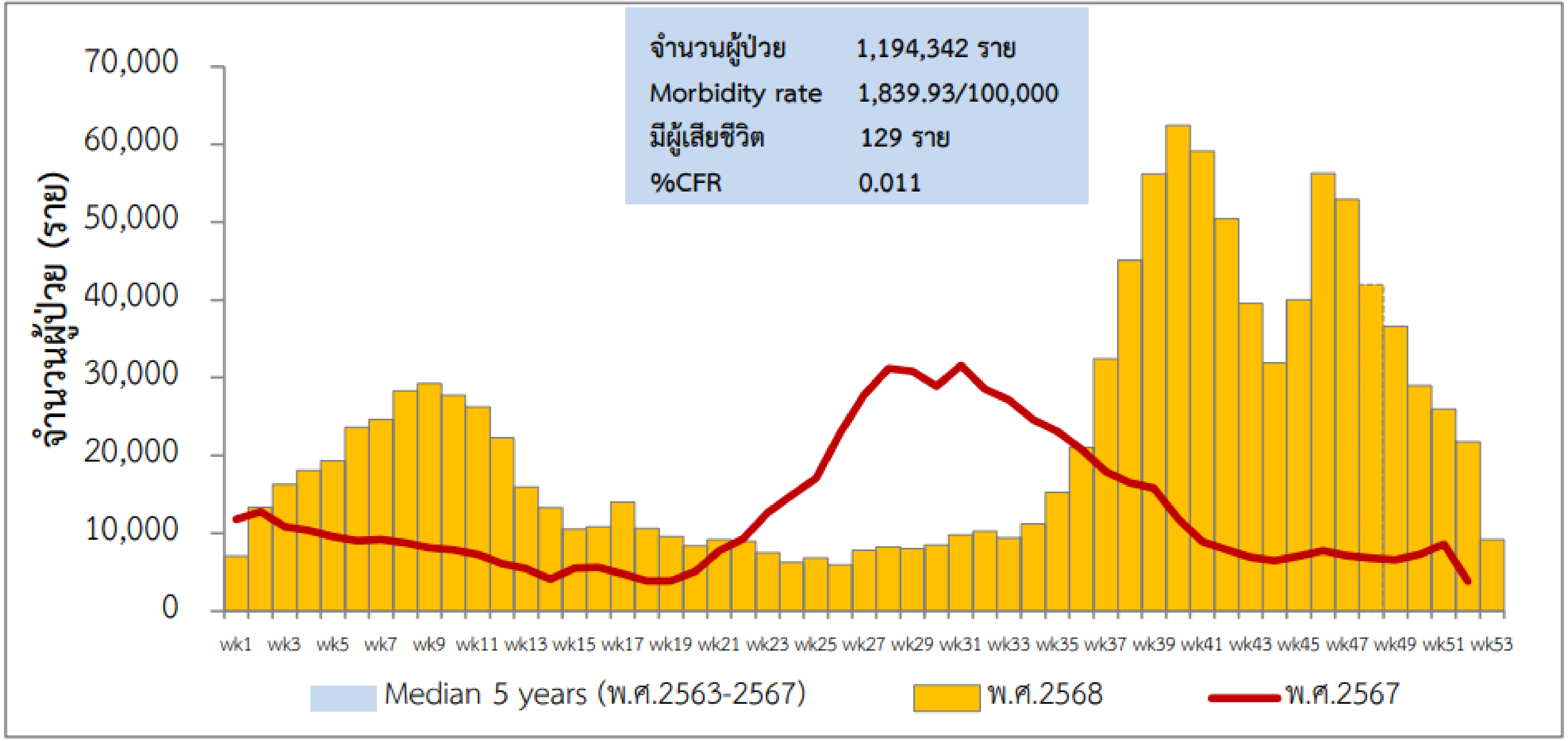
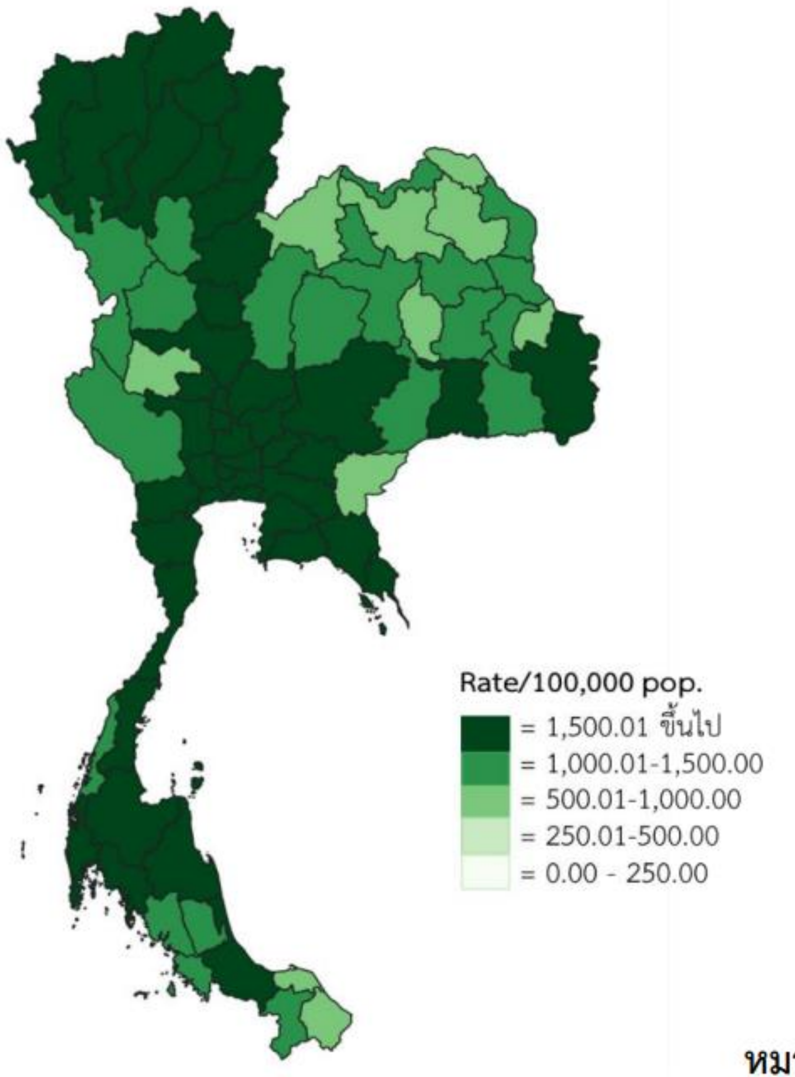


Thailand influenza situation in 2025



https://ddc.moph.go.th/doe/pagecontent.php?page=607&dept=doe

Epidemic Curve: 2025 Seasonal Trends



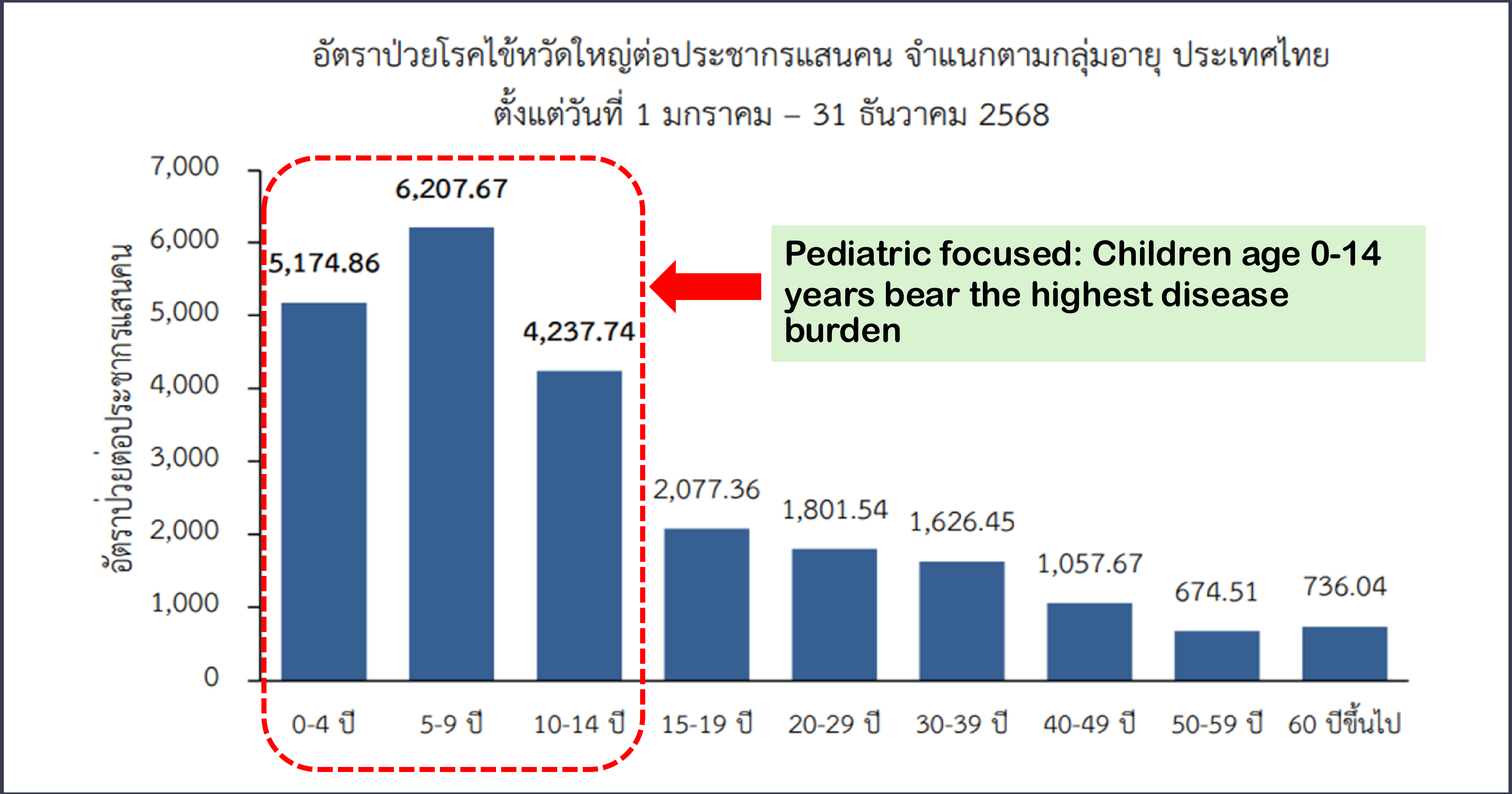
จังหวัดที่มีอัตราป่วยสูง 10 อันดับแรก

ลำดับ	จังหวัด	อัตราป่วยต่อประชากรแสนคน
1	ภูเก็ต	4,237.13
2	ชลบุรี	3,300.18
3	กรุงเทพมหานคร	3,197.20
4	ระยอง	2,842.84
5	นนทบุรี	2,738.19
6	สมุทรปราการ	2,702.57
7	ลำพูน	2,692.24
8	นครปฐม	2,676.63
9	พะเยา	2,523.53
10	สมุทรสาคร	2,522.50

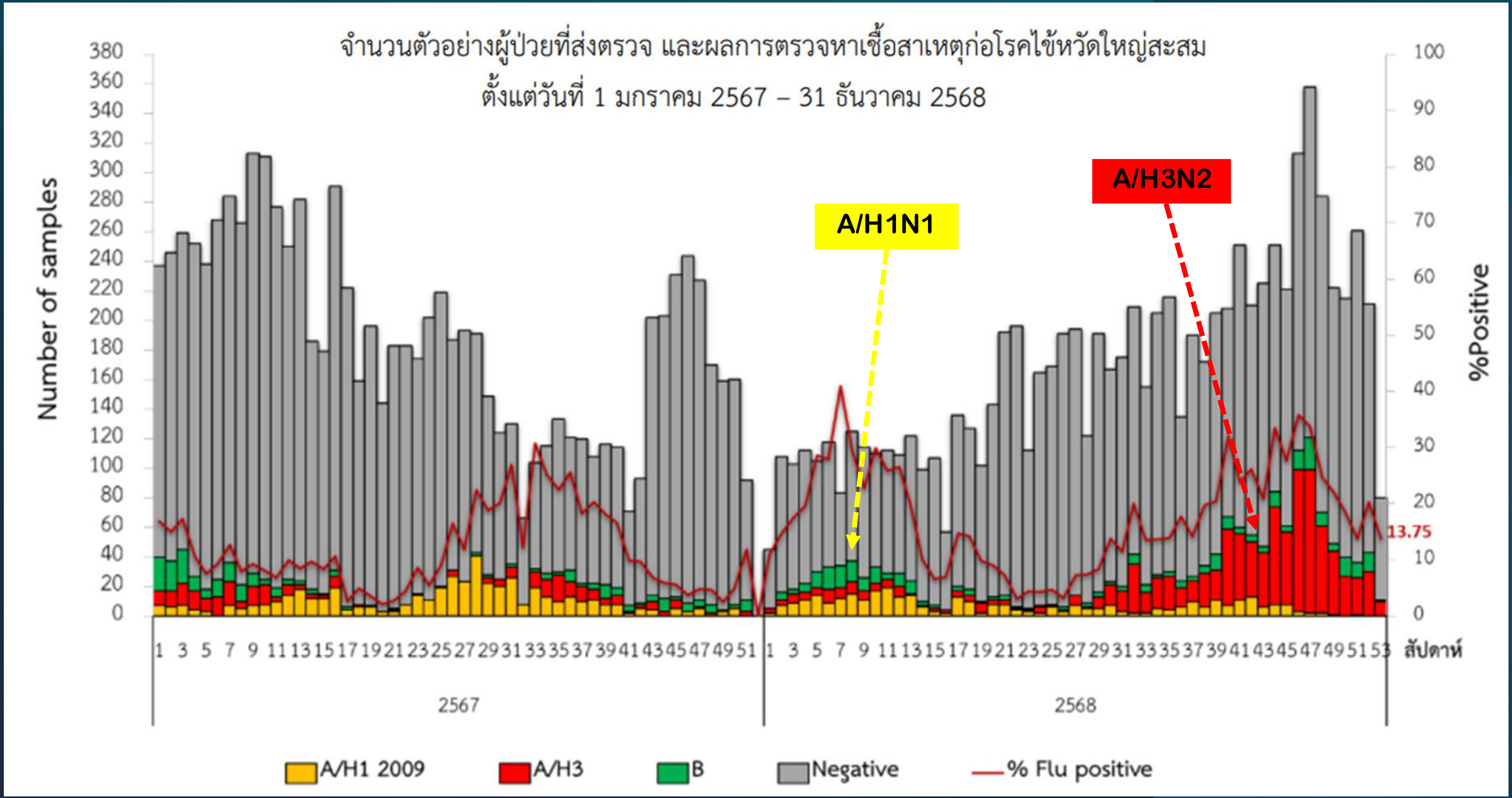
Tourist hubs

หมายเหตุ : นำเสนอข้อมูลตามวันและสถานที่เริ่มป่วย

https://ddc.moph.go.th/doe/pagecontent.php?page=607&dept=doe



https://ddc.moph.go.th/doe/pagecontent.php?page=607&dept=doe

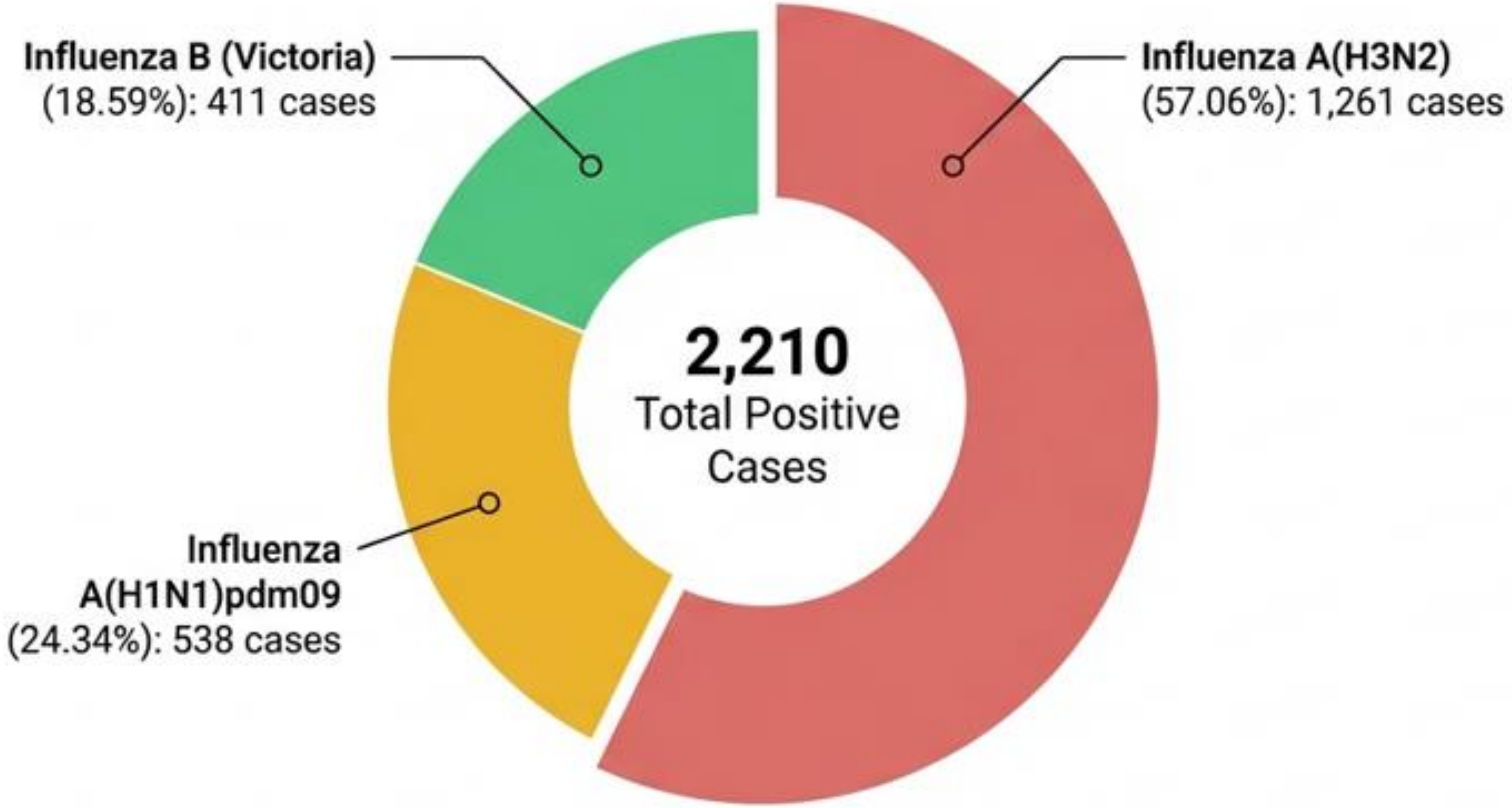


https://ddc.moph.go.th/doe/pagecontent.php?page=607&dept=doe

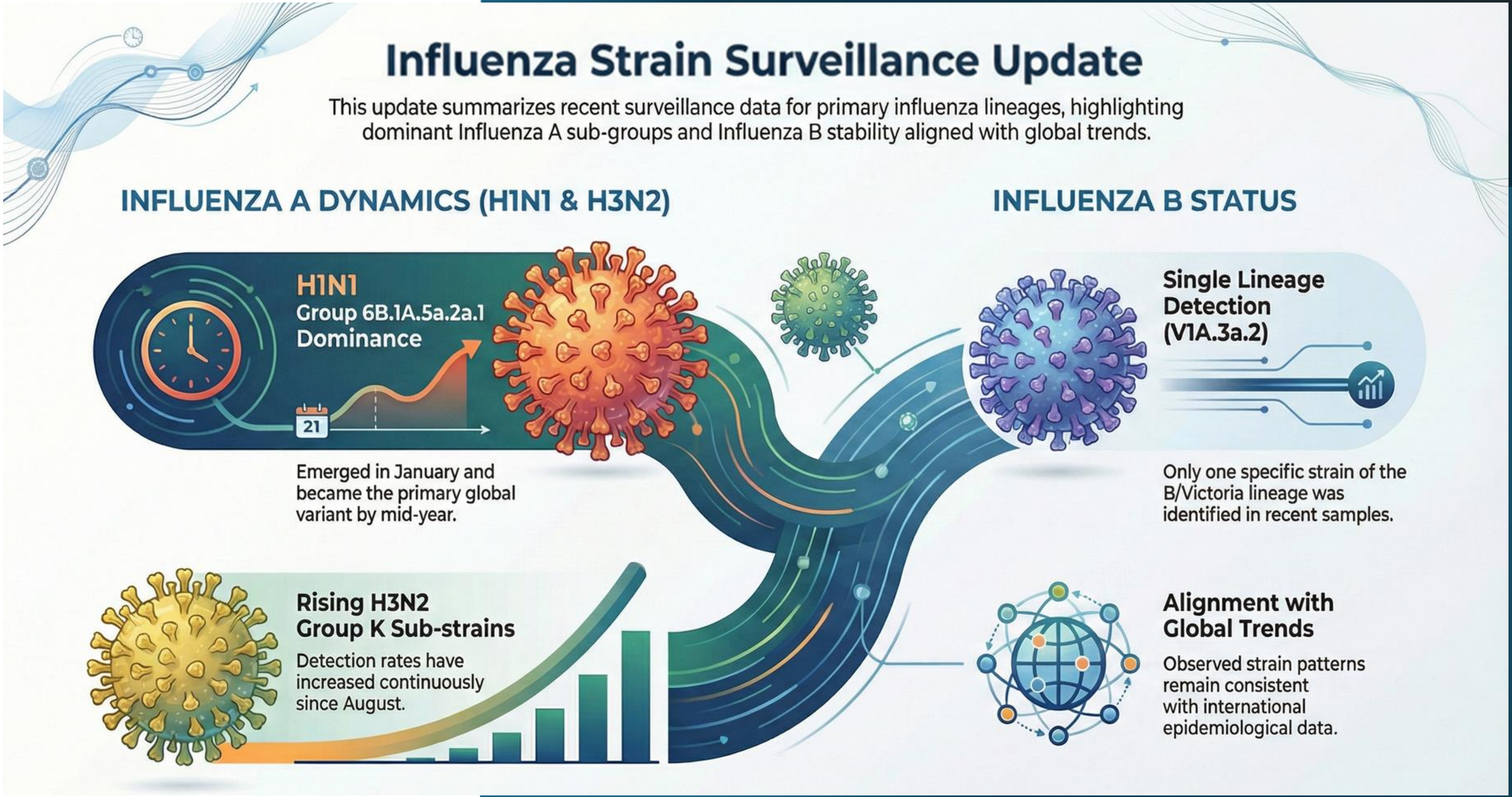
One in five samples tested positive for Influenza in 2025

10,493
Total Samples Tested

21.06%
Overall Positivity Rate



https://nih.dmsc.moph.go.th/th/detailAll/3449/abs/235



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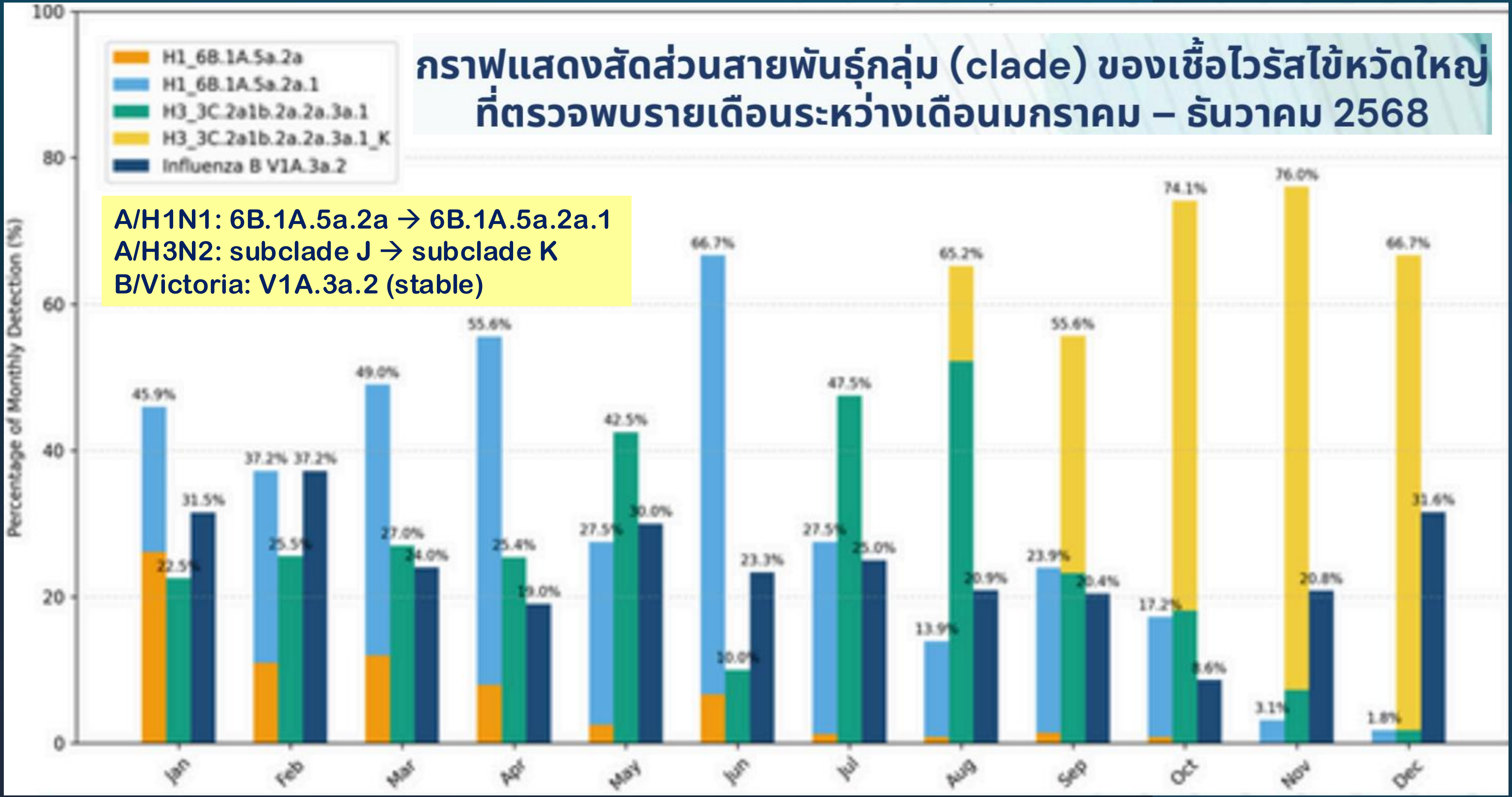


https://nih.dmsc.moph.go.th/th/detailAll/3449/abs/235

Antiviral Susceptibility: Zero drug resistance detected in 2025

Target: Neuraminidase (NA) gene analysis

H1N1 Results:
No H275Y mutation found. Oseltamivir fully effective.

H3N2 & B Results:
No resistance markers found in any samples.

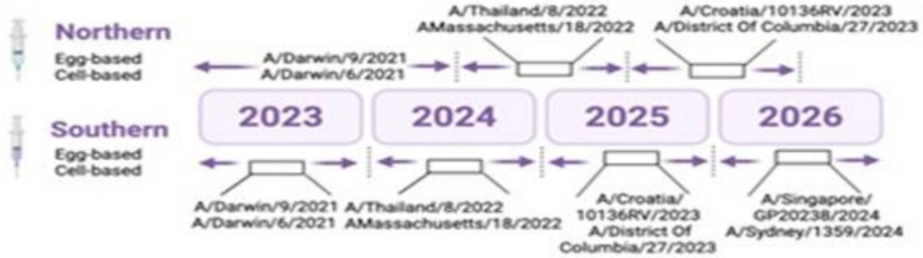
https://nih.dmsc.moph.go.th/th/detailAll/3449/abs/235

December Spotlight: Analysis of 110 specific samples from December confirmed no resistance genes.

Thai strains align with 2026 Southern Hemisphere vaccine targets

Strain Type	Thailand Dominant Strain	Vaccine Match Status
Influenza A(H1N1)	Subclade D.3.1 (82.8%)	MATCH (Compatible with 2026 Targets)
Influenza A(H3N2)	High Diversity: Subclades K and J.2	MATCH (Falls under vaccine group 3C.2a1b.2a.2a.3a.1)
Influenza B/Victoria	Subclades C.5.7 and C.5.6	MATCH

Phylogenetic Analysis
Influenza A (H3N2) 2025, Thailand



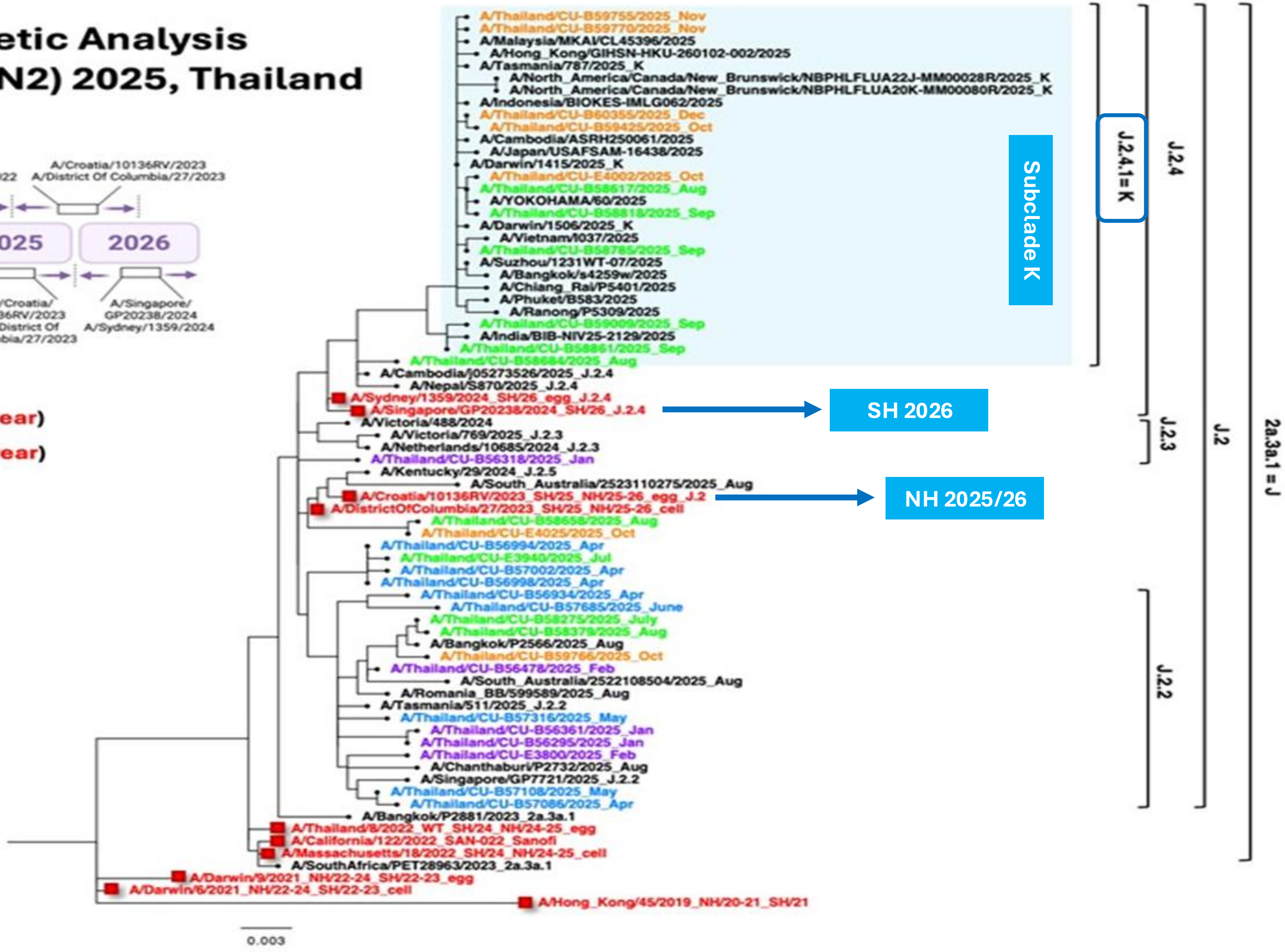
Vaccine strains:

Southern hemisphere vaccine (SH/year)

Northern hemisphere vaccine (NH/year)

Sample collection

- Jan-Mar 2025
- Apr-Jun 2025
- Jul-Sep2025
- Oct-Dec 2025



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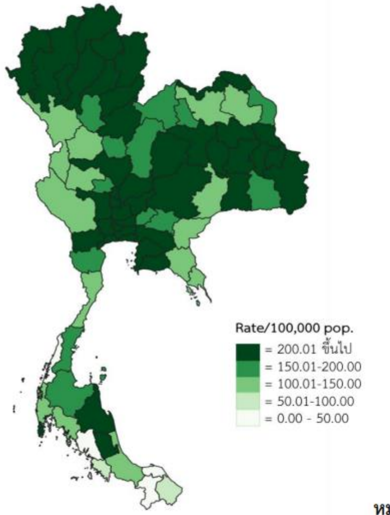
Current influenza situation in Thailand 2026

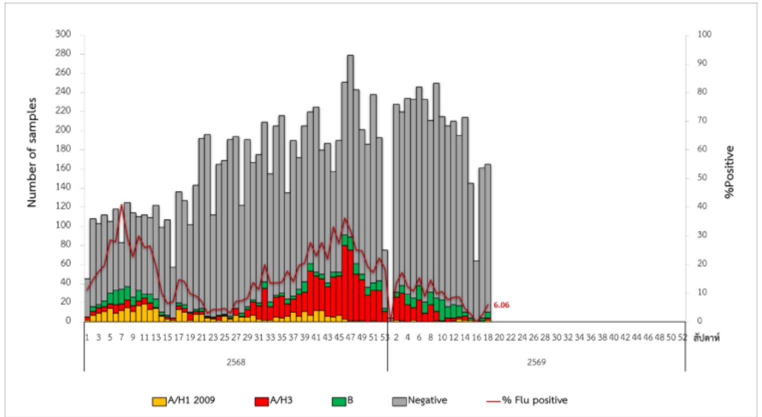


จังหวัดที่มีอัตราป่วยสูง 10 อันดับแรก

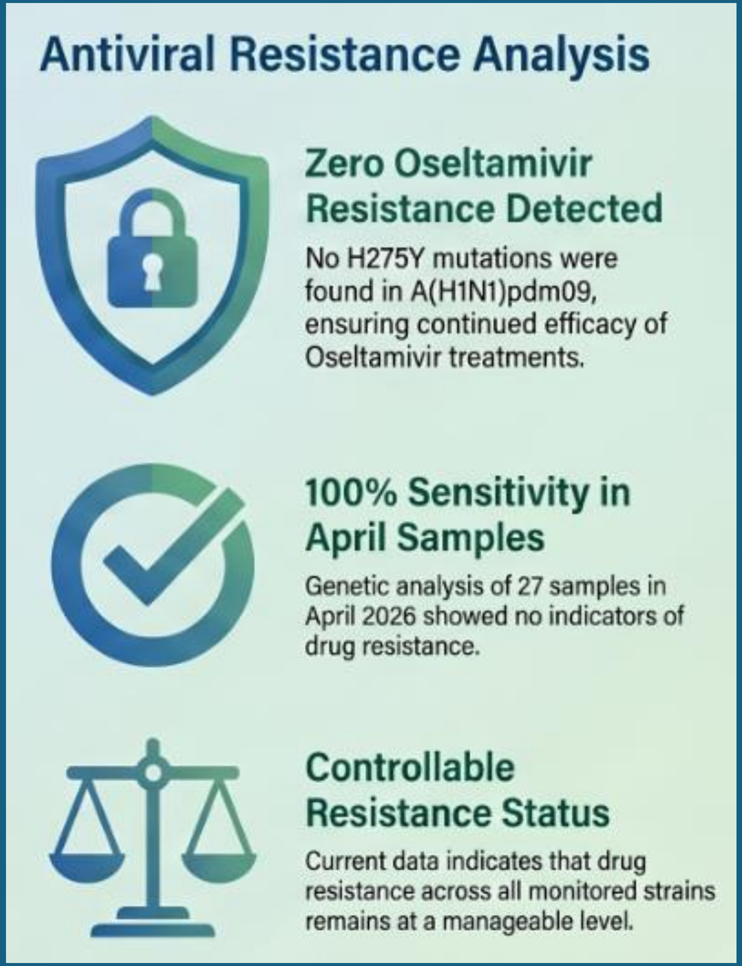
ลำดับ	จังหวัด	อัตราป่วยต่อประชากรแสนคน
1	พะเยา	555.40
2	อุบลราชธานี	478.35
3	ลำปาง	472.32
4	กรุงเทพมหานคร	445.02
5	พิษณุโลก	433.32
6	ระยอง	412.95
7	เชียงใหม่	412.77
8	ลำพูน	406.24
9	แพร่	395.06
10	ชลบุรี	386.24

หมายเหตุ : นำเสนอข้อมูลตามวันและสถานที่เริ่มป่วย

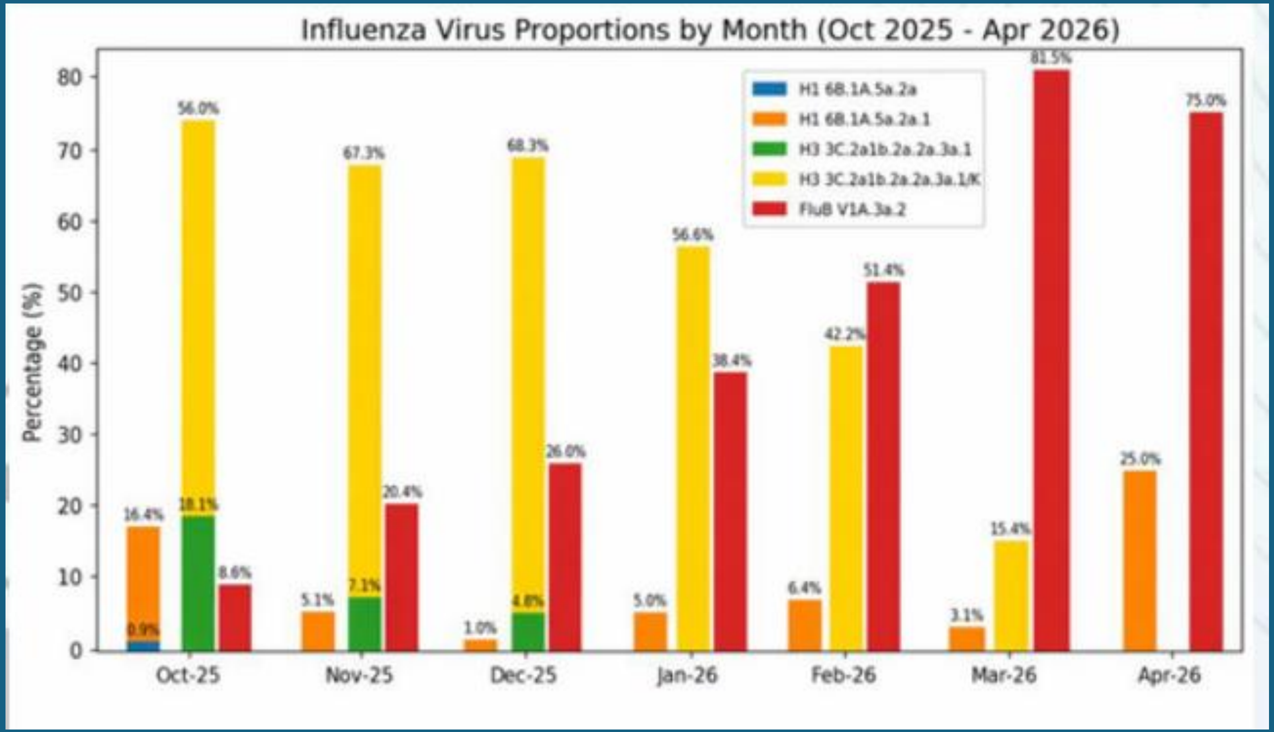
แหล่งข้อมูล : ระบบเฝ้าระวังโรคดิจิทัล (DDS) กองระบาดวิทยา กรมควบคุมโรค



แหล่งข้อมูล : ผลการเฝ้าระวังเชื้อไวรัสก่อโรคไข้หวัดใหญ่ กองระบาดวิทยา และสถาบันป้องกันควบคุมโรคเขตเมือง กรมควบคุมโรค ร่วมกับกรมวิทยาศาสตร์การแพทย์ และศูนย์ความร่วมมือไทย-สหรัฐ ด้านสาธารณสุข และสถาบันสุขภาพเด็กแห่งชาติมหาราชินี

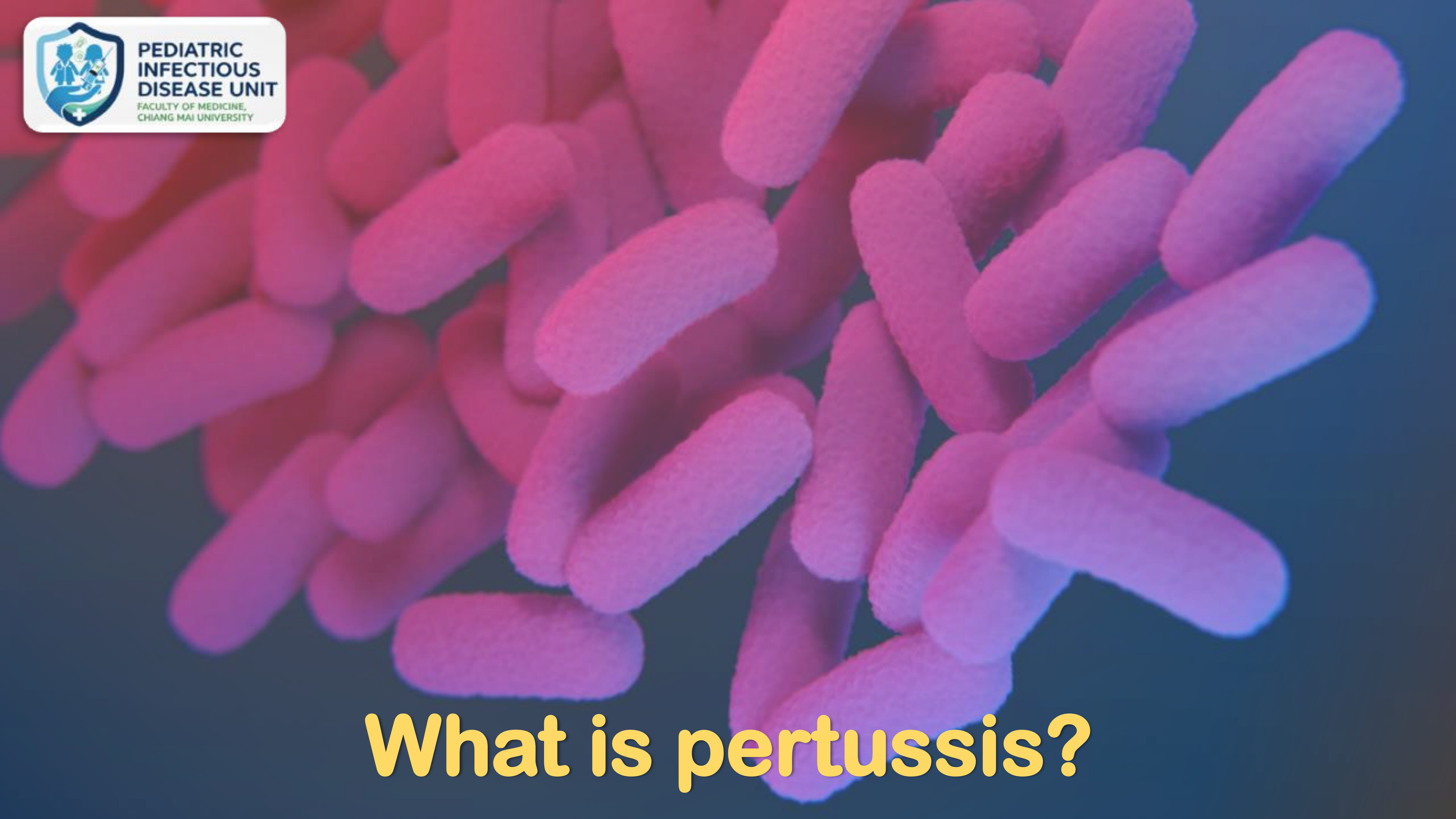


กราฟแสดงสัดส่วนสายพันธุ์กลุ่ม (clade) ของเชื้อไวรัสไข้หวัดใหญ่ที่ตรวจพบรายเดือนระหว่าง วันที่ 1 ตุลาคม 2568 ถึง 21 เมษายน 2569



A/H1N1: 6B.1A.5a.2a.1 (stable)
A/H3N2: subclade K (stable)
B/Victoria: V1A.3a.2 (stable)

https://nih.dmsc.moph.go.th/th/detailAll/3449/abs/235



What is pertussis?

Meet the Micro-Villain: Bordetella pertussis

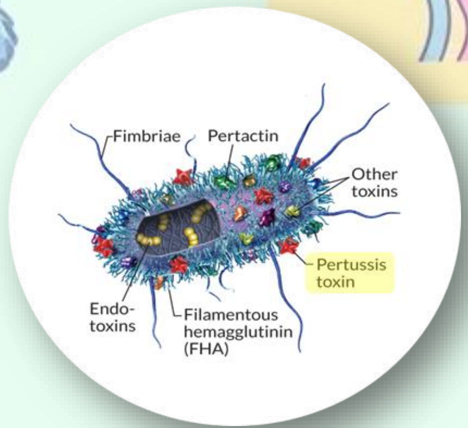
Gram-negative pleomorphic bacillus

FHA (Filamentous hemagglutinin)

Promotes adhesion. Acts as grappling hooks to latch onto human airway cilia.

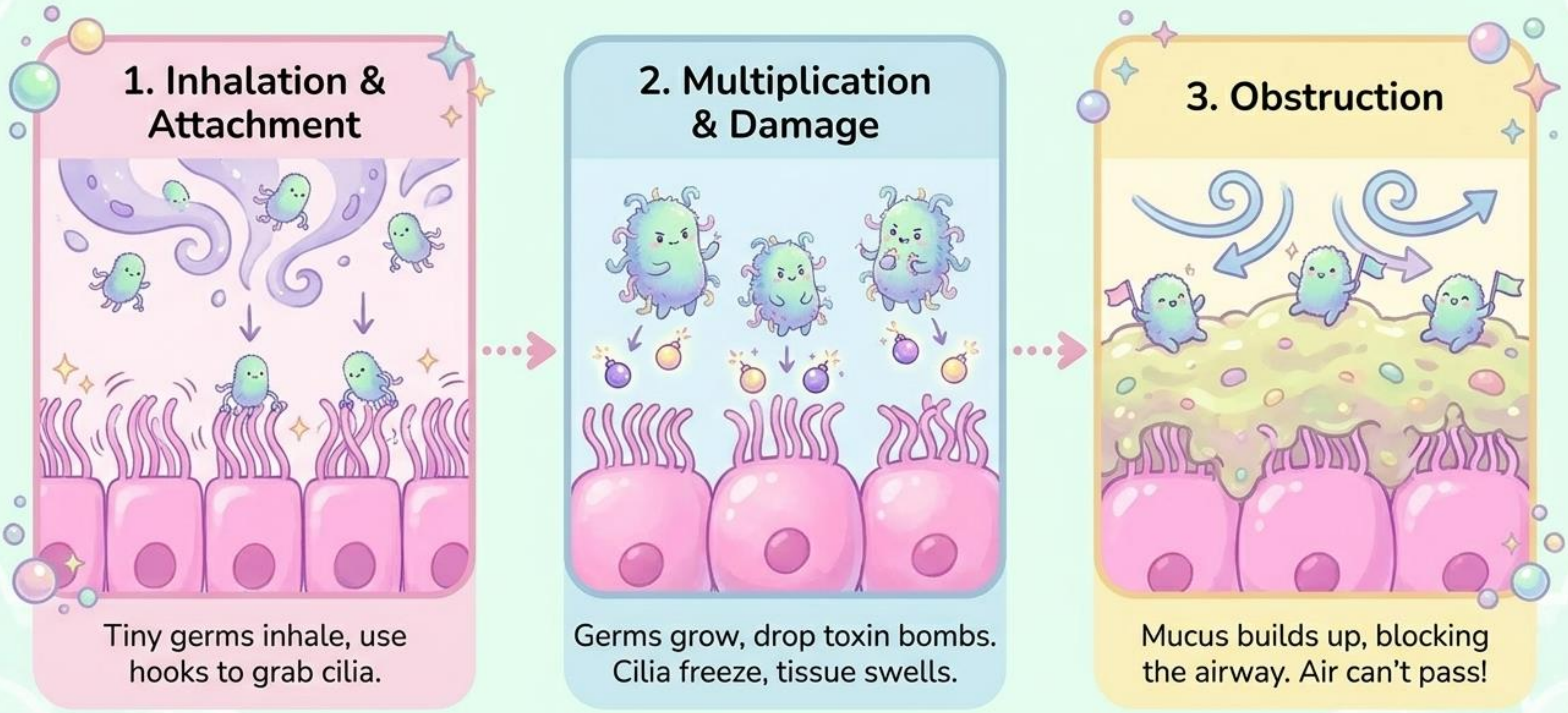
PT (Pertussis toxin)

The main weapon. Damages the cilia, causes extreme airway swelling, and triggers clinical immunity.



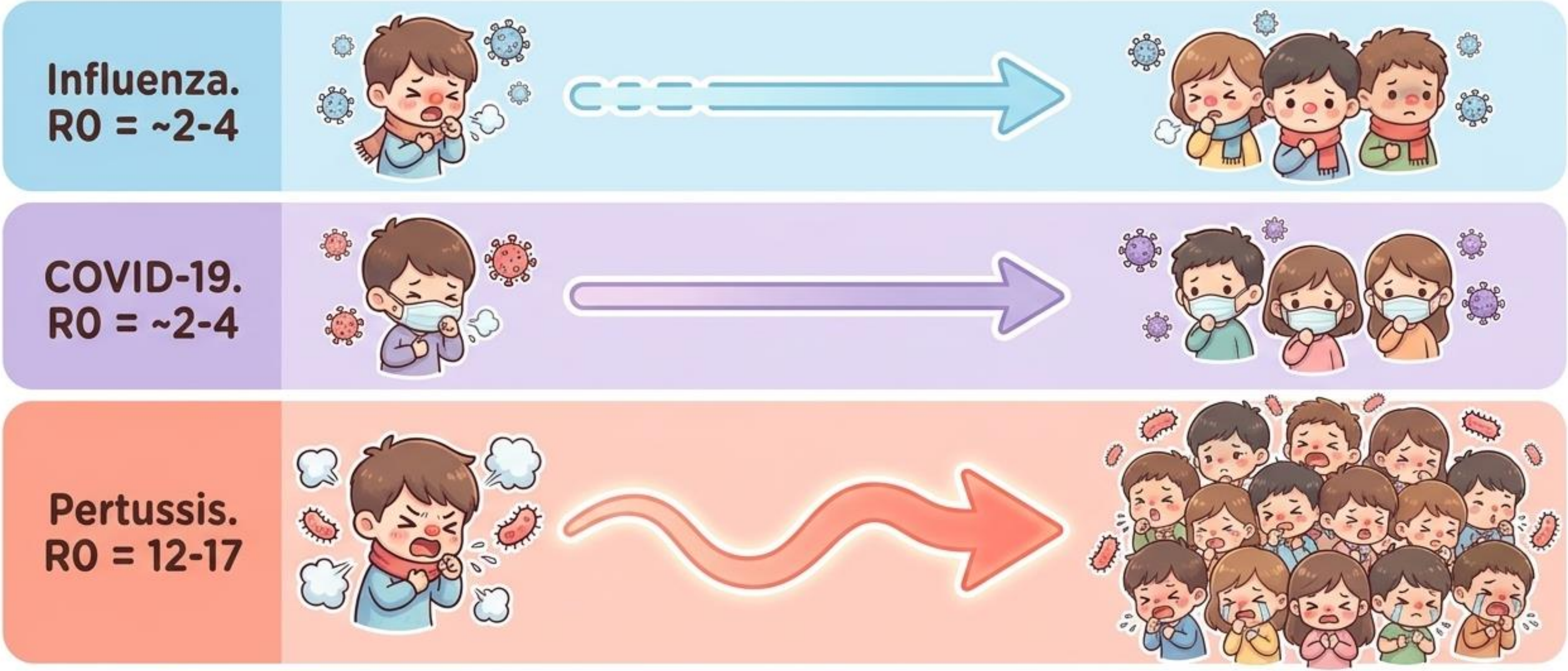
American Academy of Pediatrics. Redbook 2024.

Step-by-Step: How Pertussis Paralyzes the Airway



American Academy of Pediatrics. Redbook 2024.

The Extreme Speed of Transmission



Pertussis spreads highly efficiently through respiratory droplets, outpacing both the Flu and SARS-CoV-2.

The 100-Day Cough: A Grueling Clinical Course

Catarrhal Phase

1–2 weeks

Paroxysmal Phase

1–6 weeks

Convalescent Phase

Weeks to months

The disease unfolds over months, often completely exhausting the patient's respiratory system.

https://www.cdc.gov/pertussis/clinical/features.html

Phase 1: Catarrhal (The Disguise)

Duration: 1 to 2 weeks.

Symptoms: Mild Upper Respiratory Infection (URI), occasional low-grade fever.

Infant Risk: Can present simply as apnea (pauses in breathing) in young infants.

The Danger: Highly infectious, but looks exactly like a normal cold. This is when the bacteria spread most rapidly to others.

Phase 2: Paroxysmal (The Peak)

Duration:
1 to 6 weeks.

The Whoop:
A high-pitched sound upon forced inhalation after a severe coughing fit.

The Signature Cough:
Paroxysmal coughing (rapid, uncontrollable bursts of coughing trying to expel thick mucus from the tracheobronchial tree).

Severe Symptoms:
Difficulty breathing, Apnea, Cyanosis (turning blue from lack of oxygen), and Post-tussive vomiting.

Whooping cough

https://www.cdc.gov/pertussis/about/complications.html

Phase 3: Convalescent (The Long Recovery)

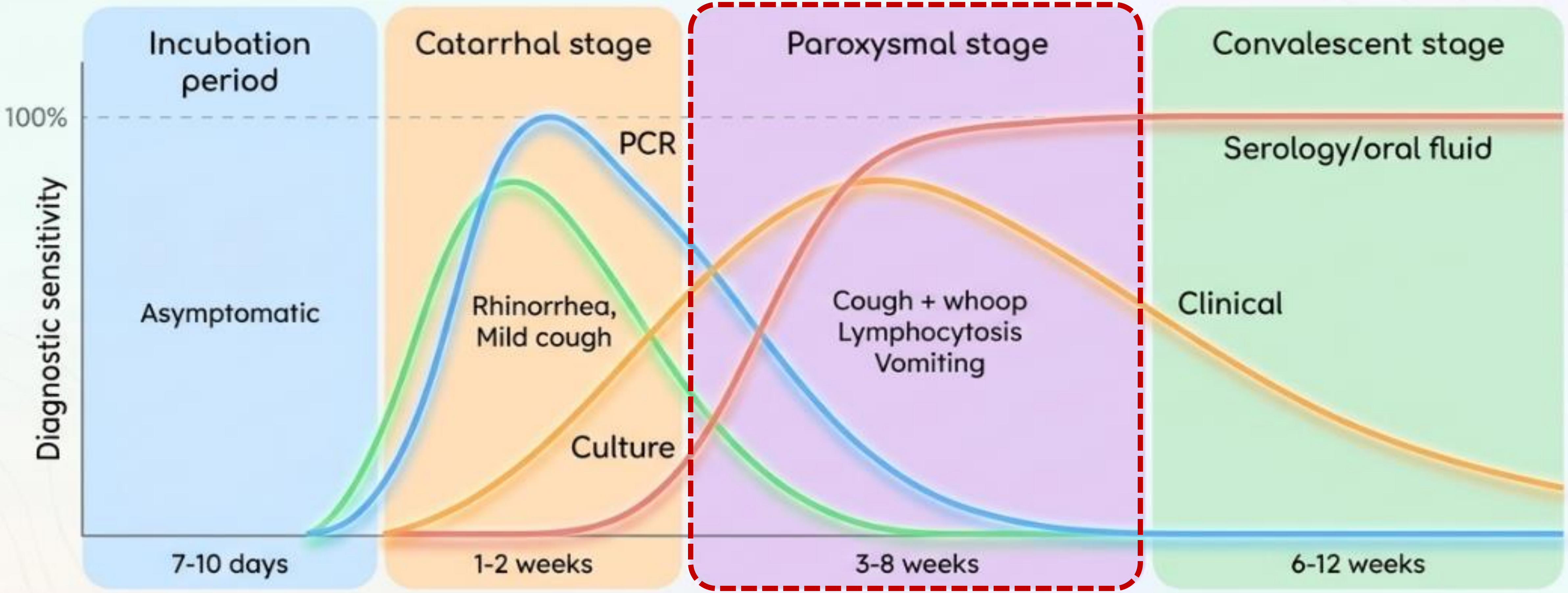
Gradual Improvement

- **Duration:** Weeks to months.
- The coughing fits slowly become less frequent and less severe over time as the respiratory tract heals.

Lingering Vulnerability

- The respiratory tract remains highly sensitive. Paroxysms (violent coughing fits) can frequently recur if the patient contracts subsequent, normal respiratory infections during this window.

Laboratory diagnosis



Key Insight: PCR and Culture offer the highest sensitivity early in the infection, while Serology becomes the necessary diagnostic tool during the prolonged Paroxysmal and Convalescent stages.

Pediatric Pertussis: Antibiotic Eradication & Timing

Eradicating *Bordetella pertussis* requires precise antibiotic dosing and strict adherence to optimal treatment windows.

Treatment Window Timeline

Clinical Rule of Thumb
Treat within 3 weeks of cough onset for most children (extend up to 6 weeks for infants or severe cases).

Weeks 1-2 (Catarrhal Phase)
Action: Start antibiotics immediately.
Impact: Highly effective; reduces both disease severity and overall duration.

Week 3+ (Paroxysmal Phase)
Action: Continue to prescribe.
Impact: Usually does not significantly change symptoms, but is critically recommended to prevent transmission.

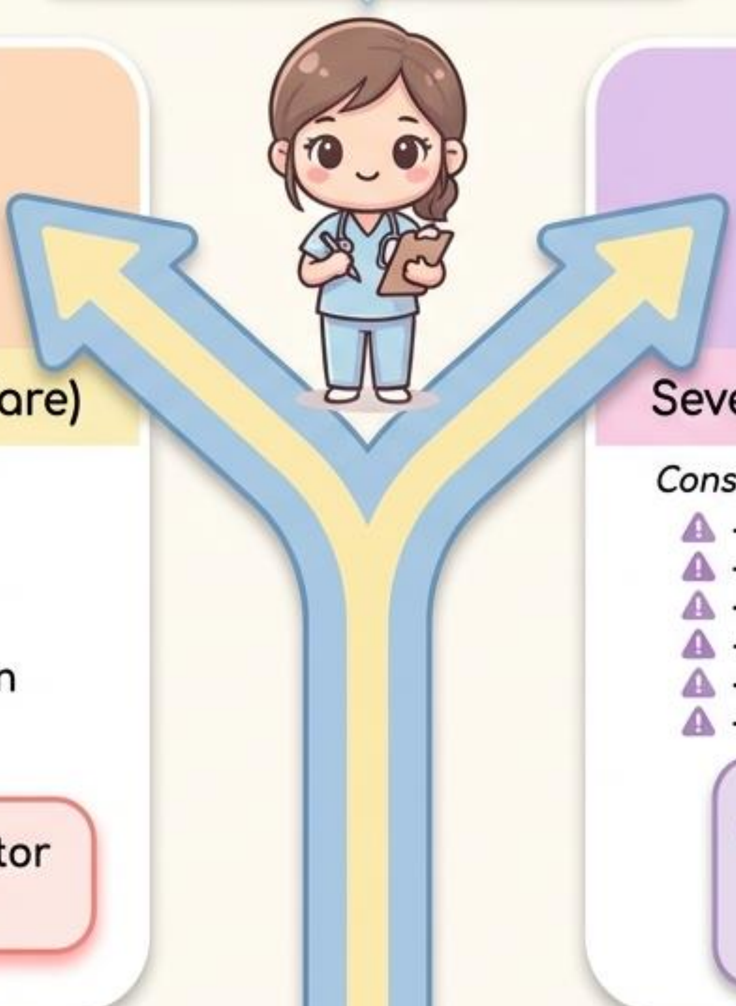
Pharmacological Decision Matrix

Antibiotic	Duration	Dosing Guidelines	Clinical Rationale
Azithromycin [PREFERRED]	5 days	**<1 month**: 10 mg/kg/day orally, once daily. **>=1 month**: Day 1: 10 mg/kg (max 500 mg). **Days 2-5**: 5 mg/kg/day (max 250 mg/day).	Most commonly used; better tolerated with a shorter duration.
Clarithromycin	7 days	15 mg/kg/day divided twice daily (max 1 g/day).	Standard alternative macrolide.
Erythromycin	14 days	40-50 mg/kg/day divided into 4 doses.	Less favored due to GI adverse effects.
Trimethoprim-sulfamethoxazole (TMP-SMX) [ALTERNATIVE]	14 days	Restricted to children >=2 months only.	Used only when macrolides cannot be tolerated or are contraindicated.

Pediatric Pertussis: Supportive Care & Admission Triage

Supportive care dictates the clinical trajectory, demanding immediate triage between mild home management and severe hospital admission.

Supportive care is critical, especially in vulnerable infant populations.



Mild Disease Management (Home Care)

- Maintain steady hydration
- Provide small, frequent feeds
- Perform nasal suctioning to maintain airway clearance

VIGILANCE WARNING: Constantly monitor for any signs of apnea or cyanosis.

Severe Disease (Hospitalization Criteria)

Consider immediate admission if the patient exhibits:

- Age <3-6 months (highest risk)
- Apnea, Cyanosis, or Hypoxemia
- Pneumonia
- Feeding difficulty leading to dehydration
- Seizures
- Leukocytosis or hyperleukocytosis

In-Patient Interventions

- If admitted, prepare to deploy:
  - Oxygen therapy
  - IV fluids or NG tube feeding
  - Advanced respiratory support
  - ICU care (specifically for severe infant pertussis)

The Dire Threat to Young Infants

Hospitalization: ~33%
1 in 3 babies require hospital care.

Apnea: 68%
Experience life-threatening pauses in breathing.

Mortality:
1% risk of death.

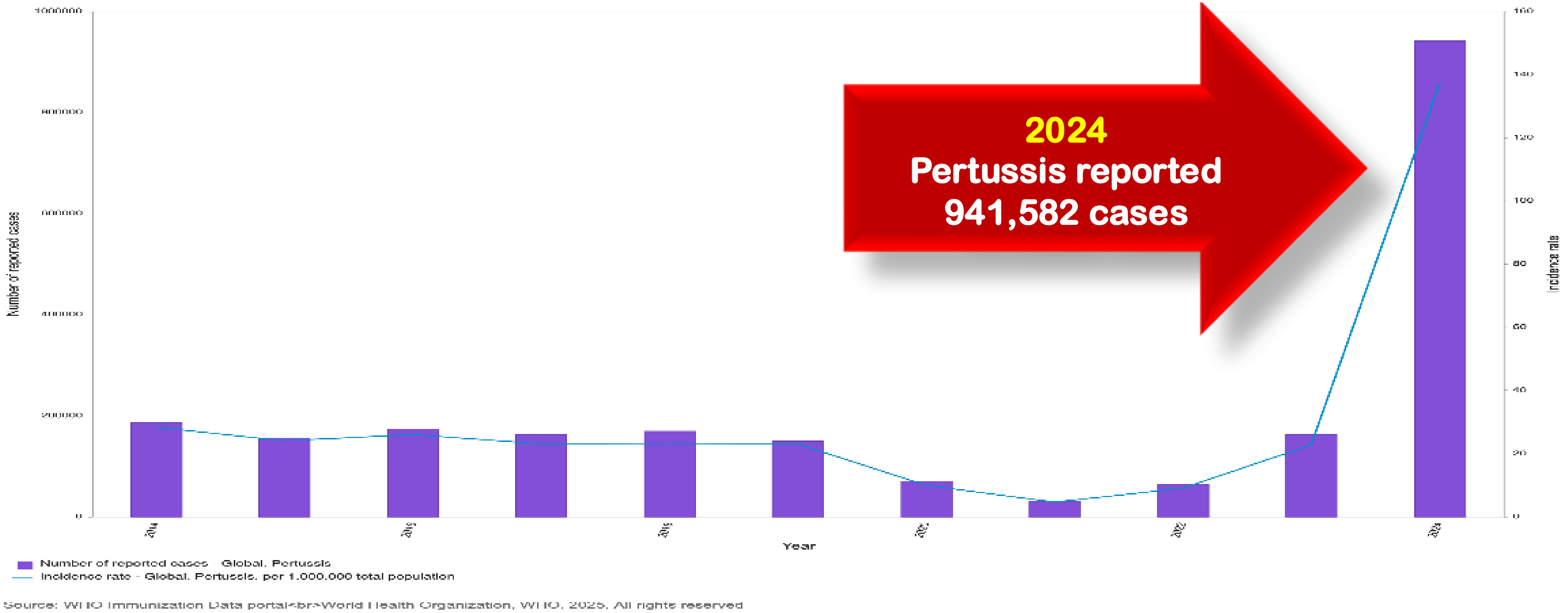
Pneumonia: 22%
Develop severe lung infections.

Neurological: 2%
convulsions |
0.6% encephalopathy

More severe complications include refractory pulmonary hypertension, pneumothorax, and rectal prolapse. For infants, Pertussis is a medical emergency.

Pertussis resurgence

Pertussis resurgence

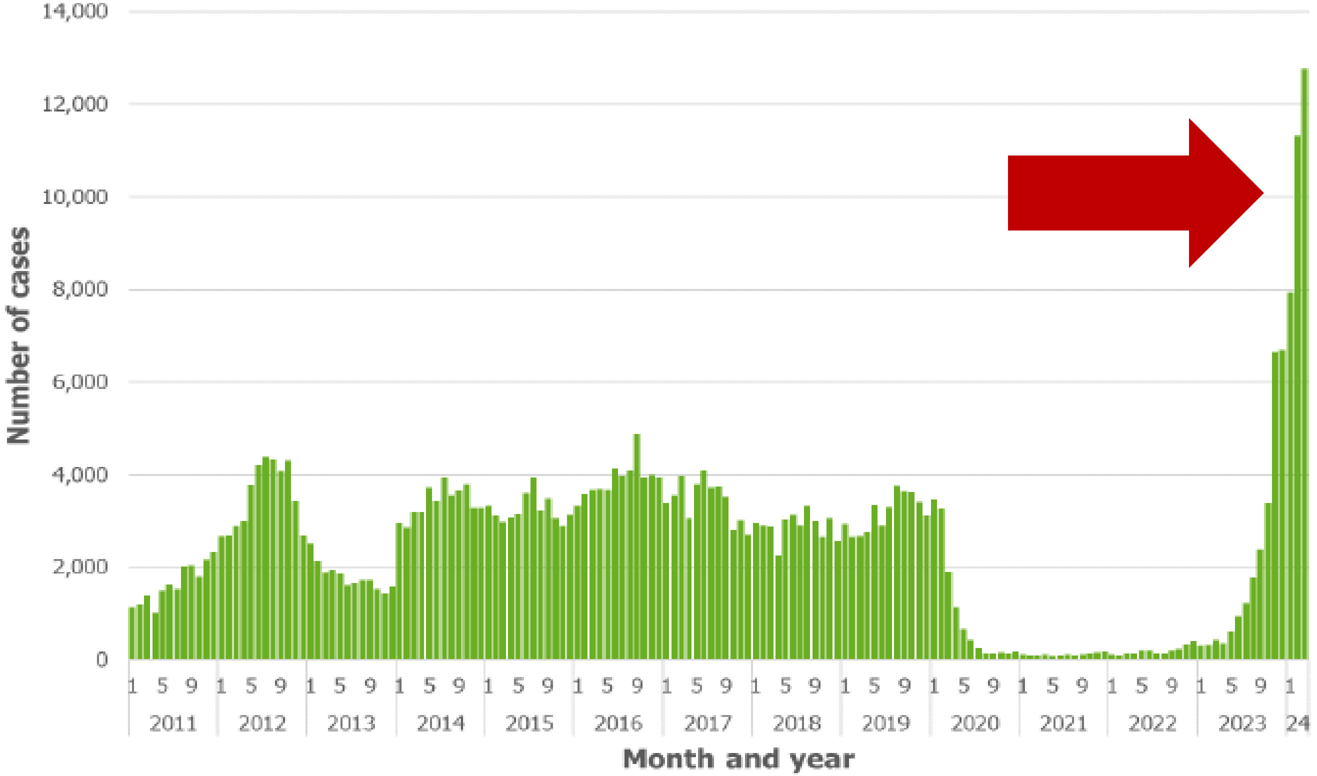


Country / Region	Disease	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Global	Pertussis	186,438	156,065	174,624	163,030	169,240	149,849	69,605	30,402	64,313	163,388	941,582

https://www.who.int/publications/m/item/vaccine-preventable-diseases-surveillance-standards-pertussis

Increase of pertussis cases in the EU/EEA

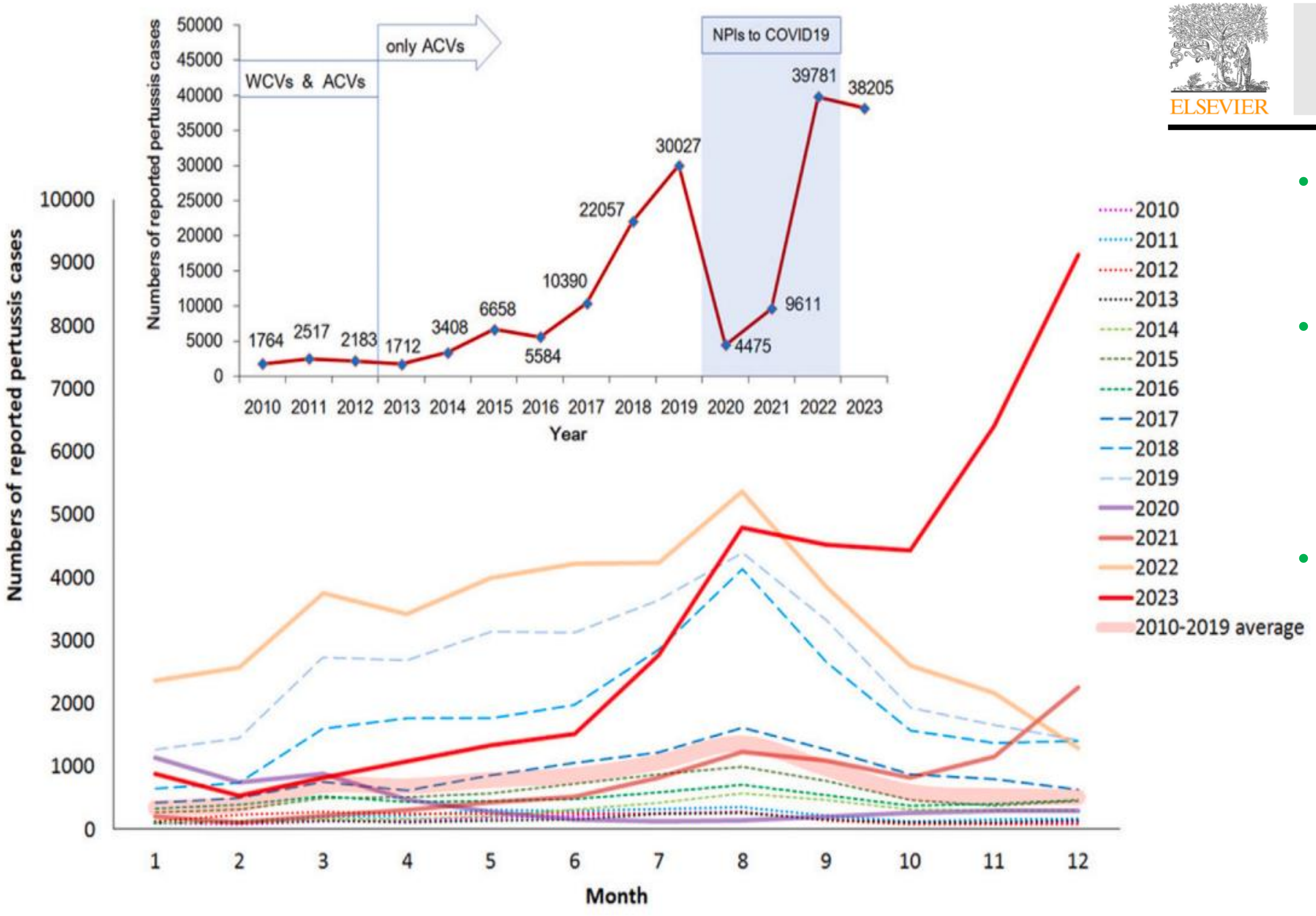
Figure 1. Number of pertussis cases reported to ECDC, by month and year, 1 January 2011 to 31 March 2024[2], EU/EEA[3]



Deaths:

- **Between 2011-2022:**
  - 103 deaths were reported
  - 69 (67%) were in infants
  - 25 (24%) were in adults ≥60 years
- **Between January 2023 and April 2024:**
  - 19 deaths were reported
  - 11 (58%) in infants
  - 8 (42%) in adults ≥60 years

Resurgence and atypical patterns of pertussis in China



Reported pertussis cases by year and by month in mainland China, 2010–2023.

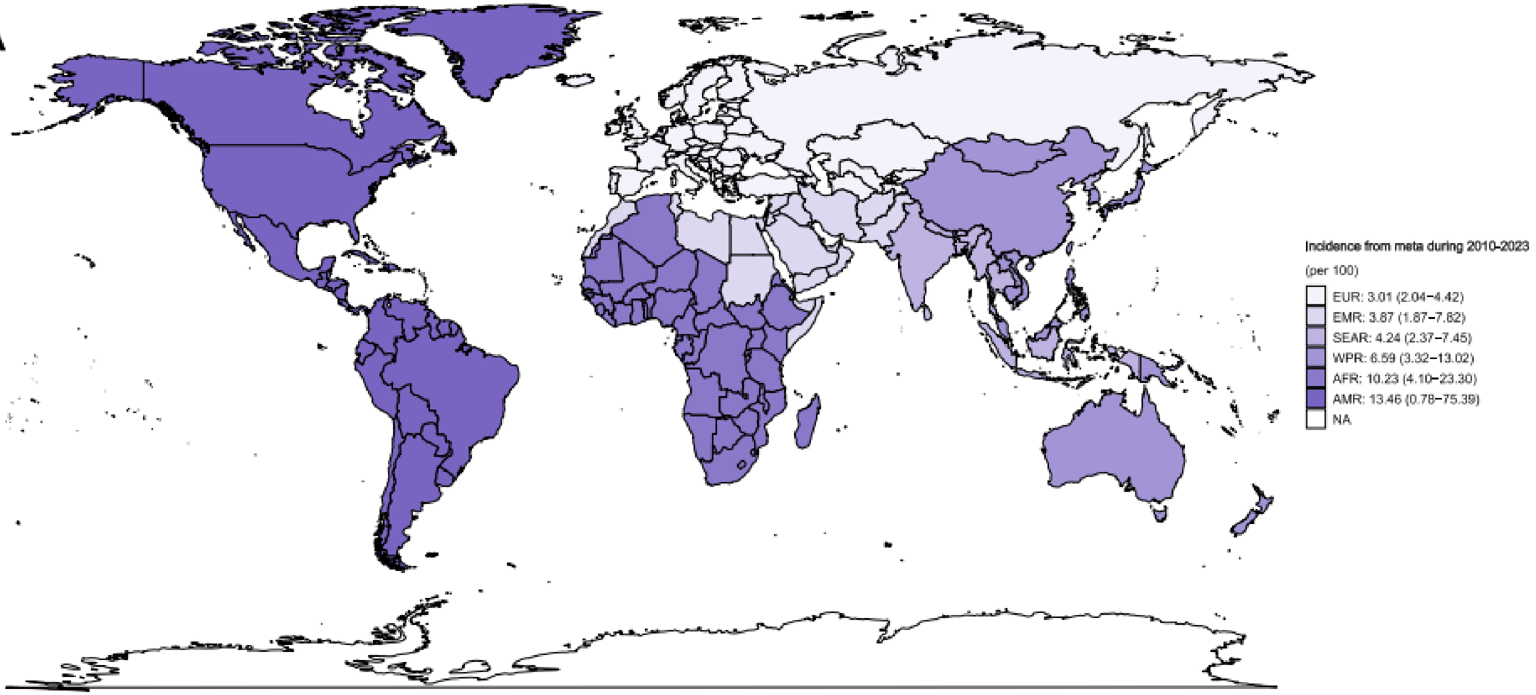
Journal of Infection 88 (2024) 106140
Contents lists available at ScienceDirect
Journal of Infection
journal homepage: www.elsevier.com/locate/jinf
ELSEVIER

- A marked increase in reported pertussis cases following the exclusive administration of aP in China's NIP
- Key factors: enhanced disease awareness, improved surveillance and diagnostic capabilities through PCR and the relatively brief duration of immunity conferred by aP
- The unusual pertussis epidemic patterns observed in the last 4 years relate to
  - Heightened attention to respiratory infections post-COVID-19 outbreak
  - Broad acceptance of PCR
  - Decreased exposure, interrupted vaccination efforts, absence of booster vaccination during the pandemic

Mengyang G, et al. J Infect. 2024;88:106140.

A global systemic literature review of *Bordetella pertussis* anti-PT IgG seroprevalence 2010-2023

A



164 studies

(systematic review and meta-analysis)

Global estimation:

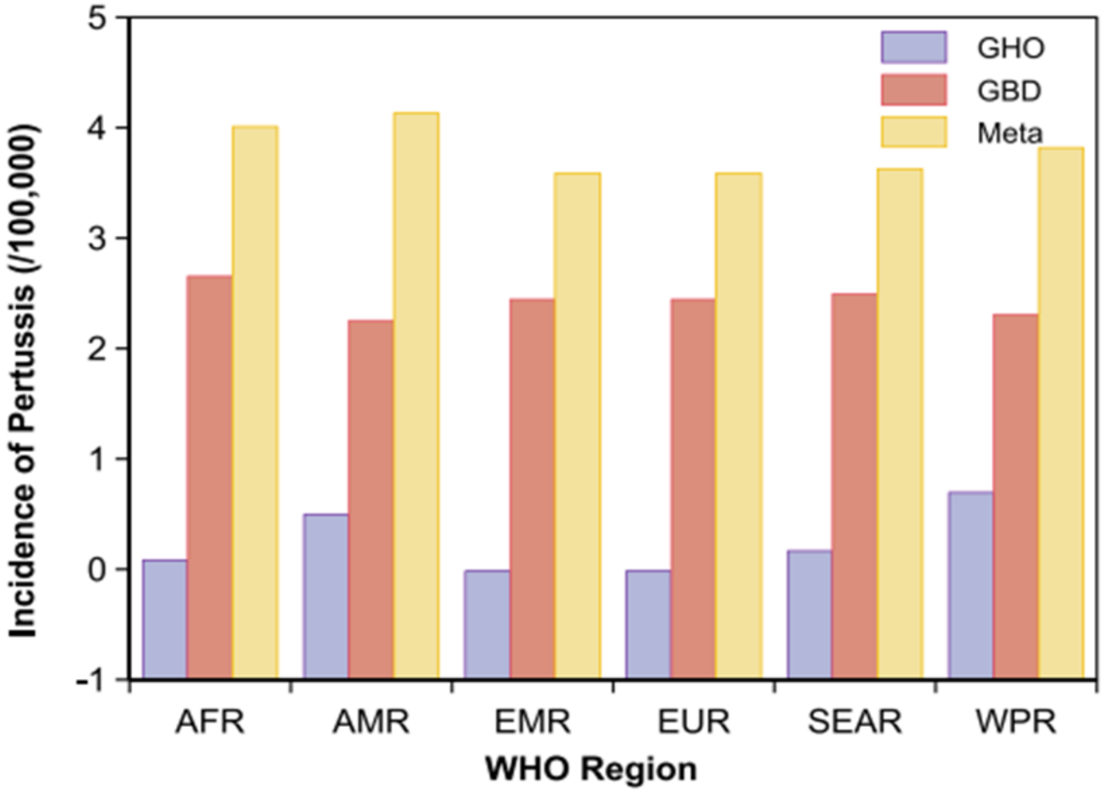
- **Incidence of pertussis infection: 5.1%** (95% CI, 3.0–8.6%)
- **Seroprevalence of anti-PT IgG: 23.4%** (95% CI, 12.63–39.20%)

Age group (yr)	<1	1-3	4-6	7-9	10-19	20-39	40-59	≥60
Estimated infection rate (%)	12.8	13.8	12.3	13.5	10.5	3.8	5.5	9.9
Estimated case (million)	17.6	56.3	50.0	53.5	131.4	87.7	93.8	93.0

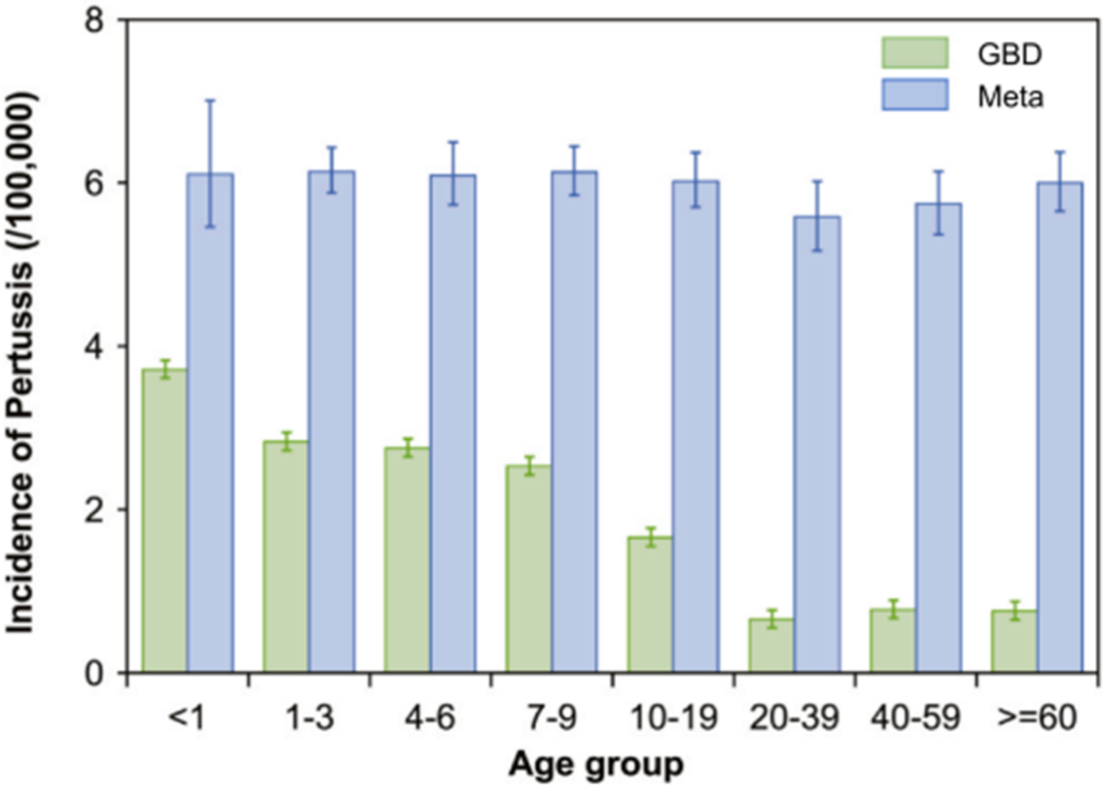
Ma QY, Li T, Gong S, Li J, Wei Y, Hao Y. J Infect. 2026;92:106708.

Burden of pertussis is always underestimated

Comparison of the incidence distribution of pertussis among six WHO regions and age groups from different source data



Comparison of incidence of pertussis by WHO region from three data sources

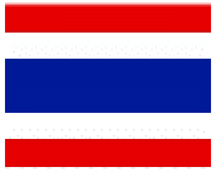


Comparison of incidence of pertussis by age group from GBD and meta

Global Burden of Disease (GBD) and Global Health Observatory (GHO)

Ma QY, Li T, Gong S, Li J, Wei Y, Hao Y. J Infect. 2026;92:106708.

Situation of pertussis in Thailand



19.6% of children [1]

- Prospective study
- Queen Sirikit National Institute of Child Health
- Age: Newborn – 15 yrs.
- Children with cough lasting >1 weeks (paroxysm, whooping cough, vomiting) - PCR

18.4% of adults [2]

- Prospective study
- King Chulalongkorn Memorial Hospital
- Age: 28 – 85 yrs.
- Adult with prolonged cough lasting >2 weeks (14-180 days) - PCR (Nasopharyngeal swabs) and Serology (IgG - PT)

1. Suntarattiwong P et al. Internation Journal of Infectious Diseases. 2019; 81:43-5,
2. Siriyakorn N et al. BMC Infectious Diseases. 2016;16:25

Situation of pertussis in Thailand

2567

จำนวนผู้ป่วย

1,283

อัตราป่วยต่อประชากรแสนคน

21.1

จำนวนผู้เสียชีวิต

1

อัตราตายต่อประชากรแสนคน

0.016

อุทกภัย	(All)
กลุ่มโรค	กลุ่มโรค...
โรค	Pertussis
เขตสุขภาพ	(All)
จังหวัด	(All)
วัน/สถานที่	(Multipl...
ปี	2567
สัปดาห์	(All)
สัญชาติ	(All)
กลุ่มอายุ	(All)
แผนกรักษา	(All)

2568

จำนวนผู้ป่วย

253

อัตราป่วยต่อประชากรแสนคน

4.2

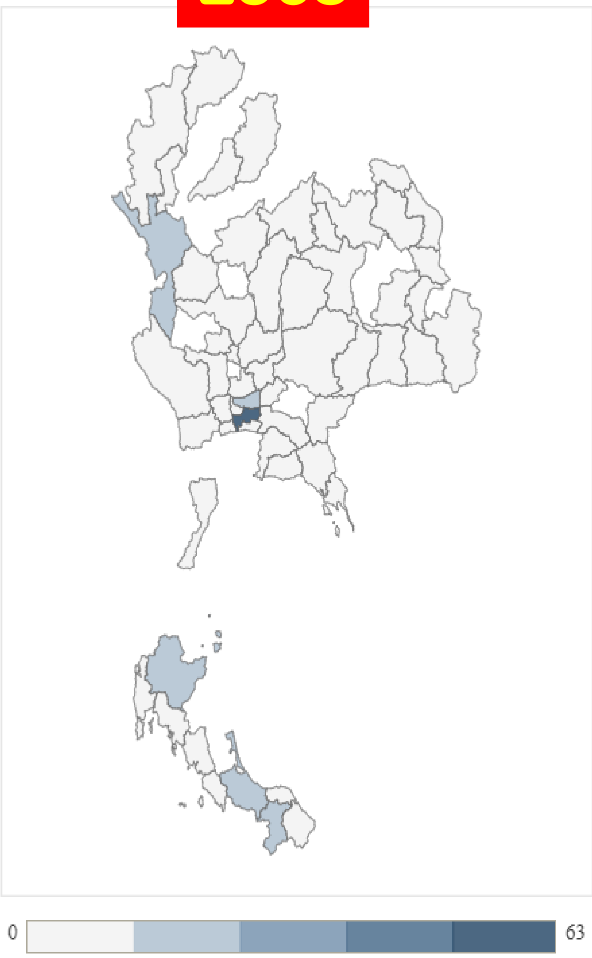
จำนวนผู้เสียชีวิต

1

อัตราตายต่อประชากรแสนคน

0.016

อุทกภัย	(All)
กลุ่มโรค	กลุ่มโรค...
โรค	Pertussis
เขตสุขภาพ	(All)
จังหวัด	(All)
วัน/สถานที่	(Multipl...
ปี	2568
สัปดาห์	(All)
สัญชาติ	(All)
กลุ่มอายุ	(All)
แผนกรักษา	(All)



2569

จำนวนผู้ป่วย

44

อัตราป่วยต่อประชากรแสนคน

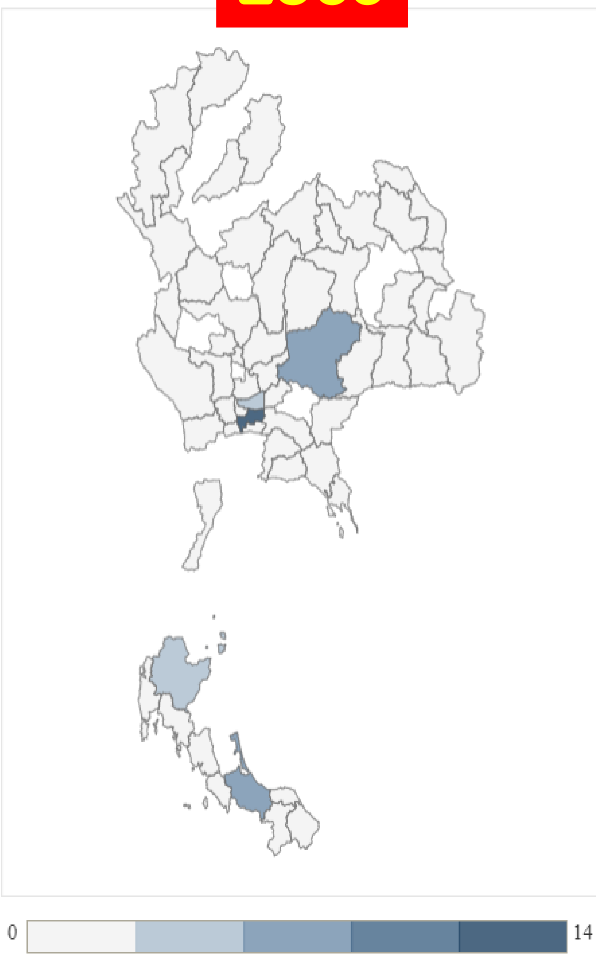
0.723

จำนวนผู้เสียชีวิต

0

อัตราตายต่อประชากรแสนคน

อุทกภัย	(All)
กลุ่มโรค	กลุ่มโรค...
โรค	Pertussis
เขตสุขภาพ	(All)
จังหวัด	(All)
วัน/สถานที่	(Multipl...
ปี	2569
สัปดาห์	(All)
สัญชาติ	(All)
กลุ่มอายุ	(All)
แผนกรักษา	(All)

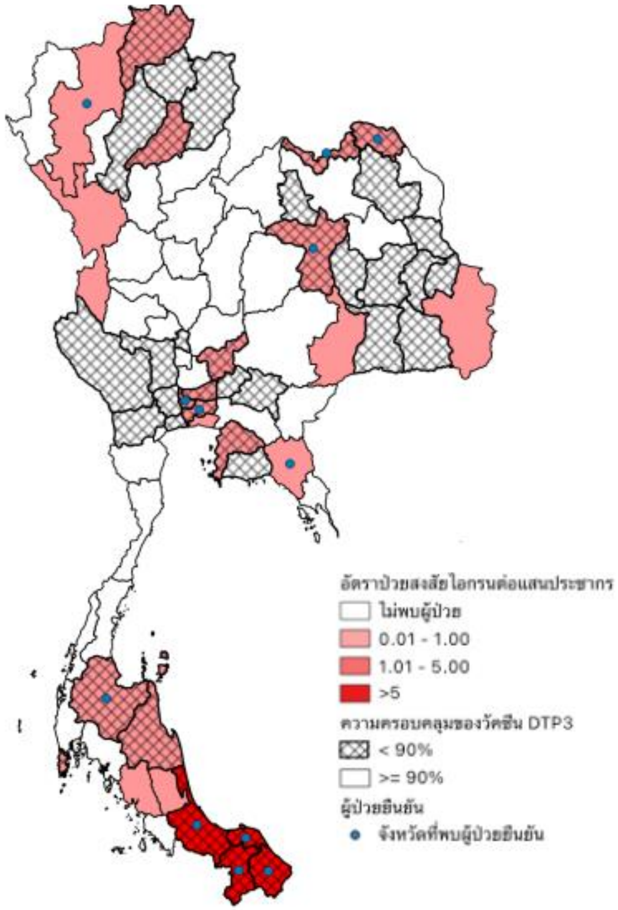


กลุ่มพัฒนาระบบข่าวกรองและเฝ้าระวังโรคไม่ติดต่อ กองระบาดวิทยา กรมควบคุมโรค โทร: 025903899

https://dvis3.ddc.moph.go.th/t/DDC_CENTER_DOE/views/DDS2/sheet35

กองระบาดวิทยา Division of Epidemiology
กรมควบคุมโรค กระทรวงสาธารณสุข
Department of Disease Control, Ministry of Public Health

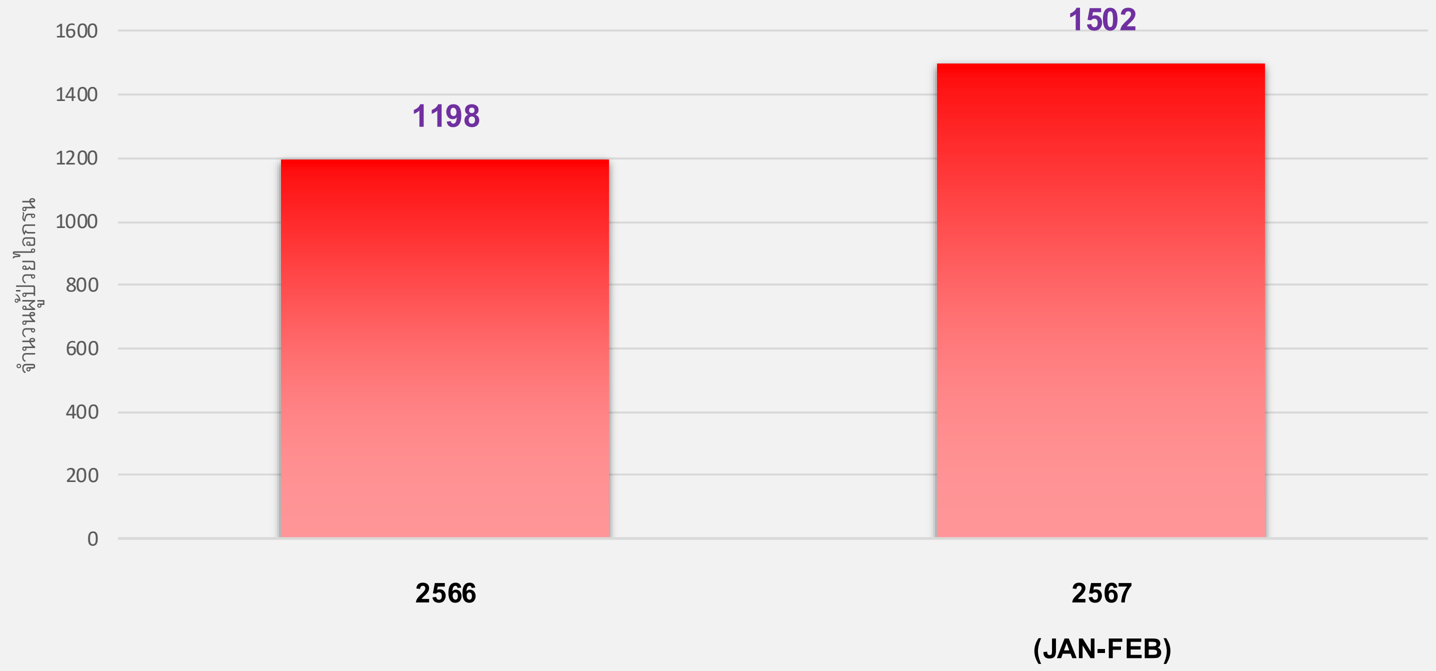
รายงานสถานการณ์โรคไอกรน (Pertussis) ณ วันที่ 1 มีนาคม 2567



รูปที่ 2 อัตราป่วยสงสัยไอกรน ความครอบคลุมของวัคซีน DTP3 จำแนกรายจังหวัด และจังหวัดที่พบผู้ป่วยยืนยัน

รายงานผู้ป่วยโรคไอกรนของประเทศไทย

1 มกราคม 2566 ถึง 28 กุมภาพันธ์ 2567



พบจำนวนผู้ป่วยสะสมรวม 2,700 ราย

อัตราป่วย 4.1 ต่อประชากรแสนคน

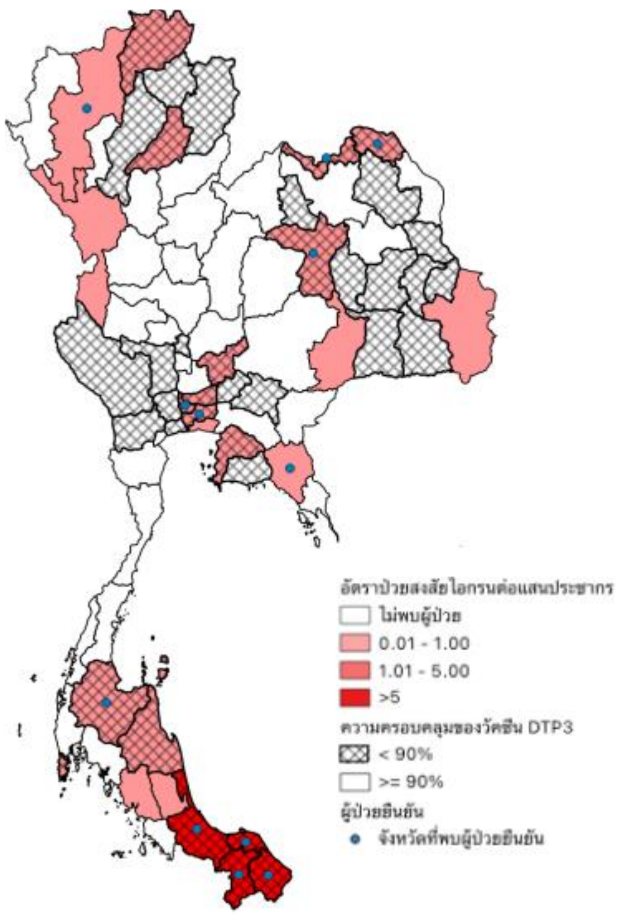
(2015: Report pertussis case rate 0.3/100,000 population)

https://ddc.moph.go.th/doe/pagecontent.php?page=1481&dept=doe

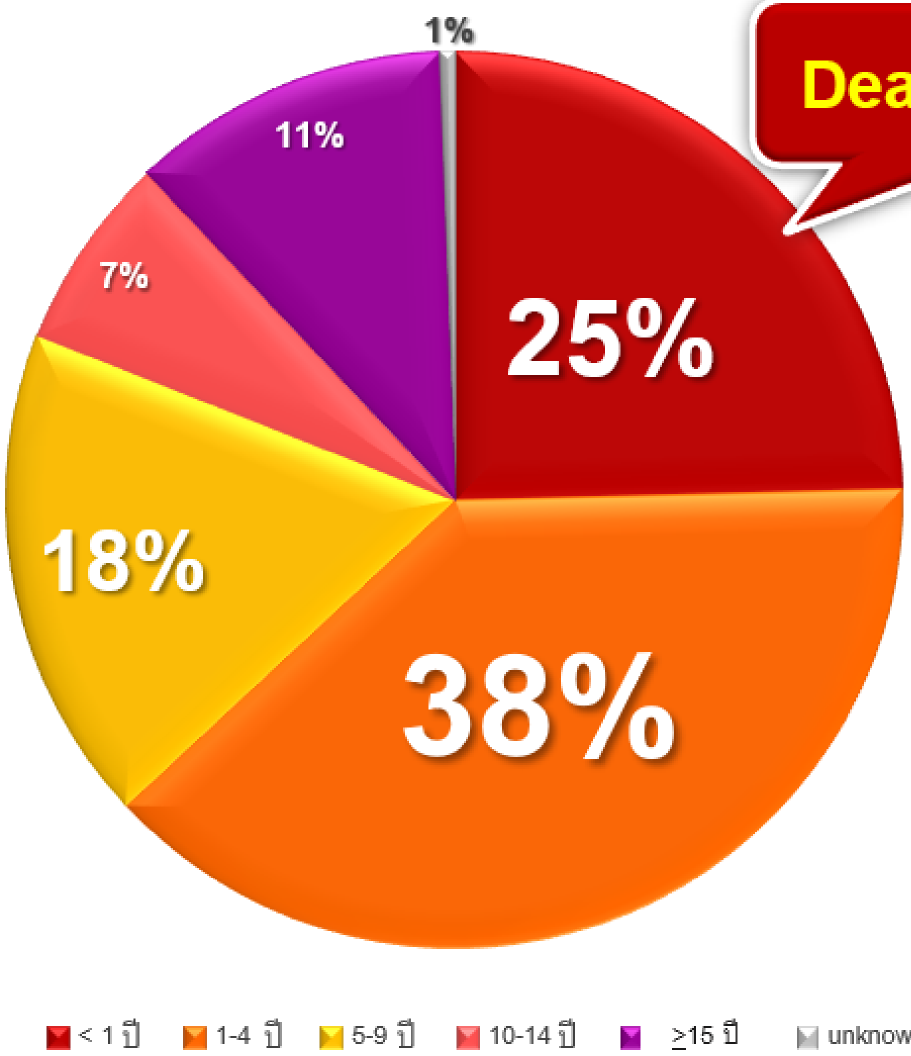
ร้อยละผู้ป่วยโรคไอกรน (แบ่งตามช่วงอายุ)

อายุระหว่าง 10 วัน – 91 ปี (มัธยฐานอายุ 3 ปี)

รายงานสถานการณ์โรคไอกรน (Pertussis) ณ วันที่ 1 มีนาคม 2567

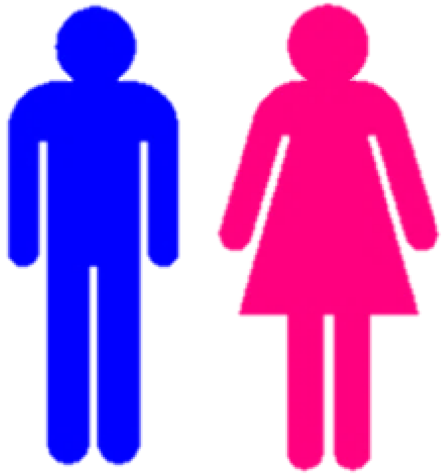


รูปที่ 2 อัตราป่วยสงสัยไอกรน ความครอบคลุมของวัคซีน DTP3 จำแนกรายจังหวัด และจังหวัดที่พบผู้ป่วยยืนยัน



Death: 7 case

Age: 18 days to 3 months

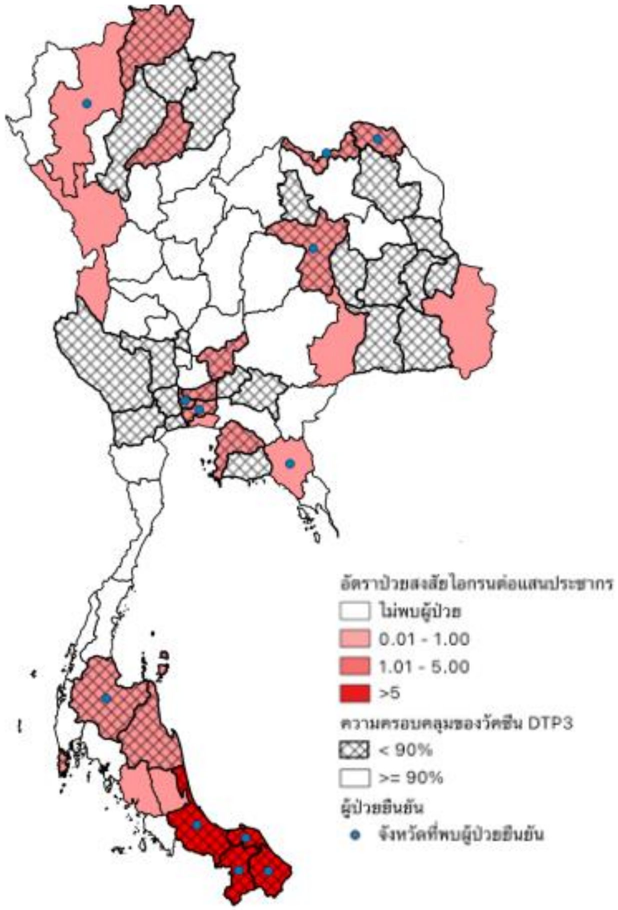


ชาย : หญิง

(1.11) (1)

https://ddc.moph.go.th/doe/pagecontent.php?page=1481&dept=doe

รายงานสถานการณ์โรคไอกรน (Pertussis) ณ วันที่ 1 มีนาคม 2567



รูปที่ 2 อัตราป่วยสงสัยไอกรน ความครอบคลุมของวัคซีน DTP3 จำแนกรายจังหวัด และจังหวัดที่พบผู้ป่วยยืนยัน

ความครอบคลุมของวัคซีน DTP3 ของประเทศไทย ปี 2566

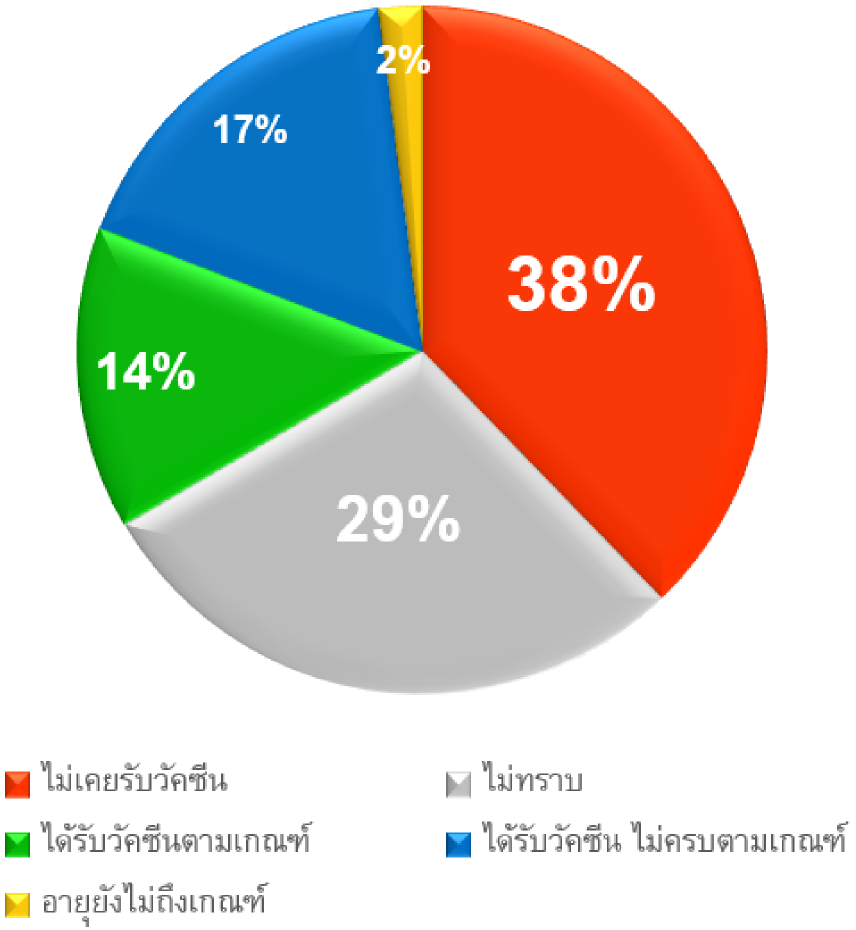
88.32%

เป้าหมาย

>90%

ประวัติการได้รับวัคซีนป้องกันไอกรน

ของผู้ป่วยทั้งหมด 2,700 ราย



https://ddc.moph.go.th/doe/pagecontent.php?page=1481&dept=doe

กลางเดือนพฤศจิกายน 2567

ไอกรนระบาด โรงเรียนสาธิต มศว.ปทุมวัน ประกาศปิด 2 สัปดาห์ ให้นักเรียนทุกคนเรียน./Retrieved 7 Dec 2024,// from/https://www.komchadluek.net

Pertussis prevention

PERTUSSIS VACCINES ARE FOR "EVERYONE"

The Master Defense Plan: Four Layers of Protection

1. Maternal Immunization: Protecting before birth.

2. Childhood Primary Series: Building the infant's own shield.

3. The Cocooning Strategy: Securing the perimeter.

4. Early Identification & Isolation: Rapid tactical response.

Because no single intervention is perfect, preventing Pertussis requires a layered, cooperative approach.

Defense Layer 1: Maternal Vaccination

The Tool: Tdap vaccine.

The Schedule: Recommended during each pregnancy, ideally between 20–32 weeks gestation.

The Mechanism: Maternal antibodies cross the placenta to provide immediate, passive immunity.

The Impact: Highly effective at preventing severe disease, hospitalization, and infant death during their most vulnerable first months of life before their own vaccines begin.

สแกนเพื่อดาวน์โหลด
คำแนะนำในรูปแบบอิเล็กทรอนิกส์

ตารางที่ 2 คำแนะนำการให้วัคซีนป้องกันโรคสำหรับผู้ใหญ่และผู้สูงอายุที่มีโรคประจำตัว หญิงตั้งครรภ์ และบุคลากรทางการแพทย์

Vaccines	Pregnancy	Healthcare personnel	Heart disease, diabetes, or chronic lung disease	Chronic kidney disease	Chronic liver disease	Asplenia	HIV	Immunocompromised	Post-transplantation	Traveler
Tetanus, diphtheria, and pertussis	1 dose of Tdap, TdaP, or aP	Boost with 1 dose of Td every 10 years Substitute one-time of Td with Tdap or TdaP								
Influenza	1 dose	1 dose annually							See text	1 dose annually
COVID-19	1 dose	1 dose annually						1–2 dose(s)		
Measles, mumps, and rubella		2 doses					2 doses if CD4 ≥200 and ≥15%		SOT HSCT	2 doses
Varicella		2 doses					2 doses if CD4 ≥200 and ≥15%		SOT HSCT	2 doses
Hepatitis A virus	2 doses				2 doses	2 doses	2 doses		2 doses	2 doses
Hepatitis B virus	3 doses	3 doses		See text	3 doses		See text	3 doses	SOT: 4 doses HSCT: 3 doses	3 doses
Human papillomavirus										
Pneumococcal										
Respiratory syncytial virus	Bivalent 1 dose		1							
Live-attenuated zoster										
Recombinant zoster										
Dengue			2 or 3 doses depending on types of vaccine							
Japanese encephalitis										
Meningococcal										
Yellow fever										
Rabies										

HSCT: hematopoietic stem cell transplantation; SOT: solid organ transplantation; Unit of CD4 is cells/mm^3

Recommended for adults with age requirement or lack evidence of protective immunity

Recommended for adults with an additional risk factor

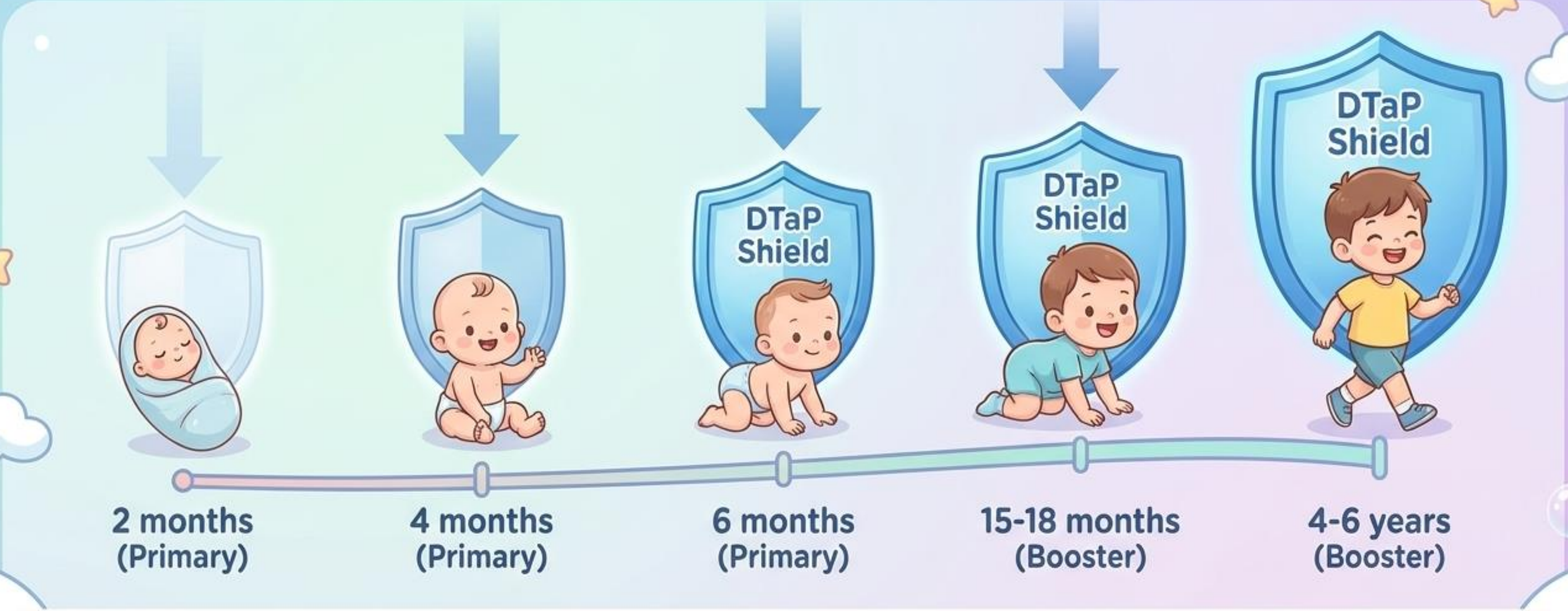
Consider (optional va
adults with age requ

วัคซีนสำหรับหญิงตั้งครรภ์และหญิงให้นมบุตร
(Vaccination for Pregnant Women and Lactating Women)

วัคซีนบาดทะยัก คอตีบ และไอกรน

- แนะนำให้ Tdap หรือ TdaP 1 โดส หรือ Td ร่วมกับ aP อย่างละ 1 โดส ในช่วงอายุครรภ์ 20–32 สัปดาห์ แต่พิจารณาให้ได้ตั้งแต่อายุครรภ์ 16 สัปดาห์ กรณีที่ไม่ได้รับวัคซีนในช่วงดังกล่าว พิจารณาให้ในช่วงอายุครรภ์หลังจากนั้นหรือหลังคลอดได้
- ถ้าหญิงตั้งครรภ์ได้รับวัคซีน Td มาไม่เกิน 10 ปี พิจารณาให้เป็น aP ได้
- กรณีไม่เคยมีประวัติได้รับวัคซีนบาดทะยักมาก่อน แนะนำให้ Td ที่ 0, 1, 6 เดือน โดยให้ Tdap หรือ TdaP หรือ Td ร่วมกับ aP ทดแทน Td 1 ครั้งที่อายุครรภ์ 20–32 สัปดาห์

Defense Layer 2: Childhood Primary Series



Protection is not complete after the first dose. Infants remain highly vulnerable before completing the primary series, which is why surrounding defenses are necessary.

ตารางการสร้างเสริมภูมิคุ้มกันโรคในเด็กและวัยรุ่นไทย แนะนำโดย สมาคมโรคติดเชื้อในเด็กแห่งประเทศไทย 2569

วัคซีนจำเป็นที่ต้องให้กับเด็กทุกคน

วัคซีน / อายุ	แรกเกิด	1 เดือน	2 เดือน	4 เดือน	6 เดือน	9-12 เดือน	18 เดือน	2-2½ ปี	4-6 ปี	11-12 ปี
บีซีจี[1] (BCG)	BCG									
ตับอักเสบบี[2] (HB)	HB1	(HB2)	DTwP-HB-Hib-1	DTwP-HB-Hib-2	DTwP-HB-Hib-3					
คอตีบ-บาดทะยัก-ไอกรนชนิดทั้งเซลล์[3] (DTwP)							DTwP กระตุ้น 1		DTwP กระตุ้น 2	Td และ ทุก 10 ปี
ฮิบ[4] (Hib)										
โปลิโอ[5] ชนิดฉีด (IPV) และกิน (OPV)			IPV1	IPV2	OPV1		OPV กระตุ้น 1		OPV กระตุ้น 2	
โรต้า[6] (Rota)			Rota1	Rota2	(Rota3)					
หัด-คางทูม-หัดเยอรมัน[7] (MMR)						MMR1	MMR2			
ไข้สมองอักเสบเจอีชนิดเชื้อมีชีวิต[8] (live JE)						JE1		JE2		
ไข้หวัดใหญ่[9] (Influenza)					Influenza ให้ปีละครั้ง (การให้ครั้งแรกในเด็กอายุต่ำกว่า 9 ปี ให้ 2 เข็ม ห่างกัน 1 เดือน)					
เอชพีวี[10] (HPV)										เด็กหญิงประถม 5 ตามแผนฯ ของ กระทรวงสาธารณสุข

วัคซีนอื่น ๆ หรือภูมิคุ้มกันสำเร็จรูป ที่อาจให้เสริม หรือทดแทน

วัคซีน / อายุ	แรกเกิด	2 เดือน	4 เดือน	6 เดือน	12-15 เดือน	18 เดือน	2-2½ ปี	4 ปี	6 ปี	9 ปี	11-12 ปี	18 ปี
คอตีบ-บาดทะยัก-ไอกรน ชนิดไร้เซลล์[3] หรือ ชนิดทั้งเซลล์ (DTaP/DTwP, Tdap, หรือ TdaP) ตับอักเสบบี โปลิโอ[5] ชนิดฉีด (IPV) ฮิบ[4] (Hib)		DTaP/DTwP-HB-IPV-Hib1	DTaP/DTwP-(HB)-IPV-Hib2	DTaP/DTwP-HB-IPV-Hib3		DTaP-IPV-(Hib4) กระตุ้น 1		DTaP-IPV หรือ Tdap-IPV หรือ TdaP กระตุ้น 2			Tdap หรือ TdaP ต่อไป Td หรือ Tdap/TdaP ทุก 10 ปี	
นิวโมคอคคัสชนิดคอนจูเกต[11] (PCV)		PCV1	PCV2	(PCV3)	PCV4							
ไข้สมองอักเสบเจอีชนิดเชื้อไม่มีชีวิต[8] (Inactivated JE)				JE1, JE2 ห่างกัน 4 สัปดาห์ และ JE3 อีก 1 ปี								
อีวี 71[12] (EV71)				EV71 2 เข็ม ห่างกัน 1 เดือน								
ตับอักเสบเอ[13] (HAV)					HAV ชนิดเชื้อไม่มีชีวิต ให้ 2 ครั้ง ห่างกัน 6-12 เดือน							
อีสุกอีใส[14] (VZV) หรือวัคซีนรวม หัด-คางทูม-หัดเยอรมัน-อีสุกอีใส (MMRV)					VZV1 (หรือ MMRV1)	VZV2 (หรือ MMRV2)						
วัคซีนไข้หวัดใหญ่[9] (Influenza)				Influenza ให้ปีละครั้ง (การให้ครั้งแรกในเด็กอายุต่ำกว่า 9 ปี ให้ 2 ครั้ง ห่างกัน 1 เดือน) ชนิดเชื้อไม่มีชีวิตฉีดเมื่ออายุ 6 เดือนขึ้นไป ชนิดเชื้อมีชีวิตพ่นจมูกเมื่ออายุ 2 ปีขึ้นไป								
เอชพีวี[10] (HPV)										HPV 2 เข็ม ห่างกัน 6-12 เดือน		
ไข้เลือดออก[15] (DEN)								DEN 2 เข็ม ห่างกัน 3 เดือน				
พิษสุนัขบ้า[16] (Rabies) ก่อนการสัมผัสโรค	2 ครั้งห่างกันอย่างน้อย 7 วัน											
วัคซีนไข้กาฬหลังแอ่นซีโรกรุ๊ปบี[17] (MenB)		MenB1	MenB2		MenB3							
โควิด-19 (COVID-19)				ดูคำแนะนำในการฉีดตามคำแนะนำของกรมควบคุมโรค กระทรวงสาธารณสุข และราชวิทยาลัยกุมารแพทย์แห่งประเทศไทย								
ภูมิคุ้มกันสำเร็จรูปป้องกันอาร์เอสวีชนิดรุนแรง[18]	แนะนำ 1 เข็มในเด็กอายุน้อยกว่า 1 ปี (ดูคำแนะนำในการฉีดตามคำแนะนำของราชวิทยาลัยกุมารแพทย์แห่งประเทศไทย)											

ฉบับวันที่ 3 กุมภาพันธ์ 2569

Defense Layer 3: The Cocooning Strategy

The Strategy:

Vaccinating all close contacts (parents, siblings, caregivers, healthcare workers) with the Tdap vaccine to create an infection-free perimeter around the baby.

The Reality Check:

While cocooning is incredibly useful for reducing exposure risk, it relies on perfect compliance. It is considered a secondary defense measure to maternal vaccination.

Defense Layer 4: Tactical Response & Control

Treatment & Post-Exposure Prophylaxis (PEP)

- Antibiotics: Early treatment with macrolides (Azithromycin, Clarithromycin, Erythromycin) decreases transmission.
- PEP Target: Most effective within 21 days of exposure. Crucial for household contacts, pregnant women, and high-risk individuals.

Isolation Protocols

- Avoid school/daycare and use droplet precautions in healthcare settings.
- Infectious Period: Maintain isolation until 21 days after cough onset (if untreated) OR until 5 days after starting effective antibiotic therapy.

Protecting Our Little Ones: Fighting RSV, Flu, & Pertussis
The Big Risk for Tiny Lungs
RSV
Flu
Pertussis
Infants under 6 months are at highest risk.
This age group faces the greatest danger of severe disease and complications.
A major cause of hospitalization and death.
These three infections remain primary drivers of severe respiratory illness in children worldwide.
A heavy burden on families.
Beyond health, these illnesses
Our Shield of Protection
Maternal and routine vaccinations.
Protecting mothers and following childhood schedules creates a strong defense for babies.
RSV Monoclonal Antibodies.
Specialized antibody treatments offer a modern way to prevent severe RSV infections.
COMMUNITY CENTER
Strengthening access is essential.
Improving immunization programs is the key to reducing the global burden of disease.